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Staged Access and Liability for Dual-Use Artificial Intelligence

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ABSTRACT

How should regulators combine staged access to a dual-use AI model with developer liability? I model a defender and an adversary searching for the same software flaws. After release, liability induces more defensive search but also makes the adversary search harder, limiting its effect on harm. Exclusive access removes this strategic response, so staged access and liability are complements in protection. Defensive effort rises towards the release date, making additional delay progressively less productive. Optimal evaluation windows are therefore bounded, and their response to greater harm saturates. Because defensive search uses real resources, the efficient liability rate can be below full internalisation of harm. That rate generally cannot induce the developer to choose the regulator's preferred release date, leaving a distinct role for a timing mandate.

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1 Introduction

Over several months beginning in early 2026, an unreleased frontier AI model was pointed at the world's open-source code. Across more than a thousand projects, it surfaced 6,202 findings initially rated high or critical severity; of those subsequently assessed, mostly by independent security firms, nine in ten were validated.¹ Seven months earlier, the same family of technology had been used offensively: a state-linked group employed an agentic coding model in what its developer described as the first largely autonomous cyber-espionage campaign ([Anthropic, 2025](#)). The capability that finds vulnerabilities for defenders to patch can also find them for attackers to exploit. Its developer has some control over who receives that capability first.

Policy has begun to respond with two instruments, neither calibrated to the contest it changes. The first governs timing. Executive Order 14409 of June 2026 establishes a voluntary framework under which frontier developers give the United States government early access to models for as long as thirty days before releasing them to other trusted partners, and creates a clearinghouse to coordinate AI-assisted vulnerability discovery and patching. The thirty-day figure is an offered heuristic rather than a calibrated mandate.² The second instrument is liability, where policy remains unsettled: current regimes range from transparency and third-party audits to civil penalties and product liability for failures to supply security updates.³ Neither has been designed using a model of the economic problem on which it acts: a contest between an adversary and a defender over the flaws that a released model can expose and find.

How long should defenders receive exclusive access, and how much of the resulting harm should the model's developer bear? The questions cannot be separated. Liability determines how intensively the defender searches, while exclusivity determines whether the adversary can answer that search. After release, stronger defensive incentives induce an adversary response. Before release, they do not. Staged access therefore increases liability's protective leverage, while liability gives the developer a reason to finance the search that makes the window useful.

¹Project Glasswing, a coordinated vulnerability research initiative built around Anthropic's Claude Mythos Preview, reported 23,019 findings on open-source code in its initial update, of which 6,202 were initially rated high or critical severity. Of the 1,752 then assessed by one of six independent security firms or by Anthropic itself, 90.6 per cent were validated and 62.4 per cent confirmed at high or critical severity ([Anthropic, 2026](#)). By late May, the count of high- or critical-severity findings across the initiative's roughly fifty partner organisations—a wider net than the open-source scan—exceeded ten thousand.

²Executive Order 14409, "Promoting Advanced Artificial Intelligence Innovation and Security", 2 June 2026. Because the framework is voluntary and expressly disclaims licensing or preclearance, it supplies an institutional basis for staged access and a calendar convention rather than a mandate. The thirty-day figure follows a longer lineage of round-number timing conventions: coordinated vulnerability disclosure has operated for two decades on a ninety-day clock. Whether a mandate improves on a voluntary arrangement is the comparison evaluated in Section 6.

³California's SB 53 deprioritises liability in favour of transparency; New York's RAISE Act attaches civil penalties; Illinois adds mandatory third-party audits; and the European Union scrapped its AI Liability Directive even as the AI Act's obligations for general-purpose models approach Commission enforcement. The revised Product Liability Directive (EU) 2024/2853, which Member States must transpose by December 2026, extends strict liability to standalone software and AI systems and makes a manufacturer answerable for damage traceable to security updates it failed to supply. It comes closest to the object modelled here: liability for realised harm from an unpatched flaw, borne by the developer rather than imposed as a penalty for a procedural violation. The liability rate should therefore be read as the expected share of realised harm borne by the developer after allowing for causation and proof.

The model considers a software system containing an unknown stock of latent flaws. During an evaluation window, a defender searches alone. After release, the defender and an adversary search the remaining stock, each at increasing marginal cost. A flaw leaves the stock when either side finds it: the defender patches it or the adversary exploits it. Those outcomes have different welfare consequences but leave the same continuation stock, reducing the dynamic problem to a contest played flaw by flaw. In Sections 3–5, the regulator commits ex ante to a deterministic release date and a liability rate; the date is not revised after findings. The developer chooses capability and whether to build, but chooses the release date only under delegation in Section 6.

At release, liability is a weak instrument on its intensive margin. Harder defensive search reduces the rent from the adversary’s existing effort and thereby raises the value of searching harder. The adversary responds too. Under quadratic costs and a fixed attack population, the defender wins a larger share of flaws, but the pass-through from a stronger defensive stake into a lower exploited share remains below one-half.

Effective exclusivity removes that response. The defender searches alone during the window, yet its effort is not constant: a flaw missed early can still be found later, so effort rises towards the committed deadline. Postponing release shifts this profile, effectively adding time at the beginning where search is weakest. Under instantaneous remediation, effective exclusivity and a lead-constrained search technology—one whose discovery rate scales with the number of flaws remaining—this response makes protection decay as a power law. For a fixed model, every optimal finite staged window is bounded, and greater harm can increase the date only towards a finite limit; when release no longer beats the outside option, the policy is withholding rather than indefinite evaluation. Remediation lags and findings-contingent stopping are separate problems.

The same distinction explains why the two instruments should be chosen jointly. After release, the adversary partly offsets the search induced by liability; before release, the induced search is uncontested. Defensive evaluation uses real resources, so a rate that fully internalises harm still induces too much search at the margin, and the efficient rate for a fixed model falls short of that benchmark. Greater protection depth—the defender’s search intensity multiplied by the calendar it is granted—raises the efficient rate along each regular stationary branch because the same liability has more protective leverage while the adversary is absent.

Endogenous capability adds the model-selection and participation margins. A more capable model can lower both sides’ search costs and expand the reachable stock of flaws. An exclusive window creates a temporary asymmetry because only the defender can use the model before release. The fixed-model regime partition and stationary liability condition are therefore analytical building blocks for the mandated package rather than its global solution. The full problem nests the developer’s capability choice, participation boundary and no-development option; it reduces to a global search over headway—the length of the exclusive window—once the feasible liability choice is solved at each date.

Delegation reveals what liability leaves out. At full appropriability and full compensation, the developer and planner choose the same date, capability and development decision. Yet the fixed-

model efficient rate is partial, and one rate then cannot generally implement both the desired search incentive and an active interior date. A very high rate need not exclude development either: the developer can delay until the expected residual stock, and with it variable evaluation expenditure, becomes small. A stock-independent programme overhead makes this calendar evasion harder and can eliminate it, but only when the limiting-surplus condition fails. Control of the date retains a margin that liability does not.

None of this identifies thirty days, or any other round number, as the appropriate window. The policy object is protection delivered while the adversary lacks comparable access, not calendar time by itself. Its value depends on benefit decay, exposed harm, search responsiveness and the frontier-fringe capability gap—the frontier model’s productivity advantage over the best openly available capability. Delegation is most plausible when developers capture most of the benefit, the desired liability rate is near full compensation, and exclusion is not at issue. A mandate has a distinct role when those conditions fail.

The analytical core combines patent-race theory with information security. The contest is a [Lee and Wilde \(1980\)](#) flow-cost race played flaw by flaw over a depleting stock. Strategic complementarity reverses the familiar response in [Loury \(1979\)](#), while no discounting gives the race–contest equivalence in [Baye and Hoppe \(2003\)](#). [Gal-Or et al. \(2026\)](#) study release timing, bounties and remediation without a depleting pool or an endogenous deadline; [Rescorla \(2005\)](#) and [Ozment and Schechter \(2006\)](#) study depletion without a strategic adversary; and [Anderson \(2002\)](#) gives an early symmetric-improvement neutrality argument.

The disclosure literature, particularly [Arora et al. \(2008\)](#), studies a planner inducing a vendor to patch. Here the liability-bearing developer is the principal that finances and selects the defensive programme’s intensity, while timing works through the defender’s effort profile—the time path of its search intensity—and the benefit-exhaustion logic of [Mortensen \(1977\)](#). [Landolt et al. \(2026\)](#) provide the closest independent analysis of AI release timing: a discrete release game in which defenders and adversaries choose costly substitute-acquisition effort, the laboratory internalises a share of harm, and capability arrival rates are elicited and release-dependent, the central comparison being a relative capability gap—the defender holds the new model while the adversary remains a generation behind. The distinction here is equilibrium feedback: the exploited share and the effort profile respond jointly to the calendar and the liability rate, which is what generates the pass-through bound and the rate-balance rule. Mapped into the present model, their capability-gap comparison holds the attack-surface, benefit, development-cost and absolute search-speed levels fixed, and an equal capability gain on the two sides then removes only the exploited-share channel (the share of resolved flaws that the adversary exploits).

The liability analysis builds on [Shavell \(1984\)](#), the judgment-proof problem in [Shavell \(1986\)](#), and the informational complementarity between regulation and liability in [Kolstad et al. \(1990\)](#). The complementarity here is instead technological: exclusivity prevents the adversary from matching induced defensive search. The partial-rate result also differs from [August and Tunca \(2011\)](#) and [Lam \(2016\)](#), since it arises from defensive search expenditure rather than user incentives.

Some models make delay valuable because information arrives while policymakers wait or because withdrawal preserves reversibility (Acemoglu and Lensman, 2024; Guerreiro et al., 2023; Henry et al., 2022; McClellan, 2022; Koh and Sanguanmoo, 2024, 2026); aggregate work instead trades growth against catastrophic risk (Jones, 2024). Gans (2025) supplies the contrasting learning-by-doing case, in which adoption can accelerate learning about harms. Delay here instead gives the defender time to convert capability into protection even when findings have no informational value. Appendix B.3 prices findings-contingent release under a fixed effort schedule and leaves the joint effort-and-stopping problem open. The governance literature supplies staged access as an institution (Shevlane, 2022; Solaiman, 2023; Anderljung et al., 2023); the model isolates effective exclusivity as the margin that changes liability’s return.

Section 2 characterises the contest. Section 3 derives the committed window, Section 4 adds capability choice, and Section 5 chooses the mandated package. Section 6 then gives the release date to the developer. Appendix A contains proofs, and Appendix B collects extensions.

2 The flaw contest

2.1 Setup

The model has a defender and an adversary searching the same software system. There is one adversary throughout and no entry.⁴ The system may contain many latent flaws, but the relevant stock consists of those within reach of the AI model at its given capability. At a fixed capability, both parties search the same effective stock; as capability rises, that stock may expand by bringing previously inaccessible flaws within reach. Every flaw either side finds leaves the pool—patched or exploited—so the two searchers are dividing a stock between them. The question the baseline model answers is how that stock gets divided, and what each side spends to divide it in its favour.

Initially, the effective stock is $N_0 \sim \text{Poisson}(m_0)$, and neither party observes its realised size. Let $n(t)$ denote the number unresolved at date t , and write it simply as n when the date is clear. Both parties search continuously. The recursion in n below is a complete-information auxiliary calculation; in the hidden-state implementation, the continuation value is the posterior expected stock times the per-flaw value.⁵

The natural way to write a search technology is to let each party buy an arrival rate: it spends, and a flaw turns up at a hazard of its choosing, whatever the size of the remaining pool. For ease of notation, I write it the other way around. Party i chooses a *per-flaw search intensity* $x_i \geq 0$ applied to each unresolved flaw, under which every live flaw is independently discovered at hazard x_i . In the observed-state auxiliary problem, defensive discoveries then arrive at rate nx_D , adversarial discoveries at rate nx_A , and flow cost is $nC_i(x_i)$. Conditional on a discovery, the adversary finds

⁴The results in the paper hold at a *fixed attack population*; Appendix B.1.4 adds N_A symmetric adversaries and free entry.

⁵The equilibrium claim is confined to stationary Bayesian strategies in public histories. Appendix B.1.1 extends uniqueness to differentiable pure public-Markov strategies of the posterior mean stock; uniqueness among arbitrary calendar- or history-dependent intensity processes is not asserted.

the flaw first with probability $x_A/(x_A + x_D)$. What each party chooses is diligence per lead rather than findings per week: it sets how hard each live flaw is hunted, and the code supplies the leads.

The per-flaw specification and its aggregate counterpart are two poles of the same technology. More generally, search can run through some number of parallel channels, each operated at intensity x_i . Any common factor multiplying both the discovery rate and flow cost cancels from the problem per flaw resolved. It changes how quickly the stock drains, not how the contest divides it.⁶ Section 2.7 nests the two by allowing a stock of n flaws to support n^α channels.

Both cost functions are increasing, strictly convex, and satisfy $C_i(0) = C'_i(0) = 0$; for closed forms I use the isoelastic family $C_i(x) = c_i x^\kappa / \kappa$ with common curvature $\kappa > 1$, and the quadratic case $\kappa = 2$ throughout the text. The curvature is the model's measure of the responsiveness of search: $\kappa - 1$ is the elasticity of marginal cost, so a large κ describes a party that searches at much the same intensity whatever is at stake, and the limit $\kappa \rightarrow \infty$ is perfectly inelastic effort. The *fixed-hazard benchmark* of the security literature (e.g., Rescorla (2005)) is that limit, and much of what follows turns on κ being finite.⁷

Information and search budgets. Let $\mathcal{F}_t$ contain the public finding history and all auditable search inputs, and let $\hat{m}_t = \mathbb{E}[N_t \mid \mathcal{F}_t]$. In the hidden-stock model, party i operates a search programme with flow budget $\hat{m}_t C_i(x_i)$, while conditional discoveries occur at rate $N_t x_i$ (the *posterior-budgeted* specification). Conditional expected discoveries and search cost are therefore both proportional to $\hat{m}_t$, which cancels from the intensity first-order condition. For every $\mathcal{F}_t$ -measurable intensity, the tower property gives $\mathbb{E}[\hat{m}_t C_i(x_i)] = \mathbb{E}[N_t C_i(x_i)]$, so under the Poisson, $\alpha = 1$ benchmark, the formulation delivers exactly the expected-cost and survival formulas used below. Expenditure adds no signal about the realised stock and does not jump to zero when an unobserved stock happens to be empty.

Contracting benchmark. To make analysis straightforward, it is assumed that the developer is the principal on the defensive side. That said, interpretations where liability falls on the developer for all costs naturally follow. The developer chooses or audits the defensive intensity, reimburses the defender's search cost, and pays a fixed fee satisfying the defender's participation constraint; I normalise the defender's reservation payoff to zero, and any policy-independent fee can be folded into the developer's fixed development outlay. Given adversarial intensity x_A , the developer therefore chooses x_D to minimise

$$\frac{x_A \gamma \lambda + C_D(x_D)}{x_A + x_D},$$

⁶The aggregate reading is standard in the patent-race literature (Lee and Wilde, 1980). With many flaws, the per-flaw reading treats each live flaw as an independent contest. Under the aggregate reading, the flaws share a search establishment; because that establishment scales arrivals and costs together, the division of the stock is unchanged.

⁷The condition $C'_i(0) = 0$ is load-bearing: it delivers interior search at every strictly positive stake, arbitrarily slow and inexpensive evaluation, and the limiting pathology at zero liability. A positive marginal cost at the origin creates a deterrence corner and may create an initial interval of zero window effort; Appendix B.1.2 develops the contest corner but not the full window problem. Asymmetric curvatures are treated in Appendix B.1.6, and many adversaries in Appendix B.1.4.

which is exactly the defender problem used below. Liability changes the developer’s desired search intensity; contractibility implements that intensity rather than making liability irrelevant. The regulator’s instrument set is (h, γ) ; direct effort regulation is outside the model.⁸

A discovery resolves the flaw it concerns, and there are two ways to be resolved. If the adversary finds it, the flaw is exploited at once: the adversary collects a private prize $v > 0$, and society bears harm $\lambda > 0$, of which the developer pays $\gamma\lambda$. Here $\gamma > 0$ is the *liability rate*: $0 < \gamma < 1$ is *partial liability*, $\gamma = 1$ is *full compensation*, and $\gamma > 1$ is *supra-compensatory liability*. Let d_L denote legally compensable damages per exploit and $\tilde{\gamma}$ the statutory multiplier. Define the effective internalisation rate by $\gamma = \tilde{\gamma}d_L/\lambda$. Thus $\gamma = 1$ means full internalisation of net social harm; it implies a unit statutory multiplier only when $d_L = \lambda$.⁹ If the defender finds the flaw, it is patched as captured by the following assumption:

Assumption 1. *The baseline has instantaneous remediation: a defensive discovery at date t neutralises the flaw at date t .*

Treating exploitation as resolution bundles into λ the whole discounted harm of an incident—a burned zero-day is patched once its use is discovered—and the adversary’s private gain, v , may be far below λ .¹⁰

The parties who deploy patches may be different again, often third-party maintainers. Assumption 1 buys the tractability of everything that follows, and what it costs is the pipeline: in practice, a found flaw takes time to become a deployed patch (in the introduction example, patch creation alone averaged two weeks), and while the patch is pending, a found-but-unpatched flaw is a third kind of object. Appendix B.4.1 relaxes it.

2.2 Dynamic outcomes

The two ways of resolving flaws have distinct welfare consequences, but whichever agent finds the flaw, it is removed and n falls by one. I call this transition property *winner-independence*; it is what simplifies the game.

Lemma 1. *Suppose payoffs are undiscounted. Then the state transition is independent of which party makes the discovery, continuation values enter both parties’ objectives additively and cancel from both first-order conditions, equilibrium intensities are independent of n , and both value functions are linear in n .*

All proofs of results stated in the main text are in Appendix A. The auxiliary recursion says that the n -flaw game is a single contest played n times, not a dynamic program. At a realised state n ,

⁸If effort is hidden and only discoveries are contractible, a moral-hazard problem replaces this benchmark, and the results below should be read as the full-contracting case.

⁹Compensable damages d_L need not equal λ : damages may be capped, harms may fall on parties who cannot sue, causation may be discounted. “Full compensation” is accordingly used throughout in the sense of full internalisation of social harm, not of a unit statutory multiplier (Section 5.2).

¹⁰The welfare account does not treat the adversary’s private prize v as a social benefit or a domestic transfer. A model that distinguishes compensable loss from net social harm requires a second loss parameter and changes the timing and liability conditions.

expected time to the next resolution is $1/(n(x_A + x_D))$, and the same factor n scales each side's aggregate flow cost $nC_i(x_i)$, so the remaining-stock size cancels from the per-resolution account and the two parties' values satisfy

$$V_A(n) = \max_{x_A} \left\{ \underbrace{\frac{x_A}{x_A + x_D} v}_{\text{prize} \times \text{win probability}} - \underbrace{\frac{C_A(x_A)}{x_A + x_D}}_{\text{flow cost} \times \text{expected wait}} \right\} + V_A(n-1), \quad (1)$$

and symmetrically for the defender, whose per-flaw problem is to minimise

$$\frac{x_A \gamma \lambda + C_D(x_D)}{x_A + x_D}.$$

The optimised terms do not depend on n . In the hidden-stock implementation, conditioning on $\mathcal{F}_t$ therefore gives the posterior expected stock times the same per-flaw continuation value. Intuitively, neither party's position improves or deteriorates according to who wins: a patched flaw and an exploited flaw leave the same board behind. Each side is playing for the current flaw and nothing else.¹¹ This reduces the dynamic game to a static one.¹²

2.3 Equilibrium

Each party's problem in (1) is a payoff *per resolution*: prize times the chance of winning, less flow cost times the expected wait. Write g for the adversary's expected gain per flaw and ℓ for the defender's expected loss per flaw. In the notation of (1), g is the maximised bracket, so that $V_A(n) - V_A(n-1) = g$ and, by Lemma 1, $V_A(n) = ng$; ℓ is the minimised value of the defender's per-flaw problem, defined as a positive loss, so that $V_D(n) = -n\ell$. These two objects are the contest's sufficient statistics, and the first-order conditions place each party's marginal cost equal to its *net stake*:

$$\underbrace{C'_D(x_D^*) = \ell}_{\substack{\text{defender: marginal cost} \\ = \text{its all-in loss per flaw}}}, \quad \underbrace{C'_A(x_A^*) = v - g}_{\substack{\text{adversary: marginal cost} \\ = \text{prize net of the rent earned anyway}}} . \quad (2)$$

Two consequences follow. The primitives determine the division of flaws through one ratio of stakes to costs, and $C'_i(0) = 0$ keeps both parties active even when that ratio is very lopsided. The next proposition makes both statements exact under quadratic costs.

¹¹The contrast is with the multi-battle contest literature, where the state *does* depend on who wins, and this is where discouragement and momentum effects come from (Harris and Vickers, 1987; Konrad, 2009). A related discrete-state precedent is Segal and Whinston (2007, Section 6.2), where affine flow profits generate state-independent value increments.

¹²Dynamics return when the transition symmetry breaks. A remediation lag makes a defensive find differ from an exploit because the flaw remains live while its patch is pending (Appendix B.4.1); discounting makes an earlier win more valuable and, plausibly, favours the adversary in deep pools—Appendix B.1.3 records that direction as a conjecture. The undiscounted benchmark instead treats an exploit as equally costly whenever it occurs, while w below measures obsolescence rather than impatience. Baye and Hoppe (2003) prove race–contest equivalence in the no-discounting limit. This occurrence valuation also fits security incidents whose harm does not depreciate with their date.

Proposition 1. *Any equilibrium with both intensities positive satisfies (2). Under quadratic costs, a unique interior stationary Bayesian equilibrium exists in which the same intensities are sequentially optimal after every public history, and the exploited share $p \equiv x_A^*/(x_A^* + x_D^*)$ is the unique root in $(0, 1)$ of*

$$\Theta = \frac{(1-p)^2(1+p)}{p^2(2-p)}, \quad \text{where} \quad \Theta \equiv \frac{\gamma\lambda c_A}{v c_D}, \quad (3)$$

with intensities $x_A^* = 2v(1-p)/(c_A(2-p))$ and $x_D^* = 2\gamma\lambda p/(c_D(1+p))$.

The two conditions in (2) have a clear interpretation. The defender searches until its marginal cost equals ℓ —not the gross liability stake $\gamma\lambda$ an exploit imposes, but the per-flaw loss that nets out the chance the defender finds the flaw itself and folds in what it spends doing so. The adversary searches until its marginal cost equals the prize net of the rent it earns anyway: what an extra unit of search buys is not v but $v - g$, since some of that prize was already coming. Both conditions place marginal cost strictly below the gross stake, so neither party ever drives the other out: the equilibrium is interior for every configuration of the primitives.¹³

The economics of the contest sits in Θ , which I call the *stake–cost ratio*: the defender’s stake times the adversary’s cost parameter, over the adversary’s prize times the defender’s. The exploited share depends on the primitives *only* through this ratio, though the search intensities, of course, do not. In other words, what determines how a system’s flaws get divided is not how valuable the target is, nor how cheap search has become, but the relative position of the two sides on both counts at once. At $\Theta = 1$ the contest is balanced and $p = \frac{1}{2}$.¹⁴

2.4 The limits of pass-through

How hard is p to move? Since the liability rate enters Θ proportionally, the answer is also the answer to how much a liability regime can accomplish on this margin alone. Write $\varphi(\Theta) \equiv -d \log p / d \log \Theta$ for the *pass-through* of the stake–cost ratio into the exploited share.

Proposition 2. *The exploited share is strictly decreasing in Θ , with $p \rightarrow 1$ as $\Theta \rightarrow 0$ and $p \sim (2\Theta)^{-1/2}$ as $\Theta \rightarrow \infty$. The pass-through is strictly increasing and satisfies*

$$0 < \varphi(\Theta) < \frac{1}{2} \quad \text{for all } \Theta > 0. \quad (4)$$

Proposition 2 bounds what the exploited share can be made to do. Doubling the liability rate cuts

¹³The interiority is exact: $c_A x_A^*/v = 2(1-p)/(2-p)$ and $c_D x_D^*/\gamma\lambda = 2p/(1+p)$, both strictly inside $(0, 1)$ at every $p \in (0, 1)$. An adversary facing an arbitrarily well-resourced defender still searches, because the defender’s own convex cost stops it short of certainty. Under quadratic costs the per-flaw stakes are $\ell = 2\gamma\lambda p/(1+p)$ and $g = vp/(2-p)$; for general curvature, $\ell = \kappa\gamma\lambda p/(\kappa - 1 + p)$. Note that x_A and x_D are Poisson intensities—discoveries per flaw per unit of time—not probabilities; the objects that are probabilities, the exploited share p and the survival factor of Section 3, are bounded automatically.

¹⁴ Θ is not a technological parameter. One of its four arguments, γ , is a policy instrument here, and another, v , could be lowered by a separate enforcement or penalty regime that lies outside the package analysed, so the offence–defence balance is something policy partly chooses. Slayton (2017) formulates the offence–defence balance in net-value terms. The contest-specific stake–cost ratio Θ is closer to the scaling question in Garfinkel and Dafoe (2019).

Θ	0.05	0.25	0.5	1	2	4	10	50
p	0.850	0.695	0.602	0.500	0.398	0.305	0.206	0.097
φ	0.08	0.18	0.24	0.30	0.36	0.41	0.45	0.48

Table 1: The exploited share and its pass-through under quadratic costs. The pass-through approaches one-half only as the contest becomes very lopsided.

that share by less than $1 - 2^{-1/2}$, or 29.3 per cent, and by even less when liability is already weak: at $\Theta = 0.25$ the pass-through is 0.18, so a doubling of γ removes about an eighth of the exploited flaws (Table 1).

Intuitively, the bound arises because the adversary matches. Raising the defender’s stake raises x_D , which is the intended effect. But the two intensities are strategic complements: a more determined defender provokes a more determined adversary.¹⁵ The adversary equates marginal cost not to the prize but to the prize net of the rent it would collect anyway. A higher x_D shrinks that rent and raises the effective marginal prize $v - g$, giving back part of what the larger defensive stake achieved.¹⁶

2.5 Welfare outcomes

Because every flaw is eventually resolved, welfare accounting at the baseline is a per-flaw multiple of m_0 (the expected number of flaws at time 0). Expected harm is λp per flaw. Both agents’ search outlays are pure dissipation: they buy no flaws that would not otherwise have been found, only a better chance of being the finder. Under quadratic costs ($\kappa = 2$), expected outlay per flaw is

$$\underbrace{\frac{\gamma \lambda p(1-p)}{1+p}}_{\text{defensive search}} \quad \text{and} \quad \underbrace{\frac{v p(1-p)}{2-p}}_{\text{adversarial search}}. \quad (5)$$

The *defence-inclusive accounting* counts harm and defensive outlay as real social costs. Thus social loss per flaw at immediate release is $\lambda p + \gamma \lambda p(1-p)/(1+p)$, and the schedule analysis of Section 3 carries both terms.¹⁷ Adversary benefits and costs are not included in welfare calculations.¹⁸

¹⁵Differentiating the adversary’s condition in (2) gives $dx_A/dx_D = (v - C'_A(x_A))/(C''_A(x_A)(x_A + x_D)) > 0$. The adversary’s intensity can rise even as its share falls.

¹⁶This is the flow-cost reversal of Lee and Wilde (1980): with flow rather than sunk costs, greater rivalry stimulates search. The bound is quadratic-specific in level but not in kind: in the defender-dominant limit $\Theta \rightarrow \infty$, pass-through approaches $1/\kappa$ (an asymptote derived in Appendix A), which falls as costs become more convex and exceeds one-half sufficiently close to linear costs; in the adversary-dominant limit $\Theta \rightarrow 0$, pass-through tends to zero. This is not a global $1/\kappa$ bound for arbitrary κ ; Appendix B.1.6 maps the distinction. With N_A symmetric adversaries, pass-through remains below one-half and, at $\Theta = 1$, falls asymptotically as $1/N_A$ (Appendix B.1.4); away from the balanced benchmark no ordering across populations is established. Entry adds an extensive margin on which both liability and the window can deter marginal adversaries.

¹⁷Each expression in (5) peaks when its side is the slight favourite—at $p = \sqrt{2} - 1$ for defence and $p = 2 - \sqrt{2}$ for offence—the familiar contest-theoretic point that spending is greatest when the contest is close.

¹⁸Whether resources burned in offence belong in the welfare calculus is the old enforcement question (Stigler, 1970). Counting adversarial outlay as social waste would hand liability a second job—suppressing the adversary’s spending as well as the harm—and with it a case for liability above full compensation (Section 5.2). A second convention, *harm-only accounting*, counts harm alone and appears here only as a benchmark: it is the right reading of a liability bill—courts

The feature of (5) that matters is that the cost parameters have vanished. Both outlays depend on c_A and c_D only through p , and hence only through the ratio c_A/c_D inside Θ . Making search cheaper for both parties in the same proportion leaves the exploited share unchanged, by (3), and then leaves both outlays unchanged. Each side simply buys more search with the saving until its equilibrium condition binds again. In other words, an equiproportional cost reduction—through a more capable model, for example—is fully dissipated.

2.6 Capability: neutrality and surface growth

Capability a may also change the parties' relative search costs and enlarge the stock of flaws within reach. Once the neutral common-cost component is set aside, these are the two channels through which capability changes baseline harm. They are not symmetric: surface growth reaches harm one for one, whereas a defensive cost advantage acts only through the pass-through of the contest. Write the corresponding elasticities as

$$\sigma_A \equiv -\frac{d \log c_A}{d \log a}, \quad \sigma_D \equiv -\frac{d \log c_D}{d \log a}, \quad \sigma_m \equiv \frac{d \log m_0}{d \log a},$$

all weakly positive, and define the *dual-use gap* $\Delta \equiv \sigma_D - \sigma_A$, positive when capability favours the defender. Since $d \log \Theta / d \log a = \Delta$, baseline expected harm $L(0) = \lambda p m_0$ responds to capability as

$$\frac{d \log L(0)}{d \log a} = \underbrace{\sigma_m}_{\text{surface growth}} - \underbrace{\varphi(\Theta) \Delta}_{\text{exploited-share shift}}, \quad (6)$$

so capability raises baseline harm whenever $\sigma_m > \varphi \Delta$, for which $\sigma_m > \frac{1}{2} \Delta$ is sufficient regardless of how defence-biased the technology is.¹⁹ A positive σ_m does not mean that capability creates new flaws: $m_0(a)$ is the expected stock within reach of a model of capability a , so a stronger model can bring previously inaccessible flaws onto the effective attack surface; if m_0 instead denotes the fixed physical stock of all flaws, then $\sigma_m = 0$.

Equation (6) separates two things that are often conflated in arguments about whether frontier capability is dangerous. The first is the *dual-use gap*: whether the model helps defenders more than adversaries. The second is the *surface elasticity*: whether the model, by being more capable, creates more to attack. The equation says the first is bounded and the second is not. Because $\varphi < \frac{1}{2}$ always, even a technology that lowers defensive search costs and does nothing whatever for the

price realised harm, not evaluation payrolls—it is the criterion the fixed-hazard security literature implicitly works with, and it delivers exact closed forms at every curvature, where defence-inclusive accounting does so only at $\kappa = 2$. Section 3.4 records the one sign it changes and Appendix B.2 collects its formulas; every other welfare statement in the paper is defence-inclusive.

¹⁹The knife-edge $\Delta = 0$ —capability lowers both costs equiproportionally—leaves the exploited share untouched and only quickens the pace at which the stock resolves, which in a model with no calendar and a fixed attack surface has no consequence; it acquires one in Section 3, and (6) adds surface growth, so $\Delta = 0$ removes only the exploited-share channel. The gap isolated by Landolt et al. (2026) is instead a relative capability gap between model generations, in a discrete release game whose elicited arrival rates are release-dependent and whose delay and defender-action costs are swept rather than held fixed.

adversary moves the exploited share by less than half of σ_D ; attack-surface growth passes through one for one. The case for defence-dominant AI at the baseline therefore requires not merely that defenders benefit more, but that they benefit more than *twice* as much as the attack surface grows. That is a higher bar than the usual framing suggests, and the timing instrument of the next section exists partly to lower it. Once a window exists, a third channel joins these two: the drawdown of the stock itself.

2.7 The stock elasticity

One choice already made deserves scrutiny before I turn to staged access. The setup gave every live flaw its own search channel, so that aggregate discovery scales one for one with the size of the remaining pool. That is the *lead-constrained* case: a team whose binding constraint is the supply of promising leads finds flaws faster when there are more of them. A red team with a fixed headcount and inference budget is instead *capacity-constrained*. A generalisation lets a stock of n flaws support n^α parallel search channels, with *stock elasticity* $\alpha \in [0, 1]$: $\alpha = 1$ is the setup above and $\alpha = 0$ is the capacity-constrained case. The common factor cancels from each per-resolution objective, so the exploited share, the pass-through bound, the dissipation expressions and (6) hold at every α . Nothing in the contest turns on it.

The clock does not inherit that invariance. Expected depletion obeys a moment equation that closes in the mean only at $\alpha = 1$ (Appendix B.1.5), so the calendar results of Section 3 depend on the elasticity that Section 2 does not. In other words, α sets the clock and the contest sets the exploited share. This is convenient, and not only for tractability. The elasticity of discovery with respect to the remaining stock is the one primitive here that is contested and that this paper cannot settle: the software-reliability literature assumes $\alpha = 1$, sceptics of vulnerability depletion argue in effect for $\alpha \approx 0$ (Rescorla, 2005; Spring, 2023), and no estimate of α exists—the empirical decay curves measure calendar time, not stock, and so do not identify it.²⁰ The design confines the contested parameter to where it does work, in valuing delay, and keeps it out of the baseline.

Assumption 2. *Search is lead-constrained: $\alpha = 1$, so a flaw’s discovery hazard does not depend on how many others remain.*

Assumption 2 is what delivers the survival law of Section 3.2 and every calendar result that rests on it. It asks less of the data than it appears to: because the per-flaw hazard is constant, those results hold flaw by flaw on an arbitrarily large stock of finite mean, and no known upper bound on the number of latent flaws is needed.²¹

²⁰The macro analogue is the fishing-out elasticity in idea production (Bloom et al., 2020; Kortum, 1997).

²¹A proper Poisson variable and finite expected harm both require $m_0 < \infty$, but nothing requires m_0 to be known or small. This answers the computability-based objection that undiscovered flaws are inexhaustible (Spring, 2023). Appendix B.1.5 records the general- α construction, and Appendix B.1.6 the case $\alpha_D \neq \alpha_A$.

3 Staged access

3.1 The committed window

Now let the regulator commit to public release at date h . The baseline is a statutory date announced and enforced *ex ante*; it is not revised in response to findings. That restriction costs less than it appears, because Assumption 2 fixes what an evaluation reveals as well as what it removes. A discovery takes a flaw out of the stock, which is good news about what remains, and reveals that there were flaws to take out, which is bad news; under a Poisson stock at $\alpha = 1$, the two exactly offset, the posterior mean of the residual stock is deterministic, and no findings-contingent rule improves on the best deterministic date under a fixed effort schedule. The offset is a knife-edge: a capacity-constrained technology makes a discovery good news on net, an overdispersed stock makes it bad. Appendix B.3 develops the overdispersed case and leaves the joint effort-and-stopping problem open. Before h , the defender searches alone: the adversary has no comparable discovery technology, a condition I call *effective exclusivity*. At h the adversary gains access and the contest of Section 2 begins on whatever is left. This arrangement is *staged access*; I call h *headway*. Appendix B.4.2 replaces exclusivity with a capability gap. The public benefit of the model at release is $e^{-wh}B$, with $B > 0$ the benefit of immediate release and $w > 0$ the rate at which it decays—obsolescence of the model relative to its successors, not time preference.²²

Because equilibrium contest intensities do not depend on the size of the pool (Lemma 1), the adversary arriving at h plays exactly the contest it would have played at date zero, on fewer flaws. Expected harm, therefore, can be decomposed as follows:

$$L(h) = \lambda \underbrace{p}_{\text{share}} \underbrace{m(h)}_{\text{surviving stock}}, \quad (7)$$

where $m(h)$ is the expected stock surviving the window. Headway is drawdown. It does not change who wins the contest on what remains; it changes how much remains to be contested.²³ The schedule uses the defence-inclusive accounting of Section 2.5: social loss counts harm and the defensive search outlay $O_D(h)$, which the developer finances. The natural intermediate object is

²²If a successor supersedes the model at Poisson rate w , then e^{-wh} is the probability it still has value at release and $1/w$ the expected time to supersession. Some cost of delay is indispensable: at $w = 0$ both harm and the defender's outlay vanish as h grows, and the optimal window is unbounded. The exponential is not the only form that supplies one—a forgone benefit flow makes the cost of delay linear in h and still delivers a finite window—but it is what gives the ceiling of Section 3.4 its shape. Memorylessness is what makes the bound a multiple of κ/w ; a benefit decaying to a positive floor, or superseded at a known date, would bound the window differently.

²³A longer window reduces what the adversary can collect in total, but not its per-flaw prize, so x_A^* is independent of h , and its response is already inside $m(h)$. Its prize elasticity of search, $1 - qp/[(1-p)(2-p)]$, lies in $(\frac{1}{2}, 1)$, but headway never moves that prize. With an entry cost, the window can deter on the extensive margin, in which case (7) holds conditional on the number of adversaries. Appendix B.1.4 illustrates that channel holding the defender's terminal-value problem fixed, identifies where the pure-strategy formulation ceases to be an equilibrium, and leaves the coupled entry-and-search equilibrium open.

Symbol	Meaning
$N_0, m_0, n, m(h), N_A$	initial flaw stock, its mean, the realised unresolved stock, expected stock after headway, and the adversary count
$x_A, x_D, C_A, C_D, c_A, c_D$	adversarial and defensive per-flaw search intensities, flow-cost functions, and isoelastic cost coefficients
ℓ, g	defender's loss and adversary's gain per resolved flaw
α, κ	stock elasticity and search-cost curvature
γ, λ, v	liability rate, social harm per exploit, and adversary's private prize
Θ, p, φ	stake-cost ratio, exploited share, and pass-through
$B(a), h, w, z$	release benefit, headway, benefit-decay rate, and protection depth
$x_D^{\text{eff}}, L_{\text{ch}}$	effective protection rate of the marginal day and chargeable loss
s, η, η_c	decay-search ratio w/x_D^* ; harm-to-benefit and chargeable-loss-to-benefit ratios
L, O_D, M, Λ	expected harm, defensive search outlay, developer burden, and defence-inclusive social loss
$\mathcal{S}, \mathcal{S}_H$	defence-inclusive and harm-only surplus
a, β, Ψ, W, Π	capability, developer appropriability, capability cost, welfare, and developer payoff
$\sigma_A, \sigma_D, \sigma_m, \Delta$	capability elasticities of adversarial cost reduction, defensive cost reduction, reachable stock, and the dual-use gap
$\sigma_L, \sigma_R, \sigma_M$	capability elasticities of baseline harm, defensive intensity, and baseline burden
Y, χ	liability elasticity of harm and the dissipation wedge
$\hat{L}, \hat{W}, \hat{\Pi}$	harm, welfare, and developer payoff evaluated at the induced capability $a^*(h, \gamma)$

Table 2: Main notation.

the developer's *burden*

$$M(h) \equiv \gamma L(h) + O_D(h) = m_0 \ell \left(1 + \frac{z}{\kappa}\right)^{-(\kappa-1)}, \quad z \equiv x_D^* h, \quad (8)$$

the developer's liability payment plus the defensive search it finances, whose closed form is derived in Appendix A; social loss is then $\Lambda(h) \equiv L(h) + O_D(h) = M(h) + (1 - \gamma)L(h)$ and the schedule surplus is $\mathcal{S}(h) = e^{-wh}B - \Lambda(h)$. I call z (the defender's contest search intensity multiplied by the calendar it is granted) the *protection depth*, and it organises every threshold in the paper.

3.2 Deadline effort and power-law protection

The defender's problem on $[0, h]$ is not the contest. There is no rival to beat, and each flaw removed is worth what that flaw would have cost in the contest to come—the per-flaw loss $\ell = C'_D(x_D^*) = c_D(x_D^*)^{\kappa-1}$ by the defender's first-order condition (2) (hence $\ell = c_D x_D^*$ when $\kappa = 2$). That equality gives the window a terminal condition in the same units as the contest: as release approaches, the shadow value of finding a live flaw rises to ℓ , and the defender's search intensity must arrive at the contest intensity exactly as the adversary arrives. The resulting path is the *effort profile*.

Proposition 3. *On $[0, h]$ the defender's optimal per-flaw search intensity $x^*(t)$ is strictly increasing, reaching the contest intensity at release: $x^*(h) = x_D^*$. For every $\kappa > 1$ the reciprocal of the search intensity*

is exactly affine in time-to-release,

$$\frac{1}{x^*(t)} = \frac{1}{x_D^*} + \frac{h-t}{\kappa}, \quad (9)$$

and the stock surviving a window of length h is

$$\frac{m(h)}{m_0} = \left(1 + \frac{z}{\kappa}\right)^{-\kappa} \xrightarrow{\kappa \rightarrow \infty} e^{-z}. \quad (10)$$

Proposition 3 has two parts. The first part says that a mandated window generates a deadline effect. Intuitively, a day early in a long window is not pivotal: if the flaw is missed today, there is a great deal of calendar left in which to catch it, so the shadow value of finding it today is low. As release approaches, that reserve of calendar runs out, the shadow value rises to the full contest stake, and effort rises with it. The mechanism is Mortensen (1977)'s: as benefit exhaustion approaches, the value of remaining search rises to its post-exhaustion level and effort rises into the deadline.²⁴ What is added here is the terminal condition. The shadow value docks onto the equilibrium contest stake, not onto an exogenous outside option.

The second part says that protection decays polynomially in headway, not exponentially, and identifies exactly what an exponential would be assuming. The power law does not come from the stock: with $\alpha = 1$, a fixed per-flaw defensive hazard x_D^{fix} would deliver $e^{-x_D^{\text{fix}}h}$ exactly. It comes entirely from the effort margin—the defender's diligence declines as the calendar lengthens—and the limit in (10) makes the identification precise: exponential survival is the case $\kappa \rightarrow \infty$ of perfectly inelastic effort, a defence team that searches at the same intensity whatever the shadow value. The fixed-hazard benchmark that the security economics literature works with is that limit, and the difference between its policy conclusions and this paper's can be traced to the finite- κ correction. Figure 1 draws the comparison.

Finite κ drives the timing results that follow: protection obeys the power law in (10), the optimal window has a ceiling proportional to κ/w , and its response to harm eventually saturates. In the fixed-hazard limit, the power law becomes exponential, the ceiling recedes to infinity, and saturation disappears.²⁵ This distinction is empirically disciplined rather than assumed: Appendix B.4.4 shows how evaluation data identify κ in two independent ways.

²⁴Bonatti and Hörner (2011) obtain a related deadline effect in a moral-hazard team model: their baseline has linear effort costs and pessimistic learning about project quality, producing a U-shaped effort path whose late rise a convex-cost extension smooths; here there is no learning about quality, and effort rises monotonically. The separation from procrastination models (O'Donoghue and Rabin, 1999) matters for welfare: there effort rises near deadlines because of present bias, and the profile is a pathology; here it rises because the continuation value of failure does, and the profile is efficient. A regulator cannot exhort the early slack away—it can change the profile only by changing γ , c_D , or the contest the window ends in.

²⁵The same limit closes the difference between the defence-inclusive and harm-only timing conditions, which differ by replacing the leading coefficient κ with $\kappa + 1$ (Appendix B.2). It also makes liability progressively less able to change the contest's division: in the defender-dominant limit, pass-through approaches $1/\kappa$ and vanishes as effort becomes inelastic (Appendix B.1.6).

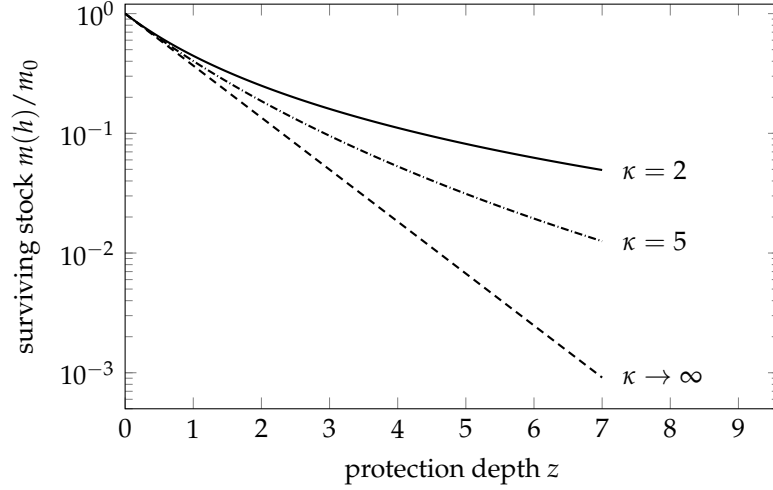


Figure 1: Survival of the flaw stock. Optimising effort (κ finite) delivers power-law decay; the exponential benchmark of the fixed-hazard literature is the limit of perfectly inelastic effort. The gap between the curves is the early slack of the effort profile.

3.3 The marginal day

Differentiating (7) and (10), the marginal day of headway removes harm at the *effective protection rate*

$$-\frac{L'(h)}{L(h)} = x_D^{\text{eff}}(h) \equiv \frac{x_D^*}{1 + z/\kappa} = x^*(0). \quad (11)$$

The equality $x_D^{\text{eff}}(h) = x^*(0)$ is the important one. Delaying release shifts the optimised effort profile towards the new deadline, so the extra day is effectively added at the *beginning* of the window, where effort is lowest, not at the end, where it is highest. Valuing an extension using effort observed just before release therefore overstates its protective value by exactly $1 + z/\kappa$. Moreover, x_D^{eff} falls towards zero as the window lengthens, so the marginal protection from further delay eventually vanishes.

The burden decays at rate $\frac{\kappa-1}{\kappa}x_D^{\text{eff}}$, whereas expected harm decays at x_D^{eff} . Differentiating $\Lambda = M + (1 - \gamma)L$ therefore gives the *chargeable protection flow* and the interior *rate-balance rule*

$$we^{-wh}B = x_D^{\text{eff}}(h) L_{\text{ch}}(h), \quad L_{\text{ch}}(h) \equiv \frac{\kappa-1}{\kappa} M(h) + (1 - \gamma)L(h), \quad (12)$$

where L_{ch} is the *chargeable loss*. Since $O_D = M - \gamma L$,

$$O'_D(h) = x_D^{\text{eff}}(h) \left[\gamma L(h) - \frac{\kappa-1}{\kappa} M(h) \right] = x_D^{\text{eff}}(h) \gamma L(h) \frac{p - (\kappa-1)z/\kappa}{\kappa-1+p}.$$

Hence

$$O'_D(0) = x_D^* \gamma L(0) \frac{p}{\kappa-1+p} > 0,$$

and defensive outlay begins to fall only after $z > \kappa p / (\kappa - 1)$. The first marginal day reduces

expected harm but raises total defensive outlay, because it creates an additional period of solo search; only at greater protection depth does stock drawdown dominate that extra search. The added expenditure is why the defence-inclusive criterion begins staging later than harm-only accounting.²⁶

At full compensation, $L_{\text{ch}} = \frac{\kappa-1}{\kappa}M$ because the uninsured term vanishes. A positive window is locally beneficial relative to immediate release if and only if $x_D^* L_{\text{ch}}(0) > wB$; global staging also requires the interior maximum to beat *withholding*, by which I mean non-release of the given model—the limit $h \rightarrow \infty$, worth zero in this fixed-capability problem.

In elasticity form, (11) reads $-d \log L / d \log h = x_D^{\text{eff}} h = z / (1 + z/\kappa)$: the calendar's effect on surviving harm rises with protection depth but never reaches the curvature κ . This is the window's counterpart of the pass-through bound. By (7), a shock can reach expected harm in only three ways: through the scale λm_0 , which passes one for one; through the exploited share, priced by the pass-through and capped below one-half (Proposition 2); and through the surviving stock, priced by the window-length elasticity and capped below κ . Equation (6) is this decomposition with no window. Every shock to harm priced below—capability in Section 4.1, the liability rate in Section 5.1—travels the same three channels.

3.4 Finite windows and harm saturation

The rate-balance rule turns into a characterisation of the schedule most cleanly at full compensation. At $\gamma = 1$ the uninsured harm vanishes, social loss is the developer's burden ($\Lambda = M$), and the chargeable loss is proportional to that burden, $L_{\text{ch}} = \frac{\kappa-1}{\kappa}M$; multiplying by x_D^{eff} gives the chargeable protection flow. Substituting (11) and (10) into (12) at $\gamma = 1$ and taking logarithms, an interior optimum satisfies, for every curvature,

$$\kappa \log\left(1 + \frac{z}{\kappa}\right) - sz = \log \frac{\eta_c}{s}, \quad s \equiv \frac{w}{x_D^*}, \quad \eta_c \equiv \frac{L_{\text{ch}}(0)}{B}. \quad (13)$$

The condition reduces the primitives to two dimensionless ratios. The first, $s = w/x_D^*$, compares the rate at which the model's benefit decays with the defender's search intensity after release. The second, $\eta_c = L_{\text{ch}}(0)/B$, compares the loss relevant to the first marginal day of protection with the benefit from immediate release. Under partial liability, uninsured harm returns to the social loss and adds one term; at $\kappa = 2$,

$$2 \log\left(1 + \frac{z}{2}\right) - \log\left[\frac{\gamma}{1+p} + (1-\gamma)\left(1 + \frac{z}{2}\right)^{-1}\right] - sz = \log \frac{\eta}{s}, \quad \eta \equiv \frac{L(0)}{B}, \quad (14)$$

with $\eta_c = \eta[1 - \gamma p / (1 + p)]$. Thus $\eta_c > s$ is the local condition for a positive window to improve on immediate release. It is not by itself a global staging condition, since the interior candidate must also beat withholding.

²⁶Harm-only accounting prices the marginal day at the full flow $x_D^{\text{eff}}L$; the remark closing Section 3.4 and Appendix B.2 record what that changes.

Note what has become of the contest. At full compensation $\ell = C'_D(x_D^*)$, so the burden is $M(h) = m_0 c_D x_D^{*\kappa-1} (1 + z/\kappa)^{-(\kappa-1)}$: the stake–cost ratio Θ , the exploited share p , the prize v and the adversary’s cost c_A reach the schedule *only* through x_D^* . The contest of Section 2 touches the calendar through one number: the defender’s contest intensity once the adversary arrives.

This is also where the sufficiency of Θ stops: the exploited share depends on the primitives through Θ alone, but the search intensities do not, and the schedule needs an intensity. By Proposition 1, $x_D^* = (2\gamma\lambda/c_D) p(\Theta)/(1 + p(\Theta))$ at $\kappa = 2$ —the defender’s stake scale multiplied by a function of the ratio. Two systems with the same stake–cost ratio can therefore warrant different calendars, and a regulator who knows only how the flaws would divide does not yet know how long to wait. What the ratio does constrain is where a system can sit in the (s, η_c) plane. Since $\eta_c = \frac{\kappa-1}{\kappa} m_0 c_D x_D^{*\kappa-1} / B$ and $s = w/x_D^*$, their product is

$$\eta_c s = \frac{\kappa-1}{\kappa} \frac{m_0 c_D w}{B} x_D^{*\kappa-2}, \quad (15)$$

which under quadratic costs does not depend on x_D^* at all. In other words, every primitive that reaches the schedule only through the contest intensity—the harm per exploit λ , the prize v , and the adversary’s cost c_A —moves a system along the fixed rectangular hyperbola $\eta_c s = m_0 c_D w / (2B)$, crossing the regime boundaries of Figure 2 but never leaving the curve. The primitives that shift the hyperbola itself are the ones outside the contest: the stock m_0 , the obsolescence rate w , the benefit B , and the defender’s cost level c_D , which does both. Section 6 returns to the curve: to a developer choosing its own release date, the liability rate is exactly such a primitive, and slides it along a hyperbola of its own.

The regulator chooses among immediate release, *staging* for a window $h > 0$, and withholding, to maximise $\mathcal{S}(h) = e^{-wh} B - \Lambda(h)$. Under partial liability, the local test against immediate release does not guarantee a non-empty staging region: withholding may overtake immediate release before an interior date becomes worthwhile. The next proposition separates those comparisons and then establishes the calendar and harm limits.

Proposition 4. *An optimal policy exists. Write h^* for the optimal window wherever staging is the optimal policy, and $z^* \equiv x_D^* h^*$ for its protection depth. Throughout, $\kappa > 1$ and $\gamma > 0$.*

- (i) Full compensation. At $\gamma = 1$, h^* is unique wherever staging is optimal, and z^* is the smaller root of (13). If $s < \frac{\kappa-1}{\kappa}$, the optimal policy is immediate release for $\eta_c \leq s$, staging for $s < \eta_c < \bar{\eta}_c(s)$, and withholding for $\eta_c > \bar{\eta}_c(s)$, where $\bar{\eta}_c(s)$ is given in closed form in the proof. If $s \geq \frac{\kappa-1}{\kappa}$, staging is never optimal, and immediate release is optimal if and only if $B \geq M(0)$.
- (ii) Liability at or below full compensation. Let $\kappa = 2$ and $\gamma \in (0, 1]$, and vary η_c by scaling the loss level while holding the post-release contest fixed. If $s < \bar{s}_\gamma(p) \equiv (1 + p - \gamma p) / (1 + p + \gamma(1 - p))$, the three-way partition of part (i) again holds, with an upper boundary constructed in the proof, and the staged optimum is the smaller positive root of (14). If $s \geq \bar{s}_\gamma(p)$, staging is never optimal, and policy passes from immediate release to withholding at $\eta_c = \bar{s}_\gamma(p)$. At $\gamma = 1$, $\bar{s}_\gamma(p) = 1/2$.

(iii) Calendar ceiling. *Wherever staging is optimal,*

$$h^* < \kappa \left(\frac{1}{w} - \frac{1}{x_D^*} \right) < \frac{\kappa}{w}. \quad (16)$$

(iv) Saturation in harm severity. *As $\lambda \rightarrow \infty$ at fixed γ and κ , the optimal window converges to the smaller root h_∞ of*

$$wB e^{-wh_\infty} h_\infty^\kappa = (\kappa - 1) \kappa^{\kappa-1} m_0 c_D, \quad (17)$$

wherever that root exists and is viable, meaning $wh_\infty < \kappa - 1$; the limit is then independent of γ , v and λ . Otherwise the limiting problem selects withholding. At $\gamma = 1$ and $\kappa = 2$ the optimal window is nondecreasing in λ across the transitions between all three policies.

An evaluation window cannot be lengthened without limit. Benefit decays exponentially, while optimised effort makes residual loss fall only polynomially, so sufficiently long release dates are dominated by withholding. If there are two stationary points, only the first is a local maximum and hence the staged choice is the smaller root. Part (ii) adds the global comparison missing from the marginal staging test: when $s \geq \bar{s}_\gamma(p)$, immediate release gives way directly to withholding.²⁷

Figure 2 shows the two comparisons. The lower boundary, $\eta_c = s$, is the local test against immediate release. The upper boundary compares the interior candidate with withholding. At $\gamma = 0.7$ and $p = 1/2$, these boundaries meet at $\bar{s}_\gamma(p) = 23/37$.

The ceiling (16) follows because any stationary point that beats withholding must satisfy $x_D^{\text{eff}} > w$.²⁸ In the fixed-hazard limit $\kappa \rightarrow \infty$, the ceiling recedes to infinity. At full compensation the bound tightens by a further $1/w$, to $h^* < (\kappa - 1)/w - \kappa/x_D^*$ (equation (40) in the proof).

The saturation result (17) qualifies the heuristic that models capable of greater harm should be withheld for longer. At $\gamma = 1$ and $\kappa = 2$, the optimal window rises monotonically with harm per exploit, but only towards a finite limit. Expected harm is driven towards zero by faster search, whose cost offsets the larger stake: at fixed h , the burden converges to $m_0 c_D \kappa^{\kappa-1} h^{-(\kappa-1)}$. The limiting date h_∞ is therefore set by benefit decay and defensive cost. The adversary's prize cancels too, so proportional scaling of prize and harm does not move it.

Convergence also requires viability. At a limiting stationary date, surplus is $Be^{-wh_\infty}[1 - wh_\infty/(\kappa - 1)]$, so the date beats withholding only if $wh_\infty < \kappa - 1$. The limiting ceiling $(\kappa - 1)/w$ is strictly inside the universal ceiling κ/w ; a stationary date can lie between them yet lose to withholding.²⁹ Figure 3 draws the approach at a representative calibration.

²⁷Existence is immediate once withholding is recorded as $S(\infty) = 0$, since S is then continuous on the compact $[0, \infty]$. The adjacent policies tie at each boundary: the staged solution converges to immediate release at the lower boundary and matches withholding at the upper one. Part (iv) describes the limiting problem and does not assert that withholding is uniquely optimal at every sufficiently large finite λ . The full-compensation identity in part (i) also returns under delegation, where the developer's conditional date problem has the same form at transformed primitives.

²⁸It does not imply that marginal surplus is negative at every later date; beyond a local minimum, surplus can approach withholding from below.

²⁹The same condition governs the date under delegation in Section 6 with B replaced by βB , which is why a developer capturing little of the benefit is pushed towards exit rather than towards a longer evaluation.

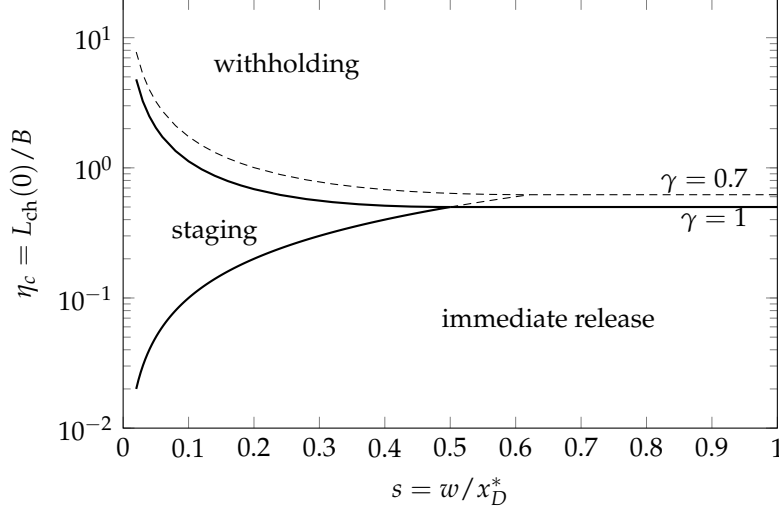


Figure 2: The regime partition in chargeable units at $\kappa = 2$ and $p = 1/2$, holding the post-release contest fixed as the loss scale varies. Solid: full compensation, with $\bar{\eta}_c(s) = e^{2s-1}/(4s)$ and closure at $s = 1/2$. Dashed: $\gamma = 0.7$. In that case, the staging region closes exactly at $\bar{s}_\gamma(p) = 23/37 \simeq 0.6216$; above that point, policy moves directly from immediate release to withholding.

The calendar ceiling survives if defensive outlay is left off the social account, but harm saturation does not. Under harm-only accounting, greater harm eventually *shortens* the optimal window because the defender's accelerating search outruns the growth in what is at stake. Under defence-inclusive accounting, that acceleration is itself a resource cost, and the date saturates instead.³⁰ Thus the proportionality heuristic fails at the top either way—by reversal if search were free, by saturation because it is not. The pipeline and fringe discussions below use the harm-only object where its closed forms are convenient, flagged as they arise.

3.5 Comparative statics

At full compensation, three comparisons organise the comparative statics. Greater harm lengthens the window, but its effect vanishes as the date approaches saturation. Harm and liability buy the same defensive contest intensity, yet only liability changes the planner's accounting. Liability and cheaper defensive search also produce the same contest response, yet only the latter saves real resources. The next proposition makes these comparisons exact. Write $\varepsilon_q \equiv d \log h^* / d \log q$ for the elasticity of the optimal window with respect to a primitive q , and set $\mathcal{D} \equiv h^* [x_D^{\text{eff}}(h^*) - w] > 0$.

Proposition 5. *At an interior optimum under full compensation, for every $\kappa > 1$,*

$$\varepsilon_{m_0} = -\varepsilon_B = \frac{1}{\mathcal{D}}, \quad \varepsilon_w = \frac{wh^* - 1}{\mathcal{D}}, \quad \frac{\partial h^*}{\partial \log m_0} = \frac{1}{x_D^{\text{eff}} - w} \leq \frac{\kappa - 1}{w}, \quad (18)$$

³⁰Formally, harm-only surplus is $S_H(h) = e^{-wh}B - L(h)$. Its interior condition is (13) with $\kappa + 1$ in place of the leading coefficient κ , and η in place of η_c ; the partition has boundaries $\eta = s$ and $\bar{\eta}(s) = (e^{s-1}/s)^\kappa$; and the ceiling (16) is unchanged. At $\kappa = 2$, $h^* \propto \lambda^{-1/6}$ with v fixed and $h^* \propto \lambda^{-1/3}$ when $v \propto \lambda$. The defence-inclusive saturation date is independent of v . Appendix B.2 records the formulas.

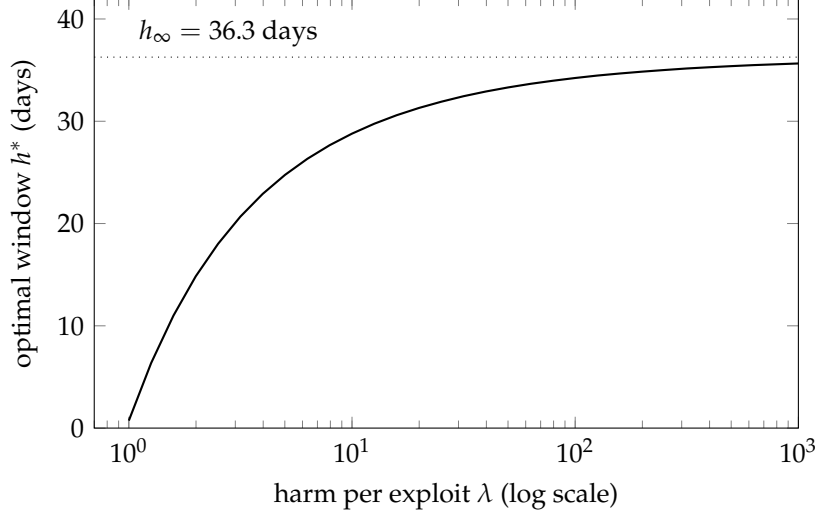


Figure 3: The fixed-capability optimal window under full compensation as harm per exploit grows ($\gamma = 1, w = 0.01$ per day, $v = m_0 = 1, c_A = 13.07, c_D = 9.15, B = 2$). The date rises and saturates at $h_\infty = 36.3$ days (Proposition 4(iv)).

and under quadratic costs, with $u \equiv x_D^{\text{eff}}(h^*)/x_D^* = (1 + z^*/2)^{-1}$,

$$\begin{aligned} \varepsilon_\lambda &= \frac{2u}{D} \left(1 - \frac{\varphi}{1+p}\right) > 0, & \varepsilon_v &= -\varepsilon_{c_A} = \frac{2u}{D} \frac{\varphi}{1+p} > 0, \\ \varepsilon_{c_D} &= \frac{1}{D} \left[1 - 2u \left(1 - \frac{\varphi}{1+p}\right)\right], & \varepsilon_\gamma &= \frac{u(1-p^2-2\varphi)}{(1+p)D} > 0, \end{aligned} \quad (19)$$

with $\varepsilon_\lambda \rightarrow 0$ as $\lambda \rightarrow \infty$.

Primitives that change the contest intensity do so along two channels.³¹ Differentiating $x_D^* = (2\gamma\lambda/c_D) p(\Theta)/(1 + p(\Theta))$ with respect to any primitive q ,

$$\frac{d \log x_D^*}{d \log q} = \underbrace{\frac{d \log(\gamma\lambda/c_D)}{d \log q}}_{\text{stake scale}} - \underbrace{\frac{\varphi(\Theta)}{1+p} \frac{d \log \Theta}{d \log q}}_{\text{contest}}, \quad (20)$$

which is the calendar's version of (6). The stake scale is how hard the defender would search without an opponent's response; the contest term removes the part offset by the change in the exploited share. For $q \in \{m_0, B, w\}$ both channels are zero, which is why (18) contains no pass-through term and holds at every curvature. The remaining primitives— λ, γ, c_D, v and c_A —can largely be read from (20).

The prize and the adversary's cost have no stake channel and enter Θ with opposite signs, so their elasticities are pure pass-through and exactly opposed. Harm per exploit carries both channels at unit weight; $\varphi < \frac{1}{2}$ ensures that the adversary gives back part, but never all, of the search bought by the larger stake. The defender's cost mirrors harm in (20), yet $\varepsilon_{c_D} \neq -\varepsilon_\lambda$ because c_D also scales

³¹Every elasticity is evaluated at $\gamma = 1$; in ε_γ the rate varies, and the criterion varies with it. The factor u is the discount the effort profile applies to the contest intensity.

the real-resource burden directly. That extra route is the $1/\mathcal{D}$ term in (19).

Liability completes the set, and it makes the same point from the other side. Both the stake scale and Θ depend on the liability rate and harm per exploit only through the product $\gamma\lambda$, so $d \log x_D^*/d \log \gamma = 1 - \varphi/(1+p)$: a one per cent increase in either primitive buys exactly the same defensive search intensity. The two elasticities differ all the same, and they differ by

$$\varepsilon_\lambda - \varepsilon_\gamma = \frac{u(1+p)}{\mathcal{D}} > 0,$$

in which the pass-through has cancelled. Nothing in the contest separates them; what separates them is the criterion. Raising γ shifts harm between the burden the developer bears and the uninsured harm $(1-\gamma)L$ that the planner still counts, and although that term vanishes at full compensation, its derivative does not. Harm severity leaves the accounting alone; the liability rate does not.

The harm elasticity is positive at every interior optimum and falls to zero as the date saturates. Proposition 4(iv) supplies the corresponding global statement across the transitions between immediate release, staging and withholding. Initial stock also lengthens the window, but by a bounded amount: doubling it adds at most $(\kappa-1) \log 2/w$ within the staging region. The fixed-hazard counterpart, $1/(x_D^{\text{fix}} - w)$, instead diverges as $x_D^{\text{fix}} \downarrow w$ because it omits the effort-profile correction.³²

At full compensation, the window-length elasticity with respect to liability is positive, but it is not uniformly small. Along a regular staged branch approaching the immediate-release boundary, $h^* \rightarrow 0$ and $\mathcal{D} = h^*(x_D^{\text{eff}} - w) \rightarrow 0$, so the elasticity can become arbitrarily large. Away from full compensation, the calendar response can turn negative because the rate also shrinks uninsured harm. That does not by itself make the instruments substitutes: calendar time is h , whereas protection depth is $z = x_D^*h$. Section 5.1 derives the protection-depth condition under partial liability.

Because γ and $1/c_D$ enter both channels of (20) identically, a proportional increase in liability and the same proportional reduction in defensive search cost produce the same exploited share and protection depth holding the calendar fixed. After the release date is reoptimised, their optimal protection depths generally differ: lowering c_D saves real resources, whereas raising γ induces more search at the existing cost. Thus ε_γ is positive, while ε_{c_D} changes sign at $u = \frac{1}{2}(1 - \varphi/(1+p))^{-1}$.

Table 3 maps the baseline results to the principal departures developed in Appendix B. It is a scope map: the final column records the policy questions that the baseline does not answer.

³²If the stock change crosses a regime boundary, the schedule jumps and no level derivative describes it. The sign of ε_w is $wh^* - 1$, since the marginal delay cost wBe^{-wh} falls in w once $wh > 1$. At full compensation, the harm-severity path follows the fixed hyperbola (15), while $\tilde{\eta}_c(s)s = e^{2s-1}/4$ falls, so regime transitions are one-way. Under partial liability, the interval property can fail: under harm-only accounting, the withholding boundary grows like λ while η grows like $\lambda^{1/2}$, allowing immediate release, staging, withholding, and then staging again as harm rises.

Departure	Result retained	What changes or remains open
Independent remediation lag	Per-flaw separability survives; effort can be characterised for fixed h	Effort may peak or fall near release; the optimal window and staging region are not signed
Productive adversarial fringe	A stronger fringe raises the burden and, at full compensation, lowers the value of a fixed window; a matched fringe makes headway worthless	Under partial liability, no welfare ranking across intermediate fringe productivities is established; intermediate optimal release dates require solving the coupled pre-release contest
Common stock elasticity $\alpha \neq 1$	The per-resolution contest and exploited-share formulas retain the common-factor cancellation	The depletion clock satisfies a moment equation; the optimal schedule is solved only at $\alpha = 1$
Findings-contingent release	At the Poisson benchmark, findings do not improve expected value under a fixed effort schedule	Stopping times may still be random; the joint effort-and-stopping problem is open
Programme overhead $f_D > 0$	The conditional effort profile is unchanged	The high-rate burden becomes $M_\infty + f_D h$; calendar evasion obeys the limiting-surplus condition
Adversary entry	The intensive pass-through bound applies conditional on the attack population	Entry adds an extensive deterrence margin; the fully coupled entry-and-window equilibrium is open

Table 3: Scope map for departures from the baseline.

4 Endogenous capability

Sections 2 and 3 take the AI model as given and ask when to release it. An access rule announced before development also changes which model is built. Indeed, when capability expands immediate-release harm but makes the window more productive, sufficient protection depth can reverse capability’s marginal effect on harm. Headway therefore does two jobs: after development it allocates exclusive access; before development it changes the return to capability.

The developer first chooses whether to build, with no model yielding payoff zero; write this discrete action as $a = \emptyset$. Conditional on building, it selects capability a from a compact interval $[a_{lo}, a_{hi}]$ with $a_{lo} > 0$.³³ Gross public benefit at release is $B(a)$, increasing and weakly concave; capability costs $\Psi(a)$, increasing and convex, with any fixed development outlay folded in. Writing $\beta \in (0, 1]$ for *developer appropriability*—the developer’s share of public benefit—welfare and the developer’s payoff conditional on building are

$$\begin{aligned}
 W(h, a; \gamma) &= \underbrace{e^{-wh} B(a)}_{\text{benefit at release}} - \underbrace{\Psi(a)}_{\text{capability cost}} - \underbrace{\Lambda(h, a, \gamma)}_{\text{harm plus defensive outlay}}, \\
 \Pi(h, a; \gamma) &= \beta e^{-wh} B(a) - \Psi(a) - M(h, a, \gamma),
 \end{aligned} \tag{21}$$

where $M = \gamma L + O_D$ is the burden of (8) and $\Lambda = M + (1 - \gamma)L$. Through the contract, the developer selects or audits flaw-level search while the distinct defender carries it out. The resulting outlay enters the developer’s build and capability calculus; the two objectives differ only in

³³The positive lower bound matters: the no-build action is a discrete alternative and not the limit of weak models, since a vestigial model still creates an attack surface while a nonexistent one does not. The notation $a = \emptyset$ distinguishes no development from $h = \infty$, which denotes non-release conditional on a model having been built.

uncaptured benefit and uninsured harm.³⁴

For closed forms, I use a specific case, choosing capability units so that the reference capability is one. Let

$$B(a) = ba, \quad \Psi(a) = \Psi_0 + \frac{c_\Psi}{2} a^2, \\ c_A(a) = c_A a^{-\sigma_A}, \quad c_D(a) = c_D a^{-\sigma_D}, \quad m_0(a) = m_0 a^{\sigma_m}, \quad \Theta(a; \gamma) = \Theta(1; \gamma) a^\Delta.$$

Here σ_A and σ_D are the capability-induced reductions in adversarial and defensive search costs, σ_m is the expansion of the reachable flaw stock, and $\Delta = \sigma_D - \sigma_A$ is the dual-use gap.

Suppressing their (a, γ) arguments, three combinations of the elasticities do the work below:

$$\sigma_L \equiv \sigma_m - \varphi\Delta, \quad \sigma_R \equiv \sigma_D - \frac{\varphi\Delta}{1+p}, \quad \sigma_M \equiv \sigma_m - \frac{\varphi\Delta}{1+p}, \quad (22)$$

the elasticities of baseline harm, of the defensive intensity, and of the baseline burden with respect to capability. The second is not σ_D : capability lowers the defender's cost, which raises x_D^* directly, but it also moves the exploited share and, through that share, the stake the defender searches for— σ_R is exactly (20) evaluated at the capability margin, a stake-scale term net of the contest's feedback.

One regularity condition is needed, because neither loss object is log-linear in capability. Fix finite bounds $\bar{h} > 0$ and $0 < \underline{\gamma} < \bar{\gamma} < \infty$ wide enough to contain the active finite policy pairs compared in Sections 4–6, and write $\mathcal{P} \equiv [0, \bar{h}] \times [\underline{\gamma}, \bar{\gamma}]$ for the resulting compact policy domain.

Assumption 3. For every $(h, \gamma) \in \mathcal{P}$ and every $a \in [a_{\text{lo}}, a_{\text{hi}}]$: (i) $\Psi''(a) + M_{aa}(h, a, \gamma) > 0$; (ii) $\Psi''(a) + \Lambda_{aa}(h, a, \gamma) > 0$.

Part (i) makes the developer's problem strictly concave throughout $\mathcal{P}$ and is all that the capability profile below requires; part (ii) does the same for welfare and is invoked only where expressly stated.³⁵ The difference between the two parts is economics rather than housekeeping: dual-use asymmetry pushes both loss objects towards *concavity* in capability—capability moves Θ , Θ moves the pass-through, and the pass-through converts the dual-use gap into a change in the exploited share—so part (i) is consistent with welfare being locally convex in capability.³⁶

³⁴An accounting in which the developer's objective charges the liability bill γL but not the search outlay—the convention a transfer view of liability suggests—changes several conclusions below, and I flag each as it arises. The *developer participation constraint* requires its maximised payoff to be nonnegative.

³⁵Imposed only at a particular policy pair, the inequalities make every differentiable capability-response claim a local statement along the corresponding regular interior branch. The calendar-evasion and high-rate arguments use continuity, compact capability and positivity directly, rather than imposing Assumption 3 on an unbounded policy domain.

³⁶At immediate release, $a^2 L_{aa}/L = -\Delta^2 \varphi_\Theta + \sigma_L(\sigma_L - 1)$, where $\varphi_\Theta \equiv d\varphi/d \log \Theta > 0$; the first term is strictly negative whenever $\Delta \neq 0$, whichever side capability favours, and the burden's curvature carries the same ingredient. The failure mode the assumption excludes is real: with a large surface elasticity and heavy liability, the developer's problem acquires a second local optimum at the capability floor—building something vestigial can beat building something good. Appendix A records a sufficient condition.

4.1 Capability and harm

Capability acts through three margins: the reachable stock of flaws, the division of that stock between adversary and defender, and the productivity of the defender's window. Only the third grows with protection depth. The next proposition separates capability's effect on harm at a given window from the effect of the announced window on the capability the developer chooses.

Proposition 6. *Under quadratic costs: (i)*

$$\frac{\partial \log L(h, a, \gamma)}{\partial \log a} = \underbrace{\sigma_m}_{\text{surface}} - \underbrace{\varphi \Delta}_{\text{exploited share}} - \underbrace{\frac{z}{1+z/2} \sigma_R}_{\text{defensive complementarity}}, \quad (23)$$

with $\sigma_R \geq 0$, and $\sigma_R = 0$ only if $\sigma_A = \sigma_D = 0$; the elasticity weakly falls in h from σ_L to $\sigma_L - 2\sigma_R$, strictly whenever $\sigma_R > 0$. (ii) At any fixed (a, γ) , when $0 < \sigma_L < 2\sigma_R$ —the crossing case—the marginal effect of capability on harm is positive below the pointwise protection-depth threshold $z_a(a, \gamma) \equiv 2\sigma_L / (2\sigma_R - \sigma_L)$ and negative above it; it is nonpositive at every protection depth if $\sigma_L \leq 0$ and nonnegative at every protection depth if $\sigma_L \geq 2\sigma_R$. (iii) Under Assumption 3(i), write $a^*(h)$ for $a^*(h, \gamma)$ at fixed γ . At an interior choice $a^*(h) \in (a_{lo}, a_{hi})$, capability rises with the announced window if and only if

$$\underbrace{\beta w e^{-wh} B'(a)}_{\text{delay tax}} < \underbrace{\frac{x_D^{\text{eff}}(h) M}{2a} \left[\sigma_M + \sigma_R \frac{1 - z/2}{1 + z/2} \right]}_{\text{burden relief}}, \quad (24)$$

with every object evaluated at $a = a^*(h)$. At any fixed (a, γ) , the bracketed burden-relief term is strictly positive at $z = 0$ and has at most one zero in protection depth: at $z_{II}(a, \gamma) = 2(\sigma_M + \sigma_R) / (\sigma_R - \sigma_M)$ when $\sigma_D > \sigma_m$, and nowhere otherwise.

Part (i) assigns one channel to each margin. The surface passes one for one, the exploited share moves at the pass-through, and protection during the window moves at the calendar elasticity $z/(1 + z/2)$, capped below two. The last channel always works towards less harm. It does not require capability to favour the defender. If capability leaves defensive costs unchanged and lowers only adversarial costs, then the exploited share and the defender's stake both rise. The defender searches harder because it has more to lose. This is an effort response, not by itself a reduction in harm, and a fixed-hazard benchmark cannot express it.³⁷

Part (ii) gives the sign reversal. At immediate release, capability changes harm with elasticity σ_L , combining the larger attack surface with the change in the exploited share. A window adds the defensive-complementarity term, whose strength rises with protection depth. When $0 < \sigma_L < 2\sigma_R$, capability at fixed (a, γ) initially raises harm but lowers it once z exceeds the pointwise threshold $z_a(a, \gamma)$. Reaching that threshold requires $h = z_a(a, \gamma) / x_D^*(a, \gamma)$.^{38,39}

³⁷Equation (23) is the decomposition of Section 3.3 with capability as the shock, and (6) is its immediate-release case.

³⁸Along the endogenous capability path both $z(h)$ and $z_a(a^*(h), \gamma)$ may move with h , so this pointwise sign change does not by itself imply a single crossing in calendar time.

³⁹The harm-protection channel is proportional to $z(1 + z/\kappa)^{-(\kappa+1)}$, which peaks at unit protection depth for every κ ;

Part (iii) turns from harm to the model induced by the announcement. Because the developer finances both liability and the defender's evaluation, a longer window relieves its burden more slowly than it reduces harm. Unless capability changes nothing ($\sigma_A = \sigma_D = \sigma_m = 0$, where the relief is zero throughout), that relief remains positive to greater protection depth and can change sign only when the defensive-cost elasticity exceeds the surface elasticity.⁴⁰⁴¹ What the developer does not internalise at its own choice is the *capability wedge*, which the next subsection prices.

4.2 The capability wedge

At the developer's interior choice, substituting its first-order condition into W_a gives the capability wedge: the social gradient in capability as a difference of two terms the developer does not internalise:

$$W_a|_{a^*} = \underbrace{(1 - \beta) e^{-wh} B'(a^*)}_{\text{uncaptured public benefit}} - \underbrace{(1 - \gamma) L_a(h, a^*, \gamma)}_{\text{uninternalised marginal harm}}. \quad (25)$$

The burden appears in both objectives and cancels, so the capability wedge is the same under either accounting even though the developer's behaviour is not. Proposition 6(ii) then gives the cases. Where $L_a < 0$ (equivalently, in the crossing case, $z > z_a(a^*, \gamma)$) and $\gamma \leq 1$, capability prevents harm at the margin, so the two signed terms in (25) are nonnegative. Their sum is strictly positive unless $\beta = \gamma = 1$; at that benchmark, the private and social capability objectives coincide. Hence, in this harm-reducing region and under Assumption 3(ii), the developer chooses less capability than the welfare-maximising level whenever $(\beta, \gamma) \neq (1, 1)$.

At $\gamma > 1$ the harm term reverses sign, because the developer then values marginal harm reduction at more than its social worth, and supra-compensatory liability can push capability past the social optimum. Underbuilding is therefore a $\gamma \leq 1$ result away from the aligned benchmark $\beta = \gamma = 1$.⁴²

Where $L_a > 0$ (equivalently, in the crossing case, $z < z_a(a^*, \gamma)$), capability creates harm at the margin. The developer has overbuilt, in the same conditional sense, when uninternalised marginal harm exceeds uncaptured marginal benefit. At a harm-neutral policy pair, $L_a = 0$ and the capability margin is aligned when $\beta = 1$; in the crossing case this locus is $z = z_a(a^*, \gamma)$. The harm-neutral locus is itself moved by policy: the regulator sets the calendar h directly and the search intensity x_D^* through liability.

Because capability responds to the announcement, the rate-balance rule of Section 3 acquires exactly one extra term. Holding γ fixed, any interior optimum of the composite $W(h, a^*(h); \gamma)$

the burden-protection channel that drives part (iii) is proportional to $z(1 + z/\kappa)^{-\kappa}$ and peaks at $z = \kappa/(\kappa - 1)$. The two humps sit at neighbouring protection depths, and neither location depends on the primitives.

⁴⁰Along the endogenous capability path the threshold itself moves, so no global one-crossing conclusion follows without an additional monotonicity condition.

⁴¹If the developer instead paid the liability bill but not evaluation costs, the flip would lie far shallower; the reading that a regulator materially weakens the model it faces by extending the window is largely an artefact of that accounting.

⁴²The positive social gradient at the private choice does not require Assumption 3(ii); ranking the private and social capability levels does.

satisfies

$$\underbrace{we^{-wh}B(a^*)}_{\text{marginal cost of delay}} = \underbrace{x_D^{\text{eff}}(h) L_{\text{ch}}(h)}_{\text{chargeable protection flow}} + \underbrace{W_a|_{a^*} a^{*'}(h)}_{\text{capability correction}}, \quad (26)$$

and at a fixed-capability optimum h_0 the composite gradient has the sign of $W_a|_{a^*} a^{*'}(h_0)$.

Capability rises only where the full inequality (24), including the delay tax, holds; z_{Π} marks where its burden-relief term alone changes sign, so protection depth between z_a and z_{Π} is necessary but not sufficient for a rising choice. Where both pointwise thresholds exist, protection depth at h_0 lies between z_a and z_{Π} , and (24) holds at h_0 , capability is harm-reducing while the capability choice is rising, so the correction favours a longer date. Past z_{Π} , where (24) necessarily fails, with $\beta = 1$ and $\gamma < 1$ it favours a shorter one.⁴³

The ceiling in Proposition 4 is also a fixed-capability result. The commitment optimum solves (26), not the rate-balance rule alone, so κ/w has not been established for the date with induced capability. Benefit decay and, at long calendars, developer exit instead discipline that date.⁴⁴

A long mandate is not prohibition. Announcing that no model will ever be released induces developer exit when development has any positive cost, giving welfare zero. Under a very long but finite mandate, the developer may still build. Where its participation constraint binds, $\max_a \Pi = 0$ and welfare is $(1 - \beta)e^{-wh}B(a^*) - (1 - \gamma)L$, which at full developer appropriability is $-(1 - \gamma)L < 0$ whenever liability is incomplete. Such a mandate is worse than prohibition even though the limit of ever-longer mandates jumps back to zero. With endogenous capability, withholding a built model, prohibition, and developer exit are distinct; only prohibition is a genuine regulatory outside option.

5 Liability

Sections 2–4 treated the liability rate γ as part of the environment. This section makes it an instrument: the regulator announces a package (h, γ) before development, and the developer is free to build or not.

In the standard framing, liability is a transfer: it moves realised liability payments between the developer and claimants without changing expected harm. Here the rate also sets the exploited share through the stake–cost ratio Θ , determines the defender’s search intensity through the per-flaw stake ℓ , and prices the harm that remains. Expected harm is therefore something the liability rule partly chooses.⁴⁵

⁴³No global ordering along the endogenous path is implied.

⁴⁴Once Ψ is sunk, a regulator re-optimising takes capability as fixed and applies the fixed-capability rule. At an interior announced date, a vanishing capability correction aligns the ex ante and ex post first-order conditions; time consistency additionally requires the corresponding second-order and global conditions. Without commitment, (26) is read with $a^{*'} = 0$.

⁴⁵The liability-bearing developer is the principal: it chooses or audits the search intensity and reimburses the operating unit or contractor that carries it out. If an independent maintainer instead chooses hidden effort and cannot be contracted on that basis, liability can shift losses without inducing the search effort that gives the window its value. The benchmark

The incentives created depend on timing. After release, liability acts on a contest: stronger defensive incentives provoke an adversarial response. During exclusive access, it acts through a monopoly defender whom the adversary cannot match. That difference drives the interaction with headway, and the defensive search outlay induced on either side of release delivers the baseline result that the optimal liability rate is partial.

5.1 Complements or substitutes

Liability statutes and evaluation mandates are typically written separately, and the economics of enforcement has licensed the separation: ex post liability and ex ante regulation are cast either as substitutes, each doing what the other cannot (Shavell, 1984), or as complements that cover each other's informational blind spots (Kolstad et al., 1990). The relationship here is technological, survives complete information, and has both signs—one attached to each of liability's two channels.

Write $Y \equiv -\partial \log L / \partial \log \gamma$ for the *liability elasticity of harm*: the proportional reduction in expected harm bought by a proportional increase in the liability rate.

Proposition 7. *At any headway $h \geq 0$,*

$$Y = \underbrace{\varphi(\Theta)}_{\text{exploited-share channel}} + \underbrace{\frac{z}{1+z/2} \left(1 - \frac{\varphi(\Theta)}{1+p}\right)}_{\text{protection-depth channel}}, \quad (27)$$

strictly increasing in h from $Y(0) = \varphi < \frac{1}{2}$ to a limit in $(\frac{3}{2}, 2)$.

Proposition 7 separates the two channels. The exploited-share channel is capped below one-half because the adversary matches the defender (Proposition 2). The protection-depth channel can approach two because no such response occurs during exclusive access. Liability that removes little harm at release can therefore remove much more across a window with large protection depth. In this sense, liability and headway are complements in protection, although they need not be complements in calendar days.

This sits awkwardly against the comparative statics of Section 3.5, where under partial liability the calendar response to the rate can turn negative. If the instruments are complements, why would more of one ever call for less of the other?

Part of the answer is units. From the closed-form defensive contest intensity $x_D^* = 2\gamma\lambda p / (c_D(1+p))$ —or from (20) at a unit stake channel—

$$\frac{\partial \log x_D^*}{\partial \log \gamma} = 1 - \frac{\varphi}{1+p} \in \left(\frac{1}{2}, 1\right) : \quad (28)$$

a higher liability rate increases the defender's search intensity. Since the protection depth an optimal window delivers is $z^* = x_D^* h^*$, the calendar and protection-depth responses to the rate

deliberately excludes that moral-hazard problem.

differ by exactly this conversion:

$$\underbrace{\frac{d \log h^*}{d \log \gamma}}_{\text{calendar response}} = \underbrace{\frac{d \log z^*}{d \log \gamma}}_{\text{protection-depth response}} - \left(1 - \frac{\varphi}{1+p}\right). \quad (29)$$

The conversion term lies between one-half and one. Thus a higher liability rate can shorten the optimal calendar even while increasing the protection depth the regulator demands: the defender supplies more protection in fewer days.

The rest of the answer is that the protection-depth response has an exact sign, in which the two channels of Proposition 7 reappear on opposite sides. Raising the liability rate does three things to the marginal value of protection depth: it makes a given protection depth arrive sooner, by (28); it raises the stake that a marginal unit of protection depth protects, which scales with ℓ ; and, through the exploited share, it removes part of the harm the window exists to protect while reweighting the remainder out of the uninsured column, charged in full, into the burden, charged at $\frac{\kappa-1}{\kappa}$. The first two effects are the protection-depth channel working for the window; the third is the exploited-share channel working against it. Collecting the three at the optimum (Appendix A), the optimal protection depth rises with the liability rate if and only if

$$\underbrace{(1-\gamma) \left[\left(1 - \frac{\varphi}{1+p}\right) (1 - wh^*) - \varphi \right]}_{\text{exploited-share channel: substitutes}} + \underbrace{\gamma \left[\chi + \frac{1 - \varphi/(1+p)}{1+p} z^* \left(1 - s - \frac{wh^*}{2}\right) \right]}_{\text{stake channel: complements}} > 0, \quad (30)$$

where $\chi \equiv (1 - p^2 - 2\varphi)/(1+p)^2 > 0$, an object the next subsection develops.

Equation (30) makes the distinction precise. The stake channel is positive at every optimum that beats withholding because the term $z^*(1 - s - wh^*/2)$ is positive strictly inside the universal ceiling of Proposition 4(iii). At full compensation, the exploited-share channel has no weight, so liability and headway are complements in protection depth throughout the staging region. At low rates, however, the exploited-share channel carries nearly all the weight and the condition approaches $(1 - \varphi/(1+p))(1 - wh^*) > \varphi$. It can fail when the pass-through and protection depth are large: a weak rate then works mainly through the exploited share and substitutes for the protection the window would otherwise provide.⁴⁶ Every stationary rate lies strictly inside the complements region (Section 5.2), so protection depth and the liability rate rise together along the optimal-rate locus.

The policy implication depends on units. When legal damages are capped or harms fall on parties who cannot sue, calendar time may substitute for the missing stake, subject to the point at which induced search becomes too weak to sustain staging.⁴⁷ Equation (30) describes a timing

⁴⁶This is a statement about small positive rates along parameter configurations that retain an interior staged optimum, not a fixed-primitives limit: as $\gamma \downarrow 0$, defensive intensity vanishes, $s = w/x_D^*$ diverges, and the staging region closes before the limit is reached (Proposition 4(ii)).

⁴⁷Under $C_D'(0) = 0$, defensive search remains strictly positive at every $\gamma > 0$; a literal no-search threshold requires a positive marginal cost at the origin (Appendix B.1.2).

response after reoptimisation; it says nothing about a calendar change when h is fixed.

The legal bill can fall while the burden rises. The expected bill is γL , so $\partial(\gamma L)/\partial\gamma = L(1 - Y)$. Once Y crosses one—at a protection depth in $(1, 2)$ for every exploited share⁴⁸—raising the rate lowers the expected legal payment because the search it finances destroys harm faster than the price rises. Yet the developer’s burden rises at every protection depth, with elasticity $(1 + z/2)^{-1}(1 - \varphi/(1 + p)) > 0$, because the rate converts legal damages into defender search outlays. The tort system observes the bill; the developer bears both.

5.2 The optimal rate

What rate should the regulator set, at a fixed date and model? The classical answer is full compensation: at $\gamma = 1$ the developer internalises every dollar of harm. Under quadratic costs, the defence-inclusive accounting, and the fixed attack population used here, the efficient rate is instead strictly partial.

Social loss is $\Lambda = M + (1 - \gamma)L$: the burden plus the harm no one is charged for. Differentiating in the liability rate, the uninsured harm responds through Y , and the burden contributes $M_\gamma - L$ —its response beyond the liability bill’s transfer content. Under quadratic costs, that excess is a function of the exploited share alone,

$$\chi \equiv \frac{M_\gamma - L}{L} = \frac{1 - p^2 - 2\varphi}{(1 + p)^2} > 0, \quad (31)$$

which I call the *dissipation wedge*: the defensive search a marginal unit of the liability rate provokes, over and above the legal payment it transfers, per unit of harm at stake. It is the same object that floors the stake channel in (30); both measure the rate’s effect on the burden net of the transfer. Its positivity is the contest inequality $2\varphi < 1 - p^2$ (Appendix A), and it is the whole marginal cost of liability here, because a transfer costs society nothing and dissipation is real. The welfare gradient in the liability rate is

$$\frac{\partial W}{\partial \gamma} = L \left[\frac{(1 - \gamma)Y}{\gamma} - \chi \right] : \quad (32)$$

the rate buys harm reduction at elasticity Y and pays for it in dissipation at rate χ .

Proposition 8. Fix $h \geq 0$ and a , and interpret $\gamma = 0$ only as a limiting benchmark. A global welfare-maximising rate is attained in $(0, 1)$. Every stationary rate satisfies

$$\gamma^* = \frac{Y}{Y + \chi} < 1, \quad (33)$$

and welfare is strictly decreasing in the rate on $[1, \infty)$. Global uniqueness is not asserted. At any regular interior optimum, $d\gamma^*/dh > 0$.

⁴⁸Setting $Y = 1$ in (27) gives the crossing depth $2(1 - \varphi)(1 + p)/[(1 + p) - \varphi(1 - p)]$; the bounds are derived in Appendix A.

Optimal liability is strictly partial, and the ratio says why. At full compensation, the developer already internalises harm. Raising the rate further pushes defensive search beyond that private harm–cost trade-off; the additional search costs more than the harm it removes, so welfare falls. The reason is distinct from the received ones: not judgment-proofness (Shavell, 1986), not user incentives (August and Tunca, 2011; Lam, 2016), but the dissipation of defensive search.⁴⁹

The result depends on the fixed attack population and on the welfare account. Adversary entry adds an extensive margin to the liability elasticity of harm that (33) does not price (Appendix B.1.4); and charging adversarial outlay to social loss would make a marginal increase at full compensation buy a reduction in offensive dissipation as well as additional defensive search, which can carry the optimum above one.⁵⁰ Partial liability is a theorem under the defence-inclusive accounting, not under every welfare convention.⁵¹ It is also a theorem about internalised social harm rather than about a statutory damages multiplier: when compensable damages fall short of λ , implementing γ^* requires a multiplier of $\gamma^* \lambda / d_L$, which can exceed one (footnote 9).

Equation (33) is an inverse-elasticity rule. Rearranged, $(1 - \gamma^*) / \gamma^* = \chi / Y$: the optimal shortfall from full compensation equals the dissipation the marginal unit of the liability rate provokes per unit of harm it removes. Both sides are equilibrium elasticities—under quadratic costs, functions of the exploited share and the protection depth alone—so the rule chooses the rate from two measurable objects rather than from the costs and prizes beneath them.

The comparative static carries the complementarity to the optimum. The dissipation wedge depends on the exploited share but not on the calendar, whereas Y rises with the window. Hence the optimal rate rises with the window along every regular branch. In the other direction, substituting the stationarity condition into (30) makes the protection-depth response strictly positive at every optimum inside the ceiling (Appendix A). Thus the package’s two margins reinforce one another in protection depth even when a higher rate shortens calendar time, and the substitutes region lies below every rate the regulator would choose. Along a regular branch continued to large protection depth, the optimal rate rises towards its large-window limit.⁵²

⁴⁹Equation (33) is a fixed point rather than a formula— Y and χ both move with the liability rate through the exploited share—though every optimum must satisfy it. Global uniqueness of the stationary rate is not proved: $(1 - \gamma)Y/\gamma$ is the product of a decreasing and an increasing function of the rate, so monotonicity does not follow from the primitives. Nothing below relies on uniqueness. Absent it, the comparative static in the window is a statement about each regular branch of stationary rates.

⁵⁰The dependence on the welfare convention can be made explicit. Write $q \equiv \gamma\lambda$ for the developer’s effective stake per exploit and $E(q) \equiv L(q)/\lambda$ for expected exploits, and let welfare place weights ω_D and ω_A on defensive and offensive expenditure and ω_G on the adversary’s gains. Writing O_A for the adversary’s expected search outlay and G_A for its expected gains, the welfare gradient in the stake is

$$\frac{\partial W}{\partial q} = E \left[\frac{\lambda}{q} Y - \omega_D (Y + \chi) \right] - \omega_A \frac{\partial O_A}{\partial q} + \omega_G \frac{\partial G_A}{\partial q}.$$

The maintained convention $(\omega_D, \omega_A, \omega_G) = (1, 0, 0)$ immediately yields $q^*/\lambda = Y/(Y + \chi)$, which is (33). Proposition 8 is in this sense a statement about the defence-inclusive convention as much as about the contest.

⁵¹Appendix A gives a supra-compensatory example under that alternative account. The delegation results below use the defence-inclusive account throughout.

⁵²The window-length elasticity with respect to liability in Proposition 5 is, in the dissipation wedge’s units, $\varepsilon_\gamma = (1 + p) u \chi / \mathcal{D}$: the dissipation wedge that prices the rate’s dissipation cost here is exactly what remains of the rate’s effect on the calendar there.

5.3 The rate when capability responds

Proposition 8 holds the model fixed, but the package is announced before development, so the rate is also a price on the capability the developer chooses. Write $a^*(h, \gamma)$ for the developer's conditional capability choice and define $\widehat{W}(h, \gamma) \equiv W(h, a^*(h, \gamma); \gamma)$. Holding the announced date fixed, the composite welfare derivative gains one term:

$$\widehat{W}_\gamma = L \left[\frac{(1 - \gamma) Y}{\gamma} - \chi \right] + \underbrace{W_a|_{a^*} a_\gamma^*}_{\text{capability correction}}. \quad (34)$$

Consider a policy pair with $L_a < 0$ (equivalently, in the crossing case, $z > z_a(a^*, \gamma)$), $\beta < 1$, and $\gamma \leq 1$. The developer then underbuilds, so $W_a|_{a^*} > 0$, but a_γ^* is not signed in general. When the harm component dominates, a higher rate raises capability because capability destroys harm at the margin and the reward $-\gamma L_a$ rises with the rate.⁵³

At the fixed-model stationary rate, a positive capability correction makes a marginal rate increase welfare-improving.⁵⁴ The correction vanishes whenever $W_a = 0$, including at $\beta = \gamma = 1$ and, when $\beta = 1$, at $L_a = 0$. Supra-compensatory liability can therefore arise here only through induced capability; it is never optimal for a fixed model.

Even a positive correction faces two limits. First, at fixed $h > 0$, the burden converges as $\gamma \rightarrow \infty$ to $m_0 c_D \kappa^{\kappa-1} h^{-(\kappa-1)}$; the stake cancels, so capability's burden relief approaches a finite limit proportional to $\sigma_D - \sigma_m$. Second, at a pointwise harm-neutral policy pair, $L_a = 0$, and the rate has no pointwise leverage over the social capability margin. In the crossing case, this locus is $z = z_a(a, \gamma)$.⁵⁵

Closing the capability wedge is consequently a fixed-point problem: the rate moves protection depth, which moves the wedge. The model neither guarantees nor rules out a feasible aligning rate.⁵⁶ The remaining instruments lie outside the model, including procurement, capability subsidies and stronger developer appropriability.

5.4 The package

A regulator holding both instruments, facing a developer that chooses its model and whether to build, announces (h, γ) . This is not a first best in which the regulator directly chooses capability.

⁵³Formally $\partial a^* / \partial \gamma = -\Pi_{a\gamma} / \Pi_{aa}$ with $\Pi_{aa} < 0$ by Assumption 3(i), so the sign is that of $-M_{a\gamma}$; decomposing $M = \gamma L + O_D$, only the term L_a in $M_{a\gamma} = L_a + \gamma L_{a\gamma} + O_{Da\gamma}$ is signed by the harm-reducing configuration.

⁵⁴A level ranking requires more: if composite welfare is strictly single-peaked on the relevant feasible interval, the endogenous-capability optimum lies above the fixed-model rate. Without that condition, only the local gradient is signed.

⁵⁵A finite rate change moves both protection depth and the harm-neutral locus. For $\beta < 1$, the algebraic aligning rate $\gamma = 1 - (1 - \beta)e^{-wh} B' / L_a$ diverges as that locus is approached from the harm-reducing side and leaves the feasible set on the other side. Once $\sigma_L \geq 2\sigma_R$, raising the rate can remove the harm-reducing region entirely; Appendix A gives the bounds.

⁵⁶Bounded assets cap the feasible rate (Shavell, 1986); saturation can make even an unbounded rate ineffective at this margin.

Define

$$\Pi^B(h, \gamma) \equiv \max_{a \in [a_{lo}, a_{hi}]} \Pi(h, a; \gamma),$$

and let $a^*(h, \gamma)$ denote the unique maximiser whenever building is active. The developer builds if and only if $\Pi^B(h, \gamma) \geq 0$, and on the active region write

$$\widehat{W}(h, \gamma) \equiv W(h, a^*(h, \gamma); \gamma).$$

At an exact build–no-build tie, the mandated value below records the better of the zero-profit build and no development.

The compact policy set $\mathcal{P}$ is economic rather than technical. With $\Psi(a_{lo}) > 0$, sufficiently long mandates induce exit uniformly over capability, so a finite $\bar{h}$ excludes no active optimum, and the upper rate $\bar{\gamma}$ is a legal, asset or administratively feasible cap.⁵⁷

Corollary 1 (Global reduction of the mandated package). *Work on the compact admissible policy set $\mathcal{P} = [0, \bar{h}] \times [\underline{\gamma}, \bar{\gamma}]$ and maintain continuity of the primitives, a unique conditional capability choice whenever the developer builds, and conditional developer profit nonincreasing in the rate—as under the quadratic baseline, where $\Pi_\gamma^B = -M_\gamma < 0$. For each h , the participation-feasible rate set $\Gamma_P(h) \equiv \{\gamma \in [\underline{\gamma}, \bar{\gamma}] : \Pi^B(h, \gamma) \geq 0\}$ is empty or a closed interval $[\underline{\gamma}, \gamma^P(h)]$. The global mandated value is therefore*

$$V(h) = \max \left\{ 0, \max_{\gamma \in \Gamma_P(h)} \widehat{W}(h, \gamma) \right\},$$

with $V(h) = 0$ when $\Gamma_P(h)$ is empty, and a global package is obtained by maximising $V(h)$ over $h \in [0, \bar{h}]$. If, in addition, $\widehat{W}(h, \cdot)$ is strictly single-peaked for every h , let $\gamma^U(h)$ denote its unconstrained maximiser on the admissible rate interval. Whenever $\Gamma_P(h)$ is non-empty, the conditional constrained rate is $\min\{\gamma^U(h), \gamma^P(h)\}$.

The full constrained package therefore reduces to a one-dimensional global search over h that covers slack participation, binding participation and no development, without relying on first-order conditions. Burden monotonicity is what makes $\Gamma_P(h)$ a lower interval; without it the feasible rate set is merely closed, the truncation rule is unavailable, and the nested maximisation proceeds over the actual feasible set. Liability tightens participation because $\Pi_\gamma^B = -M_\gamma < 0$, whereas the calendar can move the developer’s payoff either way. Under quadratic costs and an interior capability choice,

$$\frac{d}{dh} \max_a \Pi = -\beta w e^{-wh} B(a^*) + \frac{1}{2} x_D^{\text{eff}}(h) M, \quad (35)$$

so the developer trades benefit decay against burden relief.

The remaining conditions diagnose an active interior package. With slack participation, the corrected rate-balance rule (26) and rate condition (34) hold together. With binding participation,

⁵⁷The corollary makes no claim about compactifying an unbounded rate jointly with the calendar.

let $\theta \geq 0$ be its multiplier and let a_h^* and a_γ^* denote the capability responses. Then

$$L \left[\frac{(1-\gamma)Y}{\gamma} - \chi \right] + W_a a_\gamma^* = \theta \frac{\partial M}{\partial \gamma} \geq 0, \quad (36)$$

$$(1 + \theta\beta) W_h + W_a a_h^* = -\theta x_D^{\text{eff}} \left[\frac{1}{2}(1-\beta)M - \beta(1-\gamma)L \right]. \quad (37)$$

If $\theta > 0$, (36) makes the unconstrained liability gradient strictly positive at the constrained package. Equation (37) adjusts the date condition for uncaptured benefit and uninsured harm; its bracket is not signed and gives no unconditional direction for the release date.⁵⁸

6 Delegation

So far, the regulator has set the release date. Under delegation, the regulator sets only the liability rate; the developer chooses the model, the release date, and whether to build, and finances the defender's evaluation over the chosen window. Delegation resembles voluntary frontier-model evaluations and is easier to administer than a mandate whose start date and contracted search must be verified. The comparison is therefore between that administrative advantage and the policy control the regulator gives up.

6.1 When delegation reproduces the mandated date

At a fixed capability, the developer chooses the date to maximise $\beta e^{-wh} B - M$. At full appropriability and full compensation, this is also the planner's objective. That observation gives delegation its strongest case, but it does not make full compensation the right liability rate: Proposition 8 calls for partial liability because defensive search is costly. The question is whether one rate can both induce the desired search and decentralise the desired date.

Proposition 9 (Delegation versus a mandate). *Maintain the quadratic, fixed-attack-population baseline and the defence-inclusive welfare account, and let $\beta = 1$.*

- (i) *At $\gamma = 1$, $\Pi(h, a; 1) \equiv W(h, a; 1)$. Conditional on full compensation, delegation therefore implements the welfare-maximising date, capability, and development decision.*
- (ii) *Holding capability fixed, no liability rate implements an active interior mandated optimum. Alignment of the two interior dates requires $\gamma = 1$, whereas every welfare-optimal rate at a fixed date and model lies strictly below one.*
- (iii) *Suppose capability is endogenous and $a^*(h, \gamma)$ is a locally unique, differentiable interior response. Every active regular constrained mandated optimum (h^M, γ^M) again has $\gamma^M < 1$. It can be imple-*

⁵⁸Without strict complementarity, the gradient is only weakly non-negative. Where composite welfare is single-peaked in the rate, a positive gradient places the constrained rate on the rising branch, below the unconstrained optimum.

mented under delegation only if, at that package,

$$\hat{L}_h \equiv \frac{\partial}{\partial h} L(h, a^*(h, \gamma), \gamma) = -x_D^{\text{eff}} L + L_a a_h^* = 0. \quad (38)$$

Proposition 9 isolates the policy conflict. At full compensation, a developer that captures the full benefit needs no timing mandate. Once the regulator lowers the rate to balance harm reduction against search cost, the developer generally wants a different date. Implementation then fails whenever composite harm changes locally with the release date, and the exception in (38) is a knife-edge on which induced capability exactly offsets the protection an extra day buys directly. It is not a second capability condition: at a common date and rate, the same developer chooses the same model in both regimes.⁵⁹

This distinction also disciplines how much to claim for the mandate. Near full compensation, with smooth active interior choices, its welfare advantage at a common rate is only second order: delegation starts from the same date and model at $\gamma = 1$, and the mandate gains only by correcting the date displacement a small rate change causes. A mandate is therefore least valuable on the timing margin when appropriability is high and the desired liability rate is close to one.⁶⁰

6.2 When liability cannot replace the mandate

Ignoring the mandate's administrative cost, control of the date weakly enlarges the regulator's feasible set: it can always reproduce the date the developer would have selected. It is strictly useful on three distinct margins.

Timing incentives. Low appropriability by itself does not sign the developer's timing preference. At the fixed-capability planner date, the developer favours additional headway if and only if

$$(1 - \beta) \frac{\kappa - 1}{\kappa} M > \beta(1 - \gamma)L.$$

The left side is the smaller private cost of delaying benefits; the right side is the uninsured harm that the planner, but not the developer, values. Under quadratic costs and holding capability fixed, the delegated date rises with the liability rate, whereas the mandated date can fall because the same rate accelerates defensive search. With endogenous capability, neither response has an unconditional sign, and exact alignment remains exceptional under the regularity conditions of Proposition 9; Section B.5 collects the general condition.

⁵⁹For $\beta < 1$, the general condition contains the term $W_a a_h^*$ —the welfare value of the capability induced by marginal headway—rather than a separate requirement that $W_a = 0$. Section B.5 states and proves that condition, the fixed-capability aligning rate, and the regular-value sense in which implementation of a selected mandated branch is non-generic.

⁶⁰Section B.5 gives the formal expansion. Corners, regime changes and low appropriability need not share that local conclusion.

The calendar bound. Even when liability moves the delegated date in the right direction, it cannot buy an arbitrarily long evaluation. Conditional on a fixed model, the delegated problem is the planner’s full-compensation scheduling problem with benefit βB and harm $\gamma \lambda$. As the rate becomes arbitrarily large, the date converges to the finite saturation date $h_\infty(\beta)$ or the developer exits; with endogenous capability, the limiting package instead maximises the corresponding joint date–capability surplus. Thus a liability rate cannot implement a mandated window beyond the delegated calendar bound. Section B.5 gives the cases and viability condition.

Exclusion. A mandate can decline to license. Liability alone can fail earlier than the familiar balance-sheet limit because a developer that controls the calendar can delay until the posterior-scaled variable budget becomes small—*calendar evasion*. Does that mechanism survive a programme overhead? The answer depends on the limiting-surplus condition.

Proposition 10 (Calendar evasion with programme overhead). *Under the posterior-budgeted variable-cost specification, let the developer also pay a stock-independent programme overhead $f_D \geq 0$ per unit of headway. Suppose $\Psi(a_{10}) > 0$ and the capability primitives are continuous and strictly positive on the compact capability interval. Define $M_\infty^f(h, a) \equiv m_0(a)c_D(a)\kappa^{\kappa-1}h^{-(\kappa-1)} + f_D h$ and $\Pi_\infty^f(h) \equiv \max_a \left\{ \beta e^{-wh} B(a) - \Psi(a) - M_\infty^f(h, a) \right\}$. The developer declines to build at every sufficiently long window for every liability rate, and at every sufficiently short window once the rate is high enough. Moreover,*

$$\sup_{h>0} \Pi_\infty^f(h) \geq 0 \iff \text{no finite liability rate excludes development at every window.} \quad (39)$$

Call condition (39) the *limiting-surplus condition*. Calendar evasion relies on the variable budget becoming small as the posterior expected residual stock is depleted. A fixed programme overhead (the baseline is $f_D = 0$) makes evasion harder and can eliminate it, but the mechanism survives exactly when the limiting-surplus condition (39) holds. This is calendar evasion rather than judgment-proofness; asset limits provide the separate route familiar from Shavell (1984, 1986).

The comparison is therefore not simply “mandate or liability”. Delegation is most attractive when the developer captures most of the model’s benefits, the desired rate is near full compensation, the optimal package is active and interior, and exclusion is not at issue. A mandated option is most valuable when appropriability is low, when partial liability creates a material timing disagreement, when the desired window exceeds the delegated calendar bound, or when the regulator may want no development or release at all. Administrative costs can overturn a small timing gain near full compensation, but they do not restore the missing calendar or exclusion margins.

7 Conclusion

Under the instantaneous-remediation, effective-exclusivity benchmark, a regulator combines a committed release date with liability for harm after release. The mechanism is an asymmetry in

response: after release, liability strengthens both sides of a contest; before release, it strengthens only the defender. Protection depth, not elapsed time alone, is therefore the policy object.

For a fixed model, the global choice is among immediate release, a positive window and withholding. Any staged optimum obeys a finite ceiling, and greater harm can lengthen the date only towards a finite limit. Under quadratic search costs, a fixed attack population and defence-inclusive welfare, the active interior liability rate is partial. Full compensation aligns a delegated date when the developer captures all benefits, but it does not implement that fixed-model package.

When capability and participation respond, the regulator must instead compare packages after the developer chooses whether and what to build. On the compact feasible policy set, the global problem reduces to solving the participation-feasible rate choice at each date and then searching over dates, with no development as the outside option. First-order conditions diagnose an active interior solution; they do not replace that comparison. Delegation removes the regulator's direct licensing choice. Liability may still induce exit when the limiting surplus is negative; otherwise the developer can evade even a high rate through its calendar, with programme overhead governed by the limiting-surplus condition.

Table 3 records which departures preserve a result and which reopen the policy problem. The open-release externality has a separate boundary: fringe capability must persist into later generations, and the releasing developer must not internalise future policy costs. Empirically, effort paths identify search curvature, while finding histories and residual-stock measures identify protection. Both are needed for contingent release, because few late findings can mean either a depleted stock or neglected search. A calendar rule preserving neither effective exclusivity nor financed defensive effort does not create the mechanism analysed here.

Appendices

A Proofs and supporting derivations

This appendix contains only the proofs and derivations supporting the results stated in the main text. They follow the order of the main sections. Extension results are collected in Appendix B and proved where they are stated. Throughout the bilateral contest, $R \equiv x_D/x_A$ denotes the ratio of defender to adversary per-flaw search intensity, so that $p = 1/(1 + R)$.

A.1 The flaw contest

Lemma 1. For the auxiliary statewise problem, argue by induction on n , with $V_A(0) = V_D(0) = 0$. At state n , for any pair of intensities, the next resolution arrives at rate $n(x_A + x_D)$, so expected elapsed time is $1/(n(x_A + x_D))$ and expected flow cost to party i is $nC_i(x_i)/(n(x_A + x_D)) = C_i(x_i)/(x_A + x_D)$. With probability $x_A/(x_A + x_D)$, the resolution is an exploit and otherwise a patch, and in either case the state becomes $n - 1$: the continuation value entering party i 's problem is $V_i(n - 1)$ with certainty. Because payoffs are undiscounted, it enters additively and without a rate-dependent weight, so differentiating (1) with respect to x_i eliminates it, and best responses at n coincide with those at every other state. The maximised per-resolution values are therefore constants and $V_i(n) = nV_i(1)$. $\square$

Proposition 1. Differentiating $(x_A v - C_A(x_A))/(x_A + x_D)$ in x_A and clearing the denominator gives $C'_A(x_A)(x_A + x_D) = vx_D + C_A(x_A)$, which is $C'_A(x_A) = v - g$ with g the maximised value. The defender's condition $C'_D(x_D)(x_A + x_D) = x_A \gamma \lambda + C_D(x_D)$ is $C'_D(x_D) = \ell$. Since $g > 0$ and $\ell < \gamma \lambda$ at any interior point, both marginal costs lie strictly below the corresponding gross stake.

For existence and uniqueness under quadratic costs, write the two conditions as $x_D(v - c_A x_A) = \frac{1}{2} c_A x_A^2$ and $x_A(\gamma \lambda - c_D x_D) = \frac{1}{2} c_D x_D^2$. Solving each for its stake gives $v = c_A x_A(2R + 1)/(2R)$ and $\gamma \lambda = c_D x_A R(R + 2)/2$, whose ratio is $\Theta = R^2(R + 2)/(2R + 1)$. That expression is continuous and strictly increasing from 0 to ∞ on $R > 0$, so it has a unique positive root, and $p = 1/(1 + R)$ then satisfies (3), whose right-hand side is correspondingly strictly decreasing on $(0, 1)$. Substituting back gives the intensities. The pair is a mutual global best response because each party's objective is strictly quasi-concave in its own intensity given a positive intensity for the other; and $(0, 0)$ is not an equilibrium candidate, since for $x_D = 0$ the adversary's payoff $v - C_A(x_A)/x_A$ is strictly decreasing with unattained supremum, which uses $C'_A(0) = 0$. By Lemma 1, the same unique pair solves the auxiliary continuation game at every positive stock. In the actual incomplete-information game, conditional expected search costs and discovery returns are multiplied by the same posterior mean residual stock, so that factor cancels and the pair is sequentially optimal after every public history. It is therefore the unique equilibrium within the stationary Bayesian class specified in the text. The maintained domain $\gamma > 0$ is essential: at $\gamma = 0$ the defender chooses zero effort, and the adversary's supremum as $x_A \downarrow 0$ is not attained. Every zero-liability statement below is therefore a

limit as $\gamma \downarrow 0$.

For the per-flaw stakes, the defender's condition rearranges to $x_A(\gamma\lambda - C'_D) = x_D C'_D - C_D$. With $C_D(x) = c_D x^\kappa / \kappa$, the right-hand side is $(\kappa - 1)C_D$, and substituting $C'_D(x_D^*) = \ell$ and solving yields $\ell = \kappa\gamma\lambda p / (\kappa - 1 + p)$, which at $\kappa = 2$ is $2\gamma\lambda p / (1 + p) = c_D x_D^*$. The adversary's per-flaw gain under quadratic costs is $g = vp - C_A(x_A^*) / (x_A^* + x_D^*) = vp / (2 - p)$. The interiority ratios follow by dividing the closed-form intensities by v/c_A and $\gamma\lambda/c_D$. $\square$

The common-curvature contest asymptote. For later limit arguments, suppose both parties have the common cost curvature $\kappa > 1$. The first-order conditions imply

$$v = c_A x_A^{\kappa-1} \left[1 + \frac{\kappa-1}{\kappa R} \right], \quad \gamma\lambda = c_D x_D^{\kappa-1} \left[1 + \frac{\kappa-1}{\kappa} R \right],$$

and hence

$$\Theta = R^\kappa \frac{\kappa + (\kappa-1)R}{\kappa R + (\kappa-1)}.$$

The right-hand side is strictly increasing from zero to infinity. As $\Theta \rightarrow \infty$, it is asymptotic to $(\kappa-1)R^\kappa/\kappa$, so

$$R \sim \left(\frac{\kappa}{\kappa-1} \Theta \right)^{1/\kappa}, \quad p \sim \left(\frac{\kappa}{\kappa-1} \Theta \right)^{-1/\kappa}.$$

Hence the pass-through $\varphi = -d \log p / d \log \Theta$ approaches $1/\kappa$ as $\Theta \rightarrow \infty$. This establishes within Appendix A the common-curvature asymptote used below.

Proposition 2. Monotonicity and the limits follow from the strict monotonicity of the right-hand side of (3). For the asymptote, $\Theta = R^2(R+2)/(2R+1) \sim R^2/2$ for large R , so $R \sim (2\Theta)^{1/2}$ and $p \sim (2\Theta)^{-1/2}$; for small Θ , $p \sim 1 - (\Theta/2)^{1/2}$. Writing the pass-through in terms of R ,

$$\varphi(R) = \frac{R(2R^2 + 5R + 2)}{4R^3 + 11R^2 + 11R + 4},$$

the numerator of $\frac{1}{2} - \varphi(R)$ is $R^2 + 7R + 4$ and the numerator of $\varphi'(R)$ is $2R^4 + 28R^3 + 57R^2 + 40R + 8$, both strictly positive for $R > 0$ over positive denominators. Hence $\varphi < \frac{1}{2}$ everywhere and φ is strictly increasing in R , hence in Θ , with limits 0 and $\frac{1}{2}$. $\square$

An inequality used in the calendar comparative statics. The same quadratic formula proves the strict inequality invoked in Proposition 5. Since $p = (1+R)^{-1}$,

$$1 - p^2 - 2\varphi = \frac{R(R^3 + 5R^2 + 8R + 4)}{(1+R)^2(4R^3 + 11R^2 + 11R + 4)} > 0.$$

Dissipation expressions (5). Defensive outlay per resolution is $\frac{1}{2}c_D x_D^{*2} / (x_A^* + x_D^*)$; substituting the intensities and $x_A^* + x_D^* = x_D^* / (1-p)$ gives $\gamma\lambda p(1-p)/(1+p)$, and the adversary's follows

from $x_A^* + x_D^* = x_A^*/p$. The cost parameters cancel because the intensities are inversely proportional to them and the costs are quadratic. Differentiating $p(1-p)/(1+p)$ gives $p^2 + 2p - 1 = 0$, so the defensive peak is at $p = \sqrt{2} - 1$; differentiating $p(1-p)/(2-p)$ gives $p^2 - 4p + 2 = 0$ with root $2 - \sqrt{2}$. $\square$

A.2 Staged access

Proposition 3 and the burden (8). On $[0, h]$ each flaw is an independent problem under $\alpha = 1$. Let $\psi(t)$ be the defender's expected remaining loss on a flaw still live at t , so $\psi(h) = \ell$. The per-flaw Bellman equation is $-\dot{\psi} = \min_x \{C_D(x) - x\psi\}$, so $C'_D(x^*) = \psi$ and $\dot{\psi} = x^*C'_D(x^*) - C_D(x^*) \geq 0$, strictly for $\psi > 0$. Since C'_D is increasing, x^* inherits the monotonicity of ψ ; at $t = h$, $C'_D(x^*) = \ell = C'_D(x_D^*)$ by (2), so the solo intensity docks onto the contest intensity.

For general curvature, $C'_D(x) = c_D x^{\kappa-1} = \psi$ gives $x = (\psi/c_D)^{1/(\kappa-1)}$ and $C_D = x\psi/\kappa$, so $\dot{\psi} = x\psi(\kappa-1)/\kappa$. In terms of x this is $\dot{x} = x^2/\kappa$, whose backward solution from $x(h) = x_D^*$ is exactly (9): $1/x^*(t) = 1/x_D^* + (h-t)/\kappa$. Then $\int_0^h x^* dt = \kappa \log(1 + z/\kappa)$ with $z = x_D^* h$, and the survival probability $\exp(-\int_0^h x^*)$ equals $(1 + z/\kappa)^{-\kappa}$, giving (10); the limit as $\kappa \rightarrow \infty$ is standard. Evaluating (9) at $t = 0$ gives $x^*(0) = x_D^*/(1 + z/\kappa)$, and $-d \log L/dh$ from (7) and (10) equals the same quantity, which is (11). Also $\psi(0) = c_D x^*(0)^{\kappa-1} = \ell(1 + z/\kappa)^{-(\kappa-1)}$.

For the burden: $\psi(0)$ collects expected liability payments plus expected search outlay per flaw, over the window and the contest that follows, so $M(h) = m_0 \psi(0) = m_0 \ell(1 + z/\kappa)^{-(\kappa-1)}$, which is (8). That it decomposes as $\gamma L + O_D$ follows because the contest stake itself separates into expected liability plus contest outlay: at $\kappa = 2$, $\ell = \gamma \lambda p + \gamma \lambda p(1-p)/(1+p)$ by the dissipation expressions, and $m(h) \gamma \lambda p = \gamma L$. Differentiating, $\partial \log M/\partial h = -\frac{\kappa-1}{\kappa} x_D^{\text{eff}}(h)$, which with $\Lambda = M + (1-\gamma)L$ identifies L_{ch} as the chargeable loss and $x_D^{\text{eff}} L_{\text{ch}}$ as the chargeable protection flow in (12). Since $O_D = M - \gamma L$ and $L' = -x_D^{\text{eff}} L$, subtraction gives the defensive-outlay derivative stated after (12). $\square$

Section 3.1: the adversary's prize elasticity. From the closed-form rate, $\log x_A^* = \log 2 + \log v + \log(1-p) - \log c_A - \log(2-p)$, and $d \log p/d \log v = \varphi$, so $d \log x_A^*/d \log v = 1 - \varphi p/[(1-p)(2-p)]$, which tends to 1 as $\Theta \rightarrow \infty$ and to $\frac{1}{2}$ as $\Theta \rightarrow 0$, using $1-p \sim (\Theta/2)^{1/2}$ and $\varphi \sim \frac{1}{2}(1-p)$ there. $\square$

Proposition 4. Recall that $s = w/x_D^*$ is benefit decay relative to contest search intensity and $\eta_c = L_{\text{ch}}(0)/B$ is initial chargeable loss relative to release benefit. *Existence*, once for all parts. $S(h) = e^{-wh} B - \Lambda(h)$ is continuous on $[0, \infty)$, and as $h \rightarrow \infty$ the benefit decays exponentially while Λ decays polynomially, so $S(h) \rightarrow 0^-$. Extending S to $[0, \infty]$ by $S(\infty) = 0$, the value of withholding, gives a continuous function on a compact set, so the supremum is attained. Since withholding is available, $S \geq 0$ at any optimum, and the optimal policy is immediate release, staging at an interior maximiser, or withholding.

(i) At $\gamma = 1$, $\Lambda = M$ and $\mathcal{S}(h) = e^{-wh}B - M(h)$ with $M = m_0\ell(1 + z/\kappa)^{-(\kappa-1)}$, so

$$\mathcal{S}'(h) = \frac{\kappa-1}{\kappa} x_D^* m_0\ell \left(1 + \frac{z}{\kappa}\right)^{-\kappa} - wBe^{-wh};$$

dividing by wBe^{-wh} and taking logarithms, $\mathcal{S}'(h) > 0$ if and only if $G_1(z) < \log(\eta_c/s)$, where $G_1(z) \equiv \kappa \log(1 + z/\kappa) - sz$, using $\eta_c = \frac{\kappa-1}{\kappa} m_0\ell/B$. $G_1(0) = 0$ and $G_1''(z) = -\kappa/(\kappa + z)^2 < 0$, so the superlevel set $\{z : G_1(z) > \log(\eta_c/s)\}$ is an interval: $\mathcal{S}'$ changes sign at most twice—positive, negative, positive—and $\mathcal{S}$ has at most one interior local maximum, at the left endpoint z_1 , which is therefore z^* wherever staging is optimal. $z_1 > 0$ requires $\eta_c > s$; at $\eta_c \leq s$ there is no interior maximum and $\mathcal{S}(0) = B[1 - \frac{\kappa}{\kappa-1}\eta_c] > 0$ whenever $\eta_c < \frac{\kappa-1}{\kappa}$, so immediate release is optimal there.

For the partition and the sharpened ceiling: at an interior stationary point $\frac{\kappa-1}{\kappa}M = we^{-wh}B/x_D^{\text{eff}}$, so $\mathcal{S} = e^{-wh}B[1 - \frac{\kappa}{\kappa-1}w/x_D^{\text{eff}}]$, positive if and only if $x_D^{\text{eff}} > \frac{\kappa}{\kappa-1}w$, that is $z < \bar{z}_1 \equiv (\kappa - 1)/s - \kappa$, which requires $s < \frac{\kappa-1}{\kappa}$ and, on dividing by x_D^* , is the sharpened full-compensation ceiling

$$h^* < \frac{\kappa - 1}{w} - \frac{\kappa}{x_D^*}. \quad (40)$$

The maximiser of G_1 is $z_{G_1} = \kappa/s - \kappa$, and $z_{G_1} - \bar{z}_1 = 1/s > 0$, so $\bar{z}_1$ lies strictly left of the peak; hence $z_1 < \bar{z}_1$ if and only if $\log(\eta_c/s) < G_1(\bar{z}_1) = \kappa \log \frac{\kappa-1}{\kappa s} - (\kappa - 1) + \kappa s$, that is,

$$\eta_c < \bar{\eta}_c(s) \equiv \left(\frac{\kappa-1}{\kappa}\right)^\kappa s^{1-\kappa} e^{\kappa s - (\kappa-1)}, \quad (41)$$

the boundary of part (i), which equals $e^{2s-1}/(4s)$ at $\kappa = 2$. The staged interval $(s, \bar{\eta}_c(s))$ is nonempty on $s < \frac{\kappa-1}{\kappa}$: with $t \equiv \kappa s/(\kappa - 1) \in (0, 1)$, $\log(\bar{\eta}_c(s)/s) = \kappa \log(1/t) + (\kappa - 1)(t - 1) \geq \kappa(1 - t) - (\kappa - 1)(1 - t) = 1 - t > 0$. Ties: at $\eta_c = \bar{\eta}_c(s)$, $z_1 = \bar{z}_1$ and $\mathcal{S}(h_1) = 0$, the withholding value; for $s \geq \frac{\kappa-1}{\kappa}$ no interior optimum beats withholding and the comparison is $\max\{\mathcal{S}(0), 0\}$, with the tie at $\eta_c = \frac{\kappa-1}{\kappa}$, where the burden equals the benefit. These explicit comparisons establish the partition in part (i).

(ii) At $\kappa = 2$, write $u(z) = (1 + z/2)^{-1}$ and define

$$q_\gamma \equiv 1 - \frac{\gamma p}{1 + p}, \quad r_\gamma \equiv 1 - \gamma + \frac{2\gamma}{1 + p}, \quad f_\gamma(z) \equiv \frac{2\gamma}{1 + p}u(z) + (1 - \gamma)u(z)^2,$$

so that the cutoff in the statement is $\bar{s}_\gamma(p) = q_\gamma/r_\gamma$. Normalised surplus is

$$\frac{\mathcal{S}(z)}{B} = e^{-sz} - \eta f_\gamma(z) = e^{-sz} - \frac{\eta_c}{q_\gamma} f_\gamma(z).$$

The marginal condition at $z = 0$ is $\mathcal{S}_z(0) > 0$ if and only if $\eta_c > s$. Immediate-release surplus is

$$\frac{\mathcal{S}(0)}{B} = 1 - \eta r_\gamma = 1 - \frac{\eta_c}{\bar{s}_\gamma(p)}.$$

Thus immediate release and withholding tie at $\eta_c = \bar{s}_\gamma(p)$.

For the comparison with withholding, define $\mathcal{Q}(z) = e^{-sz}/f_\gamma(z)$. Since

$$\log \mathcal{Q}(z) = -sz + \log(1 + z/2) - \log[2\gamma/(1 + p) + (1 - \gamma)u(z)],$$

$\log \mathcal{Q}$ is strictly concave. In particular, for $u = (1 + z/\kappa)^{-1}$,

$$\frac{d^2}{dz^2}[-\log(a + bu)] = -\kappa^{-2} \frac{bu^2(2au + bu^2)}{(a + bu)^2} \leq 0.$$

The inequality is strict when $b > 0$; when $b = 0$, the $\log(1 + z/2)$ term makes $\log \mathcal{Q}$ strictly concave. Moreover, $(\log \mathcal{Q})'(0) = \bar{s}_\gamma(p) - s$. If $s \geq \bar{s}_\gamma(p)$, $\mathcal{Q}$ is maximised at zero, so no positive finite date can beat withholding once immediate-release surplus is non-positive. If $s < \bar{s}_\gamma(p)$, $\mathcal{Q}$ has a unique positive maximiser $\bar{z}_\gamma(s) > 0$, characterised by

$$s = -\frac{f'_\gamma(\bar{z}_\gamma)}{f_\gamma(\bar{z}_\gamma)} = \frac{u(\bar{z}_\gamma) [\gamma/(1 + p) + (1 - \gamma)u(\bar{z}_\gamma)]}{2\gamma/(1 + p) + (1 - \gamma)u(\bar{z}_\gamma)},$$

and a finite date beats withholding exactly when

$$\eta_c < q_\gamma \max_{z \geq 0} \mathcal{Q}(z) = \bar{\eta}_c(s, \gamma, p) \equiv q_\gamma \frac{e^{-s\bar{z}_\gamma(s)}}{f_\gamma(\bar{z}_\gamma(s))},$$

which is the upper boundary the statement leaves to this proof. Finally, strict concavity of the stationary-point transformation in (14) implies that surplus has at most one interior local maximum. If $\eta_c \leq s$, no such maximum can dominate immediate release. If $s < \eta_c < \bar{\eta}_c(s, \gamma, p)$, surplus initially rises and a finite date beats withholding, so the unique interior maximum dominates immediate release. At and above the upper boundary, the withholding comparison already supplies the global choice. At the lower boundary, the staged solution converges to immediate release, and at the upper boundary, the finite staged date ties withholding. Combining these comparisons gives the stated partition. At $\gamma = 1$, solving $-f'_\gamma/f_\gamma = s$ gives $u = 2s$ and recovers $\bar{\eta}_c = e^{2s-1}/(4s)$.

(iii) At an interior stationary point, $\mathcal{S} = e^{-wh}B - \Lambda$ with $we^{-wh}B = x_D^{\text{eff}}L_{\text{ch}}$ and $L_{\text{ch}} < \Lambda$ gives $\mathcal{S} = e^{-wh}B[1 - w\Lambda/(x_D^{\text{eff}}L_{\text{ch}})] < e^{-wh}B[1 - w/x_D^{\text{eff}}]$, so beating withholding requires $x_D^{\text{eff}}(h) > w$, that is $1/x_D^* + h/\kappa < 1/w$, which is (16); since (16) demands $x_D^* > w$, staging is never optimal at $s \geq 1$. (iv) For the harm monotonicity at $\gamma = 1$: then $A_2 = 0$, $\Lambda = M = m_0c_D x_D^*(1 + x_D^*h/2)^{-1}$, and

$$\mathcal{S}_h(h, \lambda) = -wBe^{-wh} + \frac{m_0c_D}{2} \left(\frac{x_D^*}{1 + x_D^*h/2} \right)^2.$$

The second term is strictly increasing in x_D^* at every h , and $x_D^* = \ell/c_D$ with $d \log \ell / d \log \lambda = 1 - \zeta\varphi > 0$, writing $\zeta \equiv 1/(1 + p)$ here and throughout the appendix, so $\mathcal{S}$ has strictly increasing differences in (h, λ) on $[0, \infty)$. The differences extend to the withholding corner: treating withholding as the point at infinity with value $0 = \lim_{h \rightarrow \infty} \mathcal{S}$, the difference $\mathcal{S}(\infty, \lambda) - \mathcal{S}(h, \lambda) = M(h, \lambda) - Be^{-wh}$

is increasing in λ because $d \log M / d \log \lambda = (1 - \zeta \varphi) / (1 + z/2) > 0$. By monotone comparative statics (Topkis, 1998), the smallest and largest optimal windows are nondecreasing in λ , and the monotonicity covers the transitions between immediate release, staging, and withholding, which is the global scope of the claim; the interior-root calculation—the marginal benefit of delay rises in λ at every h , so the smaller root rises—is the regular-branch case of the same argument. For the limit, first the intensities at general curvature: $\Theta \propto \lambda$, and by the common-curvature asymptote above, $p \sim [\kappa \Theta / (\kappa - 1)]^{-1/\kappa} \asymp \lambda^{-1/\kappa}$, so $\ell = \kappa \gamma \lambda p / (\kappa - 1 + p) \asymp \lambda^{(\kappa-1)/\kappa}$, $x_D^* = (\ell / c_D)^{1/(\kappa-1)} \asymp \lambda^{1/\kappa}$, and $\lambda p \asymp \lambda^{(\kappa-1)/\kappa}$ —at $\kappa = 2$, the quadratic intensity scalings $\lambda^{\pm 1/2}$ used elsewhere. As $\lambda \rightarrow \infty$ at fixed $h > 0$, then, $z = x_D^* h \asymp \lambda^{1/\kappa} \rightarrow \infty$, $(1 + z/\kappa)^{-(\kappa-1)} \rightarrow (\kappa/z)^{\kappa-1}$, and

$$M \rightarrow m_0 \ell \kappa^{\kappa-1} (x_D^* h)^{-(\kappa-1)} = m_0 c_D \kappa^{\kappa-1} h^{-(\kappa-1)},$$

using $\ell = c_D x_D^{*\kappa-1}$: the stake cancels. Also $|1 - \gamma| L \rightarrow 0$, since $L = \lambda p m_0 (1 + z/\kappa)^{-\kappa} \lesssim \lambda^{(\kappa-1)/\kappa} z^{-\kappa} \asymp \lambda^{-1/\kappa} \rightarrow 0$. Writing $K_0 \equiv \kappa^{\kappa-1} m_0 c_D$, the limiting surplus is $\mathcal{S}_\infty(h) = B e^{-wh} - K_0 h^{-(\kappa-1)}$ and the limiting first-order condition is $w B e^{-wh} = (\kappa - 1) K_0 h^{-\kappa}$, which is (17); its left-hand side times h^κ is single-peaked at $h = \kappa/w$, so the relevant root is the smaller one, it lies below κ/w , and it exists whenever $w B (\kappa/w)^\kappa e^{-\kappa} \geq (\kappa - 1) K_0$.

Existence of that root is not sufficient. Substituting the first-order condition into the surplus gives $K_0 h^{-(\kappa-1)} = w B e^{-wh} h / (\kappa - 1)$ at any limiting stationary date, so

$$\mathcal{S}_\infty(h) = B e^{-wh} \left[1 - \frac{wh}{\kappa - 1} \right],$$

which at h_∞ is positive if and only if the viability condition of part (iv) holds:

$$w h_\infty < \kappa - 1. \quad (42)$$

Since the smaller root approaches κ/w as the right-hand side of (17) approaches its peak, while the viability ceiling is $(\kappa - 1)/w < \kappa/w$, roots violating (42) exist; withholding is then optimal in the limit. Where (42) holds, the limiting ceiling is $(\kappa - 1)/w$, strictly inside (16).

Convergence of the optimal window needs an argument beyond pointwise convergence of the surplus, and the viability condition is what powers it. Write $x_D^{\text{eff}} = (1/x_D^* + h/\kappa)^{-1}$, so that $M = m_0 c_D (x_D^{\text{eff}})^{\kappa-1}$ and the limit object is $K_0 h^{-(\kappa-1)} = m_0 c_D (\kappa/h)^{\kappa-1}$. On any compact $[\varepsilon, H]$: $0 \leq \kappa/h - x_D^{\text{eff}} = (\kappa/h) x_D^{\text{eff}} / x_D^* \leq (\kappa/\varepsilon)^2 / x_D^* \rightarrow 0$, so $M \rightarrow K_0 h^{-(\kappa-1)}$ uniformly, and $\sup_{[\varepsilon, H]} |1 - \gamma| L \lesssim \lambda^{-1/\kappa} \varepsilon^{-\kappa} \rightarrow 0$; hence $\mathcal{S}(\cdot, \lambda) \rightarrow \mathcal{S}_\infty$ uniformly on $[\varepsilon, H]$. For the near-zero end, a bound on Λ from below must respect the sign of $1 - \gamma$. From the closed forms, $L/M = (\kappa - 1 + p) / (\kappa \gamma (1 + z/\kappa)) < 1/\gamma$, so for $\gamma > 1$,

$$\Lambda = M - (\gamma - 1)L \geq M \left[1 - \frac{\gamma - 1}{\gamma} \right] = \frac{M}{\gamma},$$

while for $\gamma \leq 1$, $\Lambda \geq M$ directly: in either case $\Lambda \geq M/\gamma_+$ with $\gamma_+ \equiv \max\{1, \gamma\}$, and γ

is fixed throughout the limit. Choose ε with $m_0 c_D (\kappa/\varepsilon)^{\kappa-1} \geq \gamma_+ (B+2)$, so that for λ large $\mathcal{S}(h, \lambda) \leq B - M(\varepsilon, \lambda)/\gamma_+ \leq -1$ on $(0, \varepsilon]$ (the burden falls in the calendar), and choose H with $Be^{-wH} < \mathcal{S}_\infty(h_\infty)/2$, so that $\mathcal{S}(h, \lambda) \leq Be^{-wh} < \mathcal{S}_\infty(h_\infty)/2$ on $[H, \infty)$, using $\Lambda \geq 0$ there. Since $\mathcal{S}(h_\infty, \lambda) \rightarrow \mathcal{S}_\infty(h_\infty) > 0$ under (42), the maximiser eventually lies in $[\varepsilon, H]$. On $(0, \infty)$, $\mathcal{S}'_\infty$ has the sign of $(\kappa - 1)K_0 - wBe^{-wh}h^\kappa$, positive near zero, with the subtracted term single-peaked; so $\mathcal{S}_\infty$ rises to h_∞ , falls to the larger root, then climbs towards 0 from below, and under (42) its unique global maximiser is h_∞ . Uniform convergence on a compact set containing a unique strict maximiser delivers $h^*(\lambda) \rightarrow h_\infty$ by the standard argmax argument. Where (17) has no root, $\mathcal{S}'_\infty > 0$ everywhere and $\mathcal{S}_\infty < 0$ everywhere; where the smaller root violates (42), $\sup_h \mathcal{S}_\infty \leq 0$; in either case the same region-by-region bounds give $\limsup_\lambda \sup_h \mathcal{S}(h, \lambda) \leq 0$, so withholding is optimal in the limit. $\square$

Proposition 5. Differentiate $G_1(z) = \log(\eta_c/s)$ at the smaller root, treating s and η_c as functions of a primitive q . Since $d(sz) = sz(d \log s + d \log z)$,

$$d \log z \mathcal{D} = d \log \eta_c + (wh^* - 1) d \log s, \quad \mathcal{D} \equiv z \left[\frac{\kappa}{\kappa + z} - s \right] = h^* [x_D^{\text{eff}} - w] > 0$$

at the smaller root. With $\varepsilon_q^x \equiv d \log x_D^*/d \log q$, the intensity elasticity decomposed into its stake and contest channels in (20) (so that $d \log s = -\varepsilon_q^x + \mathbf{1}\{q = w\}$), $y_q \equiv d \log(m_0 \ell)/d \log q$ (so that $d \log \eta_c = y_q - \mathbf{1}\{q = B\}$), and $\varepsilon_q = d \log z/d \log q - \varepsilon_q^x$,

$$\varepsilon_q = \frac{y_q - \mathbf{1}\{q = B\} + (wh^* - 1)(\mathbf{1}\{q = w\} - \varepsilon_q^x)}{\mathcal{D}} - \varepsilon_q^x. \quad (43)$$

For $q \in \{m_0, B, w\}$, $\varepsilon_q^x = 0$ and $y_{m_0} = 1$, $y_B = y_w = 0$, giving (18); the bound follows because beating withholding requires $x_D^{\text{eff}} \geq \kappa w/(\kappa - 1)$, so $\partial h^*/\partial \log m_0 = 1/(x_D^{\text{eff}} - w) \leq (\kappa - 1)/w$. At $\kappa = 2$ and $\gamma = 1$: from (3) and $\ell = 2\lambda p/(1 + p)$, the pairs (ε_q^x, y_q) are $(1 - \zeta\varphi, 1 - \zeta\varphi)$ for λ , $(\zeta\varphi, \zeta\varphi)$ for v , $(-\zeta\varphi, -\zeta\varphi)$ for c_A , and $(\zeta\varphi - 1, \zeta\varphi)$ for c_D . Substituting into (43) and using $zu = 2(1 - u)$ —so that $2 - wh^* - \mathcal{D} = 2u$ —gives the first three displays of (19).

The liability rate also perturbs the criterion, so (43) does not apply to it. Write the first-order condition as $F(h, \gamma) \equiv x_D^{\text{eff}} L_{\text{ch}} - wBe^{-wh} = 0$ with $L_{\text{ch}} = \frac{\kappa-1}{\kappa} M + (1 - \gamma)L$. Differentiating and using $F = 0$, $\partial F/\partial h = -wBe^{-wh}(x_D^{\text{eff}} - w) < 0$ at the smaller root. At $\gamma = 1$,

$$\gamma \frac{\partial F}{\partial \gamma} = \frac{\kappa-1}{\kappa} \frac{\partial (x_D^{\text{eff}} M)}{\partial \log \gamma} \Big|_h - x_D^{\text{eff}} L = wBe^{-wh} (1 - \zeta\varphi)(2 - zu) - x_D^{\text{eff}} L,$$

using $x_D^{\text{eff}} M = x_D^* m_0 \ell (1 + z/\kappa)^{-\kappa}$, whose log-derivative in the rate at fixed h is $(\varepsilon_\gamma^x + y_\gamma) - zu \varepsilon_\gamma^x = (1 - \zeta\varphi)(2 - zu)$ at $\kappa = 2$, and the first-order condition. Since $L/M = \frac{1+p}{2} u$ at $\gamma = 1$, the last term is $wBe^{-wh}(1 + p)u$, and the implicit-function theorem gives

$$\varepsilon_\gamma = \frac{(1 - \zeta\varphi)(2 - zu) - (1 + p)u}{\mathcal{D}} = \frac{u[2(1 - \zeta\varphi) - (1 + p)]}{\mathcal{D}} = \frac{u(1 - p^2 - 2\varphi)}{(1 + p)\mathcal{D}},$$

using $2 - zu = 2u$ and $2(1 + p) - 2\varphi - (1 + p)^2 = 1 - p^2 - 2\varphi$; positivity is the inequality $2\varphi < 1 - p^2$ established in the proof of Proposition 8. Finally, $\varepsilon_\lambda \rightarrow 0$ as $\lambda \rightarrow \infty$: $u \rightarrow 0$ while $\mathcal{D} \rightarrow \kappa - wh_\infty > 0$ under (42). $\square$

Computation of the main figures. The computed boundary in Figure 2 and the path in Figure 3 are generated directly from the closed forms above; replication code is available from the author.

A.3 Endogenous capability

Assumption 3: a sufficient condition. From (23) the log-derivative of harm is $\sigma_L - \sigma_R \Omega(z)$ with $\Omega(z) = z/(1 + z/2)$, and both elasticities move with a through $\varphi(\Theta(a; \gamma))$ and $p(\Theta(a; \gamma))$: $d\sigma_L/d\log a = -\Delta^2 \varphi_\Theta$ and $d\sigma_R/d\log a = -\Delta^2 [\varphi_\Theta/(1 + p) + \varphi^2 p/(1 + p)^2]$, with $\varphi_\Theta \equiv d\varphi/d\log \Theta > 0$. At $h = 0$, $a^2 L_{aa}/L = -\Delta^2 \varphi_\Theta + \sigma_L(\sigma_L - 1)$. Collecting bounds— $\max_z z\Omega'(z) = \frac{1}{2}$; $\varphi^2 p/(1 + p)^2 < \frac{1}{16}$; $\Omega \in [0, 2]$; and $\varphi_\Theta \leq \bar{\varphi}_\Theta \equiv \sup_\Theta d\varphi/d\log \Theta$, finite since φ is smooth in $\log \Theta$ with limits 0 and $\frac{1}{2}$ —gives

$$a^2 |L_{aa}| \leq L(0, a, \gamma) \left[\frac{1}{2} \sigma_R^2 + \bar{\sigma}(\bar{\sigma} + 1) + \Delta^2 (3\bar{\varphi}_\Theta + \frac{1}{8}) \right], \quad \bar{\sigma} \equiv \max\{|\sigma_L|, |2\sigma_R - \sigma_L|\},$$

and the same display with $(\sigma_M, \frac{1}{2}\Omega)$ in place of (σ_L, Ω) bounds $a^2 |M_{aa}|$ by a multiple of $M(0, a, \gamma)$. Write $B_L L(0, a, \gamma)$ and $B_M M(0, a, \gamma)$ for the two right-hand sides. Since $\Lambda_{aa} = M_{aa} + (1 - \gamma)L_{aa}$, $\Psi''(a)a^2$ exceeding $B_M M(0, a, \gamma)$ uniformly on $[a_{lo}, a_{hi}]$ is sufficient for part (i), and exceeding $B_M M(0, a, \gamma) + |1 - \gamma| B_L L(0, a, \gamma)$ is sufficient for part (ii). The Δ^2 term cannot in general be dropped, and part (i) does not imply part (ii): dual-use asymmetry contributes negative log-curvature to both loss objects, so the developer's problem can be strictly concave while welfare is locally convex in capability—the configuration discussed in the text. $\square$

Proposition 6. (i) Write $L = \lambda p(a) m_0(a) (1 + z/2)^{-2}$ with $z = x_D^*(a)h$. Since $\log x_D^* = \log(2\gamma\lambda) + \log p - \log c_D - \log(1 + p)$ and $d\log p/d\log a = -\varphi\Delta$,

$$\frac{d\log x_D^*}{d\log a} = \sigma_D - \frac{\varphi\Delta}{1 + p} = \sigma_R, \quad \frac{d\log[\lambda p m_0]}{d\log a} = \sigma_m - \varphi\Delta = \sigma_L,$$

and with $d\log(1 + z/2)^{-2}/d\log z = -z/(1 + z/2)$, (23) follows. For the sign of σ_R : if $\Delta \leq 0$ then $\sigma_R \geq \sigma_D \geq 0$; if $\Delta > 0$ then $\Delta \leq \sigma_D$ and $\varphi/(1 + p) < \frac{1}{2}$, so $\sigma_R > \frac{1}{2}\sigma_D \geq 0$ unless $\sigma_D = 0$, which contradicts $\Delta > 0$. Since $z \mapsto z/(1 + z/2)$ increases from 0 to 2, the elasticity falls weakly in h with the stated limits, strictly whenever $\sigma_R > 0$; at $\sigma_A = \sigma_D = 0$ it is constant at σ_m .

(ii) $\partial L/\partial a$ has the sign of $\sigma_L - \sigma_R z/(1 + z/2)$, weakly decreasing in z from σ_L to $\sigma_L - 2\sigma_R$, and strictly decreasing whenever $\sigma_R > 0$. When $0 < \sigma_L < 2\sigma_R$, its unique zero is $z_a(a, \gamma)$; the remaining classification follows pointwise in (a, γ) .

(iii) With $M = m_0 \ell (1 + z/2)^{-1}$, and using $\ell = c_D x_D^*$ so that $d\log \ell/d\log a = \sigma_R - \sigma_D$, the level

term has elasticity $d \log(m_0 \ell) / d \log a = \sigma_m - \varphi \Delta / (1 + p) = \sigma_M$. Then $\partial M / \partial h = -\frac{1}{2} x_D^{\text{eff}} M$ and

$$a M_{ah} = -\frac{1}{2} x_D^{\text{eff}} M \left[\sigma_M + \sigma_R \frac{1 - z/2}{1 + z/2} \right].$$

Since $\Pi_{ah} = -\beta w e^{-wh} B'(a) - M_{ah}$, the cross-partial is positive exactly when (24) holds, and $a^{*'}(h) = \Pi_{ah} / (-\Pi_{aa})$ under Assumption 3(i) at an interior choice, where the first-order condition holds with equality and the implicit-function theorem applies. The interiority is not redundant: strict concavity on a compact interval does not rule out a boundary optimum, and at (say) $a^* = a_{hi}$ a positive cross-partial leaves the optimiser at the bound, so the “if and only if” is a statement about interior choices; boundary choices follow from the Kuhn–Tucker conditions and move weakly in the direction the cross-partial indicates. Holding a and γ fixed, multiplying the bracket by $(1 + z/2) > 0$ gives $(\sigma_M + \sigma_R) + \frac{z}{2}(\sigma_M - \sigma_R)$, affine in z with slope $(\sigma_M - \sigma_R)/2$. If $\sigma_A = \sigma_D = \sigma_m = 0$, it is identically zero. Otherwise its intercept is strictly positive; if $\sigma_M \geq \sigma_R$ it remains positive throughout, and if $\sigma_M < \sigma_R$ its unique zero is $z_{\Pi}(a, \gamma) = 2(\sigma_M + \sigma_R) / (\sigma_R - \sigma_M)$. Since $\sigma_R - \sigma_M = \sigma_D - \sigma_m$, the sign flip requires $\sigma_D > \sigma_m$. Along $a^*(h)$, however, the coefficients and $z_{\Pi}(a^*(h), \gamma)$ can move with h , so the affine argument is pointwise and does not establish a pathwise one-crossing result. The corresponding harm-side hump is $z(1 + z/\kappa)^{-(\kappa+1)}$, maximised at $z = 1$ for every κ ; the burden-side hump $z(1 + z/\kappa)^{-\kappa}$ is maximised at $z = \kappa / (\kappa - 1)$. $\square$

Section 4.2: the capability wedge, the rule, and the developer’s build–no-build margin. At an interior a^* , $\Pi_a = 0$ gives $\Psi'(a^*) = \beta e^{-wh} B' - M_a$; substituting into $W_a = e^{-wh} B' - \Psi' - \Lambda_a$ and using $\Lambda = M + (1 - \gamma)L$ gives (25)—the burden term cancels, so the capability wedge is convention-free. Differentiating the composite gives

$$\frac{d}{dh} W(h, a^*(h); \gamma) = W_h + W_a a^{*'}(h), \quad W_h = -w e^{-wh} B + x_D^{\text{eff}} L_{ch},$$

and setting this derivative to zero yields (26); at a fixed-capability optimum $W_h = 0$. At the developer’s build–no-build margin, $\max_a \Pi = 0$ gives $\Psi(a^*) = \beta e^{-wh} B(a^*) - M$, so $W = (1 - \beta) e^{-wh} B(a^*) - (1 - \gamma)L$, which at $\beta = 1$ is $-(1 - \gamma)L < 0$ for $\gamma < 1$. Finally, as $\gamma \downarrow 0$, $\Theta \downarrow 0$, $p \rightarrow 1$, $\ell \rightarrow 0$, $x_D^* \rightarrow 0$, $z \rightarrow 0$, $m(h) \rightarrow m_0$, and $L \rightarrow \lambda m_0(a)$ at every fixed h . These are limiting statements: at $\gamma = 0$ the adversary’s supremum against zero defence is not attained, so the baseline contest has no equilibrium. $\square$

A.4 Liability

Proposition 7. From $L = \lambda p m_0 (1 + z/2)^{-2}$: $\partial \log p / \partial \log \gamma = -\varphi$ since $\Theta \propto \gamma$, and $\log x_D^* = \log(2\gamma\lambda/c_D) + \log p - \log(1 + p)$ gives (28), hence $\partial \log z / \partial \log \gamma = 1 - \varphi / (1 + p)$ at fixed h . Differentiating $-2 \log(1 + z/2)$ and collecting gives (27). Since $\varphi < \frac{1}{2}$ and $p > 0$, the bracket lies in $(\frac{1}{2}, 1)$; $z \mapsto z / (1 + z/2)$ increases from 0 to 2; and neither φ nor p depends on h , so Y increases strictly in h from φ to $2 - \varphi(1 - p) / (1 + p) \in (\frac{3}{2}, 2)$. $\square$

Section 5.1: the conversion identity and the sign of the protection-depth response. The identity (29) is definitional: $z^* = x_D^* h^*$ with x_D^* independent of the calendar, so $d \log z^* / d \log \gamma = d \log h^* / d \log \gamma + \partial \log x_D^* / \partial \log \gamma$, which is (29) given (28).

For the sign of the protection-depth response, note that maximising over h at fixed γ is maximising over z , so work in (z, γ) coordinates: $z^*(\gamma)$ maximises $\tilde{W}(z, \gamma) \equiv e^{-sz} B - m_0 \ell u - (1 - \gamma) \lambda p m_0 u^2$, where $u = (1 + z/2)^{-1}$ and $s = w/x_D^*(\gamma)$, and at an interior optimum with a strict second-order condition $\text{sign}(dz^*/d\gamma) = \text{sign}(\tilde{W}_{z\gamma})$. Now $\tilde{W}_z = -se^{-sz} B + u L_{\text{ch}}$ with $L_{\text{ch}} = \frac{1}{2}M + (1 - \gamma)L$. Differentiate in γ at fixed z and use the first-order condition $se^{-sz} B = u L_{\text{ch}}$ to eliminate B : the benefit term contributes $-s_\gamma(1 - sz)e^{-sz} B = (1 - \frac{\varphi}{1+p})(1 - sz) u L_{\text{ch}}/\gamma$ by (28); the burden term contributes $(u^2/2) \partial(m_0 \ell)/\partial \gamma = (1 - \frac{\varphi}{1+p}) u M/(2\gamma)$; and the uninsured term contributes $\partial[(1 - \gamma) \lambda p m_0 u^3]/\partial \gamma = -u L[1 + (1 - \gamma)\varphi/\gamma]$. Collecting, with $M/L = 2\gamma/[(1 + p)u]$,

$$\frac{\gamma \tilde{W}_{z\gamma}}{u L} = \left(1 - \frac{\varphi}{1+p}\right)(1 - sz) \left[\frac{\gamma}{(1+p)u} + 1 - \gamma \right] + \left(1 - \frac{\varphi}{1+p}\right) \frac{\gamma}{(1+p)u} - \gamma - (1 - \gamma)\varphi,$$

and grouping by insurance status, using $2(1 - \varphi/(1+p))/(1+p) - 1 = \chi$ and $(2 - sz)(1 + z/2) - 2 = z(1 - s - sz/2)$, gives (30); write E for its left side. The factor $z(1 - s - sz/2)$ vanishes at $z = 0$ and at $z = 2(1 - s)/s$ —the universal ceiling (16) at $\kappa = 2$, written in protection-depth units—and is positive strictly between, and every optimum that beats withholding lies strictly inside; hence the γ -bracket of (30) strictly exceeds χ at every such optimum with $z > 0$, which proves the full-compensation claim. For the stationary-rate claim of Section 5.2, substituting $(1 - \gamma^*)Y = \gamma^* \chi$ from (33) and $Y - \varphi = (1 - \varphi/(1+p))z/(1 + z/2)$ from (27) gives

$$E|_{\gamma^*} = (1 - \gamma^*) \left(1 - \frac{\varphi}{1+p}\right) \left(1 - sz + \frac{z}{1 + z/2}\right) + \gamma^* \frac{1 - \varphi/(1+p)}{1+p} z \left(1 - s - \frac{sz}{2}\right).$$

The second term is positive inside the ceiling as above. So is the first, because $sz < 1 + z/(1 + z/2)$ everywhere inside the ceiling: for $z < 2$ the ceiling gives $sz < \min\{2s, 2(1 - s)\} \leq 1$, and for $z \geq 2$, $z/(1 + z/2) \geq 1$ while the ceiling gives $sz < 2$. Hence $E|_{\gamma^*} > 0$ at every interior optimum that beats withholding. As $\gamma \downarrow 0$ the weight shifts entirely to the $(1 - \gamma)$ -bracket, whose sign is that of $(1 - \varphi/(1+p))(1 - wh^*) - \varphi$: where the pass-through is large, and the window deep, the expression turns negative and the protection-depth response reverses. $\square$

Section 5.1: the bill and the burden. For the bill, $\partial(\gamma L)/\partial \gamma = L(1 - Y)$. Setting $Y = 1$ and writing $\mathcal{A} \equiv (1 - \varphi)(1 + p)/(1 + p - \varphi)$, the crossing protection depth solves $2z/(2 + z) = \mathcal{A}$, so it equals $2\mathcal{A}/(2 - \mathcal{A}) = 2(1 - \varphi)(1 + p)/[(1 + p) - \varphi(1 - p)]$; $\mathcal{A} < 1$ reduces to $\varphi p > 0$, placing the crossing depth below 2, and $\mathcal{A} > \frac{2}{3}$ reduces to $1 + p - \varphi - 3\varphi p > 0$, which exceeds $(1 - p)/2 > 0$ since $\varphi < \frac{1}{2}$, placing it above 1. For the burden, $M = m_0 \ell (1 + z/2)^{-1}$ with $\partial \log \ell / \partial \log \gamma = \partial \log z / \partial \log \gamma = 1 - \varphi/(1 + p)$ gives

$$\frac{\partial \log M}{\partial \log \gamma} = \left(1 - \frac{\varphi}{1+p}\right) \left[1 - \frac{z/2}{1 + z/2}\right] = \frac{1}{1 + z/2} \left(1 - \frac{\varphi}{1+p}\right) > 0. \quad \square$$

Proposition 8. Social loss is $\Lambda = M + (1 - \gamma)L$, so $\partial\Lambda/\partial\gamma = M_\gamma - L + (1 - \gamma)L_\gamma$. Substituting $M = m_0\ell(1 + z/2)^{-1}$, $L = \lambda pm_0(1 + z/2)^{-2}$ and $\ell = 2\gamma\lambda p/(1 + p)$ gives the ratio $M = \gamma L \frac{2}{1+p}(1 + z/2)$, and with $\gamma M_\gamma = M(1 + z/2)^{-1}(1 - \varphi/(1 + p))$ from the burden elasticity above,

$$M_\gamma - L = L \frac{2(1 + p - \varphi) - (1 + p)^2}{(1 + p)^2} = L \frac{1 - p^2 - 2\varphi}{(1 + p)^2} \equiv L\chi,$$

which is (31). Since $(1 - \gamma)L_\gamma = -(1 - \gamma)Y_L/\gamma$, the gradient $\partial W/\partial\gamma = -\partial\Lambda/\partial\gamma$ is (32), and setting it to zero gives (33). Positivity of χ : it has the sign of $1 - p^2 - 2\varphi$, and from (3),

$$\frac{1}{\varphi} = 2 + p \left[\frac{2}{1 - p} - \frac{1}{1 + p} - \frac{1}{2 - p} \right],$$

so $2\varphi < 1 - p^2$ reduces, after multiplying by $(1 - p)(1 + p)(2 - p) > 0$, to $p(1 + p) > 0$. Hence every stationary rate lies below one. For existence of a global optimum, first extend welfare continuously to the limiting benchmark $\gamma = 0$: as $\gamma \downarrow 0$, $p \rightarrow 1$, $x_D^* \rightarrow 0$, $M \rightarrow 0$, and $L \rightarrow \lambda m_0$, so the limit is finite even though the contest has no equilibrium at exactly zero. The limit of the gradient needs a rate, not just a sign. As $\gamma \downarrow 0$, $\Theta \downarrow 0$ and $\varphi(\Theta) \downarrow 0$, while $1 - p \sim (\Theta/2)^{1/2}$ gives $\varphi \sim \frac{1}{2}(\Theta/2)^{1/2}$. Thus $Y \geq \varphi$ is of order $\gamma^{1/2}$ and $(1 - \gamma)Y/\gamma \rightarrow \infty$, whereas $\chi = (1 - 2\varphi - p^2)/(1 + p)^2$ is bounded. The gradient in (32) is positive for all sufficiently small positive rates and equals $-L\chi < 0$ at $\gamma = 1$. Continuity therefore gives a stationary rate in $(0, 1)$ and rules out both endpoints as global maximisers of the continuous extension on $[0, 1]$. Weierstrass' theorem yields a global maximiser in $(0, 1)$. For $\gamma \geq 1$, $\partial\Lambda/\partial\gamma = L\chi + (1 - \gamma)L_\gamma \geq L\chi > 0$ since $L_\gamma \leq 0$, so welfare is strictly decreasing on $[1, \infty)$ and no supra-compensatory rate is preferred. For the comparative static, note that Y and χ both move with the rate, so (33) cannot be differentiated as though its right-hand side were exogenous. Define instead

$$\mathcal{R}(\gamma, h) \equiv \frac{(1 - \gamma)Y(\gamma, h)}{\gamma} - \chi(\gamma),$$

so that $W_\gamma = L\mathcal{R}$ and an interior stationary rate solves $\mathcal{R}(\gamma^*, h) = 0$. Regularity means the second-order condition holds strictly, and it delivers the sign that the naive differentiation cannot: $0 > W_{\gamma\gamma} = L_\gamma\mathcal{R} + L\mathcal{R}_\gamma = L\mathcal{R}_\gamma$ at $\mathcal{R} = 0$, so $\mathcal{R}_\gamma < 0$. Neither φ nor p depends on the calendar, so χ is independent of h and $Y_h > 0$ by Proposition 7; hence $\mathcal{R}_h = (1 - \gamma^*)Y_h/\gamma^* > 0$ because $\gamma^* < 1$, and the implicit-function theorem gives

$$\frac{d\gamma^*}{dh} = -\frac{\mathcal{R}_h}{\mathcal{R}_\gamma} > 0.$$

For the supra-compensatory example under the alternative accounting: charging the adversary's outlay $O_A(h) = m(h)vp(1 - p)/(2 - p)$ to the social account, at $\kappa = 2$, $h = 1$ and unit primitives ($\lambda = v = c_A = c_D = m_0 = 1$, so that $\Theta = 1$, $p = \frac{1}{2}$ and $\varphi = \frac{3}{10}$ exactly), $\partial W/\partial\gamma|_{\gamma=1} = -3/160$ under the defence-inclusive accounting but $\partial(W - O_A)/\partial\gamma|_{\gamma=1} = +9/320 > 0$: a rate above one is

then locally preferred. $\square$

Section 5.3: the capability correction and its limits. Differentiating $W(h, a^*(h, \gamma); \gamma)$ in the rate gives $\hat{W}_\gamma = W_\gamma|_{a^*} + W_a|_{a^*} a_\gamma^*$, which is (34) with W_γ from (32). For the response, $a_\gamma^* = -\Pi_{a\gamma}/\Pi_{aa}$ with $\Pi_{aa} < 0$ by Assumption 3(i), and $\Pi_{a\gamma} = -M_{a\gamma}$. Decomposing the burden as $M = \gamma L + O_D$, $M_{a\gamma} = L_a + \gamma L_{a\gamma} + O_{Da\gamma}$: where capability reduces harm at the margin, the term $-L_a > 0$ contributes positively to a_γ^* , and the remaining terms are not signed in general. For the saturation, fix $h > 0$ and a . As $\gamma \rightarrow \infty$, the common-curvature asymptote gives $p \asymp \gamma^{-1/\kappa}$, $\ell \asymp \gamma^{(\kappa-1)/\kappa}$, and $x_D^* \asymp \gamma^{1/\kappa}$, so $z \rightarrow \infty$. By the computation in the proof of Proposition 4, the burden converges to $m_0 c_D \kappa^{\kappa-1} h^{-(\kappa-1)}$, whose capability elasticity is $\sigma_m - \sigma_D$; hence the relief $-a M_a$ converges to a finite limit proportional to $\sigma_D - \sigma_m$ and a further multiplier does nothing at this margin. For the harm-neutral locus: at fixed (γ, a) , the capability elasticity of harm $\sigma_L - \Omega(z)\sigma_R$ is weakly decreasing in h from σ_L to $\sigma_L - 2\sigma_R$, strictly whenever $\sigma_R > 0$, so a harm-neutral protection depth exists if and only if $0 < \sigma_L < 2\sigma_R$ and equals $z_a(a, \gamma)$; there $L_a = 0$, the capability wedge (25) reduces to $(1 - \beta)e^{-wh}B' \geq 0$, and for $\beta < 1$ the rate solving the alignment condition $(1 - \beta)e^{-wh}B' = (1 - \gamma)L_a$ is $\gamma = 1 - (1 - \beta)e^{-wh}B'/L_a$, which tends to $+\infty$ as $L_a \uparrow 0$ (the harm-reducing side, where alignment needs $\gamma > 1$) and to $-\infty$ as $L_a \downarrow 0$ (the harm-increasing side, where the required γ leaves the feasible set $\gamma > 0$ altogether). The harm-neutral locus moves with the liability rate—the rate shifts both the harm-neutral protection depth and the actual protection depth—and is removed entirely once $\sigma_L \geq 2\sigma_R$, since capability is then harm-increasing at every protection depth. $\square$

Corollary 1. By Berge’s maximum theorem, Π^B is continuous; uniqueness gives a continuous active capability response and hence continuity of $\hat{W}$ on the active region. The maintained monotonicity of Π^B in the rate—under the quadratic baseline, the envelope theorem and the burden elasticity give $\Pi_\gamma^B = -M_\gamma < 0$ —makes each non-empty feasible rate set an interval with an upper participation boundary. Compactness then gives an inner rate maximiser and an outer headway maximiser, with the zero outside option covering prohibition or no development. Under strict single-peakedness, truncating the unconstrained maximiser at the participation boundary gives the conditional constrained maximiser. $\square$

Section 5.4: the developer participation constraint and the second best. By the envelope theorem $\partial \max_a \Pi / \partial \gamma = -\partial M / \partial \gamma|_{a^*}$, strictly negative at every (h, γ) by the burden elasticity of Section 5.1 above; if the developer paid the liability bill but did not finance the defender’s evaluation, the same derivative is $-\partial(\gamma L) / \partial \gamma = L(Y - 1)$, positive once $Y > 1$. Equation (35) is the envelope derivative in the date at $\kappa = 2$, using $\partial M / \partial h = -\frac{1}{2}x_D^{\text{eff}}M$; evaluating it at the planner’s date by substituting (12) gives $d \max_a \Pi / dh = x_D^{\text{eff}} [\frac{1}{2}(1 - \beta)M - \beta(1 - \gamma)L]$, the sign claimed in the text. For the second-best conditions, form the Lagrangian $W(h, a^*(h, \gamma); \gamma) + \theta \max_a \Pi$. Differentiating in γ and using (32) gives (36); since $\partial M / \partial \gamma > 0$, the right-hand side is strictly positive where $\theta > 0$, so the unconstrained gradient (34) is strictly positive at the constrained package, and

where composite welfare is single-peaked in the rate, the constrained rate lies strictly below the unconstrained optimum. Differentiating in h , using (35) and $W_h = -we^{-wh}B + x_D^{\text{eff}}L_{\text{ch}}$, and adding and subtracting $\theta\beta W_h$, rearranges to (37). $\square$

A.5 Delegation

Proposition 9. At $\beta = \gamma = 1$,

$$\Pi = e^{-wh}B - \Psi - M = e^{-wh}B - \Psi - \Lambda = W$$

identically in (h, a) because $\Lambda = M$ at full compensation. This proves part (i). Holding capability fixed, the developer's date condition is $we^{-wh}B = \frac{\kappa-1}{\kappa}x_D^{\text{eff}}M$, whereas the planner's is $we^{-wh}B = x_D^{\text{eff}}[\frac{\kappa-1}{\kappa}M + (1-\gamma)L]$. Since $L > 0$, a common interior date requires $\gamma = 1$. Proposition 8 places every welfare-optimal rate at a fixed date and model strictly below one, proving part (ii).

For part (iii), let $\hat{L}(h, \gamma) = L(h, a^*(h, \gamma), \gamma)$. At $\beta = 1$,

$$\hat{W}(h, \gamma) - \hat{\Pi}(h, \gamma) = (\gamma - 1)\hat{L}(h, \gamma).$$

At a common regular interior mandated and delegated date, $\hat{\Pi}_h = \hat{W}_h = 0$ by the argument in the proof of Proposition B.9(iv), whether or not participation binds. Differentiating the preceding identity consequently gives $(\gamma - 1)\hat{L}_h = 0$. It remains to locate the constrained mandated rate. For every fixed h and every $\gamma > 1$, Proposition 8 gives

$$\hat{W}(h, \gamma) < W(h, a^*(h, \gamma); 1) \leq W(h, a^*(h, 1); 1) = \hat{W}(h, 1),$$

where the weak inequality uses $\Pi(h, a; 1) \equiv W(h, a; 1)$. If the first package satisfies developer participation, the package at the same date and $\gamma = 1$ also does because its burden is lower. Thus no supra-compensatory rate is constrained-optimal. At $\gamma = 1$, $W_a = 0$ and

$$\hat{W}_\gamma = W_\gamma + W_a a_\gamma^* = -L\chi < 0.$$

A small reduction in the rate raises welfare and weakly relaxes participation, so every active regular constrained optimum has $\gamma < 1$. Hence implementation requires $\hat{L}_h = 0$. Finally, $\hat{L}_h = L_h + L_a a_h^* = -x_D^{\text{eff}}L + L_a a_h^*$, which gives (38). $\square$

Proposition 10. Write $M^f(h, a, \gamma) = M(h, a, \gamma) + f_D h$. As $h \rightarrow \infty$, $e^{-wh}B(a) \rightarrow 0$ uniformly on the compact capability interval and $M^f \geq 0$, so profit is eventually at most $-\Psi(a_{10})/2 < 0$ for every rate. At $h = 0$, the overhead vanishes and $M^f(0, a, \gamma) = m_0(a)\ell(a, \gamma)$. Uniformly over capability, $\Theta(a, \gamma) \asymp \gamma$, and the common-curvature asymptote gives

$$p \asymp \gamma^{-1/\kappa}, \quad \ell = \frac{\kappa\gamma\lambda p}{\kappa - 1 + p} \asymp \gamma^{(\kappa-1)/\kappa}, \quad x_D^* = \left(\frac{\ell}{c_D}\right)^{1/(\kappa-1)} \asymp \gamma^{1/\kappa}.$$

Thus $M^f(0, a, \gamma) \rightarrow \infty$ uniformly in a , while $\max_a[\beta B(a) - \Psi(a)]$ is bounded. The developer therefore refuses development at $h = 0$ for all sufficiently large γ and, by continuity, on a neighbourhood of zero.

For $h > 0$, write $x_D^{\text{eff}} = (1/x_D^* + h/\kappa)^{-1}$. Then $M = m_0 c_D(x_D^{\text{eff}})^{\kappa-1}$ and

$$M_\infty^f(h, a) = m_0(a) c_D(a) (\kappa/h)^{\kappa-1} + f_D h.$$

At every finite rate, $x_D^{\text{eff}} < \kappa/h$, so $M^f < M_\infty^f$ strictly; as $\gamma \rightarrow \infty$, $M^f \rightarrow M_\infty^f$ uniformly on compact sets bounded away from $h = 0$. Suppose first that $\sup_{h>0} \Pi_\infty^f(h) \geq 0$. Continuity, divergence of M_∞^f as $h \downarrow 0$, and $\Psi(a_{10}) > 0$ as $h \rightarrow \infty$ imply that the supremum is attained at some finite $h_0 > 0$ and capability a_0 . At any finite rate,

$$\Pi(h_0, a_0; \gamma) - f_D h_0 > \beta e^{-wh_0} B(a_0) - \Psi(a_0) - M_\infty^f(h_0, a_0) = \Pi_\infty^f(h_0) \geq 0.$$

Thus even equality in the limiting condition leaves development strictly profitable at every finite rate.

Conversely, suppose $\sup_{h>0} \Pi_\infty^f(h) \leq -c < 0$. Choose H so that $\beta e^{-wH} \max_a B(a) < \Psi(a_{10})$; every $h \geq H$ is then unprofitable, independently of the overhead. Choose $\varepsilon > 0$ so that the variable component $\min_a m_0(a) c_D(a) (\kappa/\varepsilon)^{\kappa-1}$ exceeds $\max_a[\beta B(a) - \Psi(a)] + 2$. Uniform convergence at $h = \varepsilon$ and the fact that the variable burden falls with h make every $h \in (0, \varepsilon]$ unprofitable once γ is large enough. On the compact middle region $[\varepsilon, H]$, $M^f \rightarrow M_\infty^f$ uniformly, so maximised profit is eventually at most $-c/2$. A sufficiently high finite rate therefore excludes development at every window. This proves both directions of (39). Setting $f_D = 0$ gives the variable-cost baseline. $\square$

B Extensions

This appendix collects every result that changes a baseline assumption, adopts an alternative accounting convention, or adds a policy margin beyond the main analysis. Its order follows the main text: robustness of the flaw contest, alternative welfare accounting, learning from findings, extensions to staged access, and additional delegation results. Each formal extension is proved where it is stated.

B.1 Contest assumptions and structure

B.1.1 Uniqueness within Markov strategies

Proposition 1 establishes uniqueness within stationary Bayesian strategies. Under the Poisson, $\alpha = 1$ benchmark, discoveries do not move the posterior mean of the residual stock (Section 3.1); it evolves deterministically as $\dot{m} = -(x_A + x_D)m$, so the public state is m , and a *public-Markov*

strategy is an intensity that depends on the history only through it.⁶¹

Proposition B.1. *Under quadratic costs, the post-release equilibrium is unique within differentiable pure public-Markov strategies: both intensities are constant at the levels of Proposition 1, and the value functions are linear in m .*

Proof. Let $x_A(m)$ and $x_D(m)$ be differentiable Markov strategies with associated values $V_A(m)$ for the adversary and expected continuation loss $K_D(m)$ for the defender, both vanishing at $m = 0$. The undiscounted Hamilton–Jacobi–Bellman equations are

$$\begin{aligned} 0 &= \max_x \{m[xv - C_A(x)] - (x + x_D(m)) m V'_A(m)\}, \\ 0 &= \min_x \{m[x_A(m)\gamma\lambda + C_D(x)] - (x_A(m) + x) m K'_D(m)\}. \end{aligned}$$

Dividing by $m > 0$, the first-order conditions are $C'_A(x_A(m)) = v - V'_A(m)$ and $C'_D(x_D(m)) = K'_D(m)$; substituting the optimised intensities back into the two equations gives $V'_A(m) = [x_A(m)v - C_A(x_A(m))]/[x_A(m) + x_D(m)]$ and $K'_D(m) = [x_A(m)\gamma\lambda + C_D(x_D(m))]/[x_A(m) + x_D(m)]$. At every $m > 0$ the pair $(x_A(m), x_D(m))$ therefore satisfies exactly the per-resolution conditions (2), whose solution under quadratic costs is unique by Proposition 1. Both intensities are constant, and integrating from the boundary gives $V_A(m) = gm$ and $K_D(m) = \ell m$. $\square$

B.1.2 Fixed and per-period costs

The baseline’s cost functions satisfy $C'_i(0) = 0$, so both parties always search, and there is no deterrence margin. Proposition 10 adds a stock-independent programme overhead; this subsection takes the two remaining amendments, a positive marginal search cost and adversary entry.

A positive marginal cost at the origin. Let $C_A(x) = c_0x + c_Ax^2/2$ with $c_0 > 0$. For any $x_D > 0$ the adversary’s marginal per-resolution payoff at $x_A = 0$ is $(v - c_0)/x_D$, so the adversary is deterred if and only if $v \leq c_0$ —independently of x_D , of the window, and of every other primitive; if $v > c_0$, the equilibrium is that of Proposition 1 with $v - c_0$ in place of v throughout—in Θ , in the intensity $x_A^* = 2(v - c_0)(1 - p)/[c_A(2 - p)]$, and in the gain $g = (v - c_0)p/(2 - p)$.⁶² The defender’s search intensity drops out of the corner condition because the per-resolution objective multiplies prize and flow cost by the same expected waiting time, so a harder-searching defender scales the adversary’s benefit and cost equally. A linear cost component therefore delivers a statement about the penalty regime—deterrence is achieved by driving the net prize to zero—and no statement about the calendar. The symmetric amendment on the defence side gives a *defender search threshold*: with marginal cost c_{0D} at the origin, the defender searches only if $\ell \geq c_{0D}$, a lower bound on the liability rate below which the contest approaches the $\gamma \downarrow 0$ degeneration.

⁶¹A strategy conditioning on whether the last discovery was offensive or defensive is not a function of m —the stock transition is winner-independent—so trigger-like constructions remain outside the class, and the qualification in Section 2.1 for arbitrary history-contingent strategies stands.

⁶²At the corner a convention is needed, since with no adversary there are no resolutions for the per-flaw account to divide: I take the deterred outcome to be $x_A = x_D = 0$ with nil contest value, the limit of the interior equilibria as $v \downarrow c_0$.

Adversary entry. Suppose instead the adversary pays a fixed *adversary entry cost* $c_{\text{entry}} > 0$ once to direct its capacity at the released model at all. Conditional on the defender's window effort path, the adversary enters if and only if $m(h)g \geq c_{\text{entry}}$, which inverts to a finite deterrence date whenever $c_{\text{entry}} < m_0g$. That date is policy-responsive—the liability rate shrinks the residual stock against which c_{entry} is compared, and it is exactly what the baseline's power costs cannot produce. But it is not an equilibrium object as computed: the survival law it is evaluated on comes from a defender problem whose terminal value per surviving flaw is ℓ , and once entry depends on the residual stock that terminal value is discontinuous—zero if the adversary is deterred, ℓ otherwise—in exactly the state the defender controls. A defender close to deterring the adversary altogether has a reason to search that the contest never supplies. Appendix B.1.4 treats the version of this margin that can be computed—entry by many adversaries, with the defender's window problem held fixed—where the marginal entrant's indifference varies smoothly with the residual stock.

Derivations. The adversary's per-resolution payoff $f(x_A) = [x_A v - C_A(x_A)]/(x_A + x_D)$ has $f(0) = 0$ and $f'(0) = (v - c_0)/x_D$ for any $x_D > 0$; strict quasi-concavity gives $x_A^* = 0$ if and only if $v \leq c_0$, and for $v > c_0$ the condition with $C_A = c_0 x + c_A x^2/2$ reduces to the pure-quadratic condition with prize $v - c_0$, so every equilibrium object is evaluated at the effective prize. For an adversary entry cost $c_{\text{entry}} > 0$: by Lemma 1 the adversary's value from n flaws is ng , so the adversary enters at release if and only if $m(h)g \geq c_{\text{entry}}$, and inverting (10) gives the threshold date; the circularity discussed there is that the survival law used in that inversion presumes the terminal value ℓ , which entry-dependence makes discontinuous. $\square$

B.1.3 Discounting

Lemma 1 requires undiscounted payoffs, and the requirement is substantive. With a common discount rate $r > 0$, a party that wins sooner is better off, continuation values enter with rate-dependent weights, and equilibrium intensities depend on the remaining-stock size. The two continuation values suggest a direction: discounting makes the adversary impatient to harvest its prizes and makes the defender's deferral of losses attractive, so I conjecture that it tilts the contest towards the adversary in large pools—precisely the setting, a large freshly exposed attack surface, where the baseline is most likely to be applied—but the discounted equilibrium is not solved here. Two observations keep the benchmark honest. Under standard regularity and local equilibrium uniqueness, an interior equilibrium branch through $r = 0$ varies continuously with the discount rate, so strict inequalities at fixed interior primitives survive on that branch for small r . Uniform statements are another matter: the bound $\varphi < \frac{1}{2}$ is approached as $\Theta \rightarrow \infty$ rather than attained, so pointwise continuity does not protect it, and the global bounds would have to be re-established rather than inherited. For substantial rates, the welfare accounting would need rebuilding from the two-state recursion, and I leave that open.

B.1.4 Many adversaries, with and without entry

The baseline contest is bilateral. Real adversary populations are not, and aggregation matters for the paper’s headline bound: if adding adversaries made aggregate adversarial search more elastic, the pass-through bound of Proposition 2 might be an artefact of bilateralism. It is not.

Fixed attack population (N_A). Let $N_A \geq 1$ symmetric adversaries each choose an individual intensity x_A^{ind} at cost $C_A(x_A^{\text{ind}})$ per channel, with the defender facing their aggregate $N_A x_A^{\text{ind}}$; each adversary internalises only its own rent. In this subsection, redefine $R \equiv x_D / (N_A x_A^{\text{ind}})$ as the ratio of defender intensity to aggregate adversarial intensity, and write $p_{N_A} = 1 / (1 + R)$ for the exploited share. The equilibrium is again a single-ratio problem characterised below: the defender’s stake retains the form $\ell_{N_A} = 2\gamma\lambda p_{N_A} / (1 + p_{N_A})$, and the whole apparatus of the paper—the effort profile, the survival law, the protection-depth thresholds, the liability elasticity of harm—goes through with p_{N_A} in place of p .

Proposition B.2. *With N_A symmetric adversaries under quadratic costs: the equilibrium is unique; at $\Theta = 1$ the exploited share is exactly $p_{N_A} = N_A / (N_A + 1)$ and the pass-through is exactly $\varphi_{N_A}(1) = N_A(2N_A + 1) / [(N_A + 1)(2N_A^2 + 2N_A + 1)]$, strictly decreasing in N_A and asymptotically $1/N_A$; and the pass-through $\varphi_{N_A} \equiv -d \log p_{N_A} / d \log \Theta$ satisfies $\varphi_{N_A}(\Theta) < \frac{1}{2}$ at every $\Theta > 0$ and every $N_A \geq 1$, approaching $\frac{1}{2}$ as $\Theta \rightarrow \infty$ for each N_A .*

Proof. First, symmetry is without loss for equilibrium. Let adversary i choose x_{Ai} and let $X = x_D + \sum_j x_{Aj}$ denote total intensity. Positive stakes and $C'_i(0) = 0$ make every equilibrium interior. Under quadratic costs, adversary i ’s first-order condition can be written $q_X(x_{Ai}) \equiv \frac{c_A}{2} x_{Ai}^2 - (v + c_A X)x_{Ai} + vX = 0$. For any fixed equilibrium total $X > 0$, $q_X(0) > 0 > q_X(X)$ and q_X is strictly convex. It therefore has exactly one root in $(0, X)$, while its other root lies above X . Every feasible $x_{Ai} < X$ must equal that same root, ruling out asymmetric equilibria. Write the common intensity as x_A^{ind} . Aggregate arrivals then occur at rate $X = N_A x_A^{\text{ind}} + x_D$, and each adversary’s per-resolution problem is $\max_{x_A^{\text{ind}}} (x_A^{\text{ind}} v - C_A(x_A^{\text{ind}})) / X$, giving $C'_A(x_A^{\text{ind}}) = v - g_{N_A}$ with $g_{N_A} = (x_A^{\text{ind}} v - C_A(x_A^{\text{ind}})) / X$; the defender solves $\min (N_A x_A^{\text{ind}} \gamma \lambda + C_D(x_D)) / X$, giving $C'_D(x_D) = \ell_{N_A} = (N_A x_A^{\text{ind}} \gamma \lambda + C_D(x_D)) / X$. With R and $p_{N_A} = N_A x_A^{\text{ind}} / X = 1 / (1 + R)$ as defined above, the two conditions solve for the stakes and their ratio gives

$$\Theta = R \left(1 + \frac{R}{2} \right) \frac{(N_A - 1) + N_A R}{1 + R - \frac{1}{2N_A}}, \quad (44)$$

which at $N_A = 1$ reduces to the expression in Proposition 1; the right-hand side is continuous and strictly increasing from 0 to ∞ in R , so the equilibrium exists and is unique, and $\ell_{N_A} = 2\gamma\lambda p_{N_A} / (1 + p_{N_A})$ retains its form with p_{N_A} in place of p . At $\Theta = 1$, (44) gives $R = 1/N_A$ exactly, so $p_{N_A} = N_A / (N_A + 1)$; evaluating $\varphi_{N_A} = [R / (1 + R)] / [d \log \Theta / d \log R]$ at $R = 1/N_A$ gives $\varphi_{N_A}(1) = N_A(2N_A + 1) / [(N_A + 1)(2N_A^2 + 2N_A + 1)]$, whose derivative in N_A has numerator $-(4N_A^4 + 4N_A^3 - 2N_A^2 - 4N_A - 1)$; writing $N_A = 1 + t$ with $t \geq 0$ this equals $-(4t^4 + 20t^3 + 34t^2 + 20t + 1) < 0$, so the $\Theta = 1$ pass-through is strictly decreasing in N_A , and it is asymptotically $1/N_A$

by inspection. For the pass-through, $p_{N_A} = 1/(1+R)$ gives $d \log p_{N_A} / d \log R = -R/(1+R)$, so $\varphi_{N_A} = [R/(1+R)] / [d \log \Theta / d \log R]$ and the bound $\varphi_{N_A} < \frac{1}{2}$ is equivalent to

$$\frac{d \log \Theta}{d \log R} = 1 + \frac{R}{2+R} + \frac{N_A R}{(N_A - 1) + N_A R} - \frac{R}{1+R - \frac{1}{2N_A}} > \frac{2R}{1+R}.$$

Clearing denominators, the difference of the two sides has denominator $(R+1)(R+2)(N_A R + N_A - 1)(2N_A R + 2N_A - 1)$; writing $N_A = 1 + j$ with $j \geq 0$, its numerator is $(1+j)[R^3 + (4j+7)R^2 + 4(2j+1)R] + 2j(2j+1) > 0$, every coefficient being nonnegative and those in R strictly positive, which proves $\varphi_{N_A} < \frac{1}{2}$ at every $\Theta > 0$ and every $N_A \geq 1$. The bound is tight: $\Theta \sim N_A R^2/2$ as $R \rightarrow \infty$, so $p_{N_A} \sim (2\Theta/N_A)^{-1/2}$ and $\varphi_{N_A} \rightarrow \frac{1}{2}$ for each N_A . Away from $\Theta = 1$ the ordering in N_A is not established, and it must compress eventually, since every φ_{N_A} approaches the common asymptote as $\Theta \rightarrow \infty$. $\square$

The one-half bound is therefore a property of the contest form rather than bilateralism. At the balanced benchmark $\Theta = 1$, a more diffuse attack population raises the exploited share and lowers pass-through exactly: five adversaries take five-sixths of the stock, and the pass-through is about half its bilateral value. Away from $\Theta = 1$ no global ordering across populations is established; all of the elasticities approach the same square-root limit as $\Theta \rightarrow \infty$.

Free entry: a fixed-population diagnostic. Now let adversaries pay a once-off cost $c_{\text{entry}} > 0$ to direct capacity at the released model, and let the number of entrants adjust. A count N_A is consistent with entry only if $m_{N_A}(h) g_{N_A} \geq c_{\text{entry}} > m_{N_A+1}(h) g_{N_A+1}$, where g_{N_A} is an individual adversary's per-flaw gain and $m_{N_A}(h)$ is the stock surviving the defensive path associated with that count: the survival law itself moves with the population through the defender's terminal stake ℓ_{N_A} , so the entry condition is a fixed point in N_A rather than the inversion of a single survival law. Because g_{N_A} falls in N_A and each $m_{N_A}(h)$ falls in the window, the entry margin gives the timing instrument a channel the model with a fixed attack population rules out: the window drains the stock, and by draining it can deter marginal adversaries, so the exploited share can fall with the calendar instead of staying fixed.

The margin has a boundary of its own. A window long enough to deter the last adversary destroys the defender's reason to search, for the reason flagged in Appendix B.1.2: the defender's terminal value per surviving flaw is the contest stake, and with no contest to come that value is zero, so the stock is not drawn down and entry is profitable again. The deterrence margin therefore does not deliver a clean "patch until it is not worth attacking" date; beyond a threshold calendar, the model has no pure-strategy entry outcome of this form, which is a statement about the formulation rather than about policy.⁶³ The non-existence itself requires entry against the undepleted stock to be profitable, $c_{\text{entry}} < m_0 g_1$ under the lone-entrant continuation (where $g_1 = g$); with c_{entry} large enough, zero entry is self-consistent even though anticipated deterrence eliminates pre-release

⁶³Two further caveats: the defender's window problem is solved taking N_A^* as given, so the formulation withholds from it any value of tipping the marginal entrant out; and the integer margin moves the exploited share discontinuously in the calendar.

search. Solving the joint game with entry-dependent terminal values is the right formulation and is open.

The extensive margin also changes what liability buys. With entry, a higher rate reduces harm through the intensive channel bounded by Proposition B.2 and through a second: raising γ raises defensive search, which lowers each adversary's per-flaw gain and can drive marginal adversaries out. Two of the paper's claims have to be read with that in mind. The bound of Proposition 2 is a bound on the intensive margin, so liability is weak at release *conditional on the attack population*. And the window acquires a deterrence multiplier, so the qualification in Section 3.1—that headway is drawdown and nothing else—holds only at a fixed attack population, with the value of the window then exceeding its drawdown value.⁶⁴

B.1.5 The stock elasticity

Section 2.7 lets a stock of n flaws support n^α parallel search channels, with $n^\alpha = 0$ at $n = 0$. In the hidden-stock implementation, the common budget and expected-arrival scale is $\mathbb{E}[N_t^\alpha \mid \mathcal{F}_t]$, not a realised N_t^α inferred from expenditure. That factor multiplies each party's discovery rate and flow cost alike, so it cancels from every per-resolution objective and leaves the exploited share, the pass-through bound, the dissipation expressions and (6) unchanged at every α .

Depletion is another matter. The observed-state auxiliary process jumps down at rate $(x_A^* + x_D^*)n^\alpha$, so $\frac{d}{dt}\mathbb{E}[N_t] = -(x_A^* + x_D^*)\mathbb{E}[N_t^\alpha]$, a moment equation rather than a differential equation in $\mathbb{E}[N_t]$: it closes in the mean only at $\alpha = 1$, where $\mathbb{E}[N_t^\alpha] = \mathbb{E}[N_t]$. Away from that case, the survival factor of Section 3.2, and with it the effort profile, the burden and every calendar threshold, would have to be solved from the full distribution of the residual stock. That problem is not solved here.

B.1.6 Asymmetric capabilities

The main text lets capability affect the two sides asymmetrically through the dual-use gap Δ , which is the empirically relevant asymmetry. Two structural asymmetries remain.

Unequal cost curvatures. Returning to the bilateral contest, let $R = x_D/x_A$ denote the defender-to-adversary intensity ratio. With common curvature κ on both sides, the equilibrium is characterised by

$$\Theta = R^\kappa \frac{\kappa + (\kappa - 1)R}{\kappa R + (\kappa - 1)}, \quad p = \frac{1}{1 + R}, \quad \ell = \frac{\kappa \gamma \lambda p}{\kappa - 1 + p},$$

which at $\kappa = 2$ reduces to (3); the characterisation and its asymptote are derived in Appendix A. As $\Theta \rightarrow \infty$ the pass-through approaches $1/\kappa$: the asymptotic pass-through falls as costs become more convex because both parties' responses to any change in stakes are smaller. This is an asymptotic comparison, not a global bound for every curvature: when $1 < \kappa < 2$ pass-through can be non-monotone and exceed $1/\kappa$ at intermediate stake–cost ratios, and the quadratic case is special in that Proposition 2 proves its one-half asymptote is also the global supremum. As $\kappa \downarrow 1$

⁶⁴Whether entry margins of this kind are quantitatively important for AI-enabled attack is an empirical question about the fixed cost of directing capacity at a target, on which I know of no evidence.

the asymptote rises to one, so the linear-cost limit removes the bound altogether; the identification exercise below shows that κ is measurable from effort and survival rather than assumed. With *unequal* curvatures, the conditions (2) are unchanged, but the exploited share ceases to depend on the primitives through a single ratio, and the displayed argument no longer applies. The natural expectation—not derived here—is that equilibrium remains unique and interior, with the more convex side responding less, so that a defender with rigid capacity (κ_D large) would make liability a weaker instrument at release and the window relatively more valuable; I record this as a conjecture.

Unequal stock elasticities. If the two sides' search technologies scale differently with the pool— $\alpha_D \neq \alpha_A$ in the notation of Appendix B.1.5—the factor n^α no longer cancels from the ratio of intensities, and the exploited share depends on the remaining-stock size. The window then acquires a second effect the baseline rules out: by draining the pool it shifts the exploited share on what remains, and the shift can run *against* the defender—if defensive search scales with abundance (α_D near one) while adversarial search does not (α_A near zero), the flaws surviving a long window are contested by a defence working leads that have thinned and an attack whose capacity has not. The factorisation (7) fails in that case. The natural conjecture is that the asymmetry weakens the window relative to the drawdown benchmark when it is the defence that is lead-constrained, but once the intensities and the exploited share become state-dependent, the defender's endogenous response can itself change, and establishing the direction requires solving the state-dependent contest; I record this too as a conjecture. Which case describes AI-assisted search is the same unmeasured elasticity flagged in Section 2.7.

B.2 Alternative welfare accounting: harm only

Several exercises—the pipeline, fringe and access-allocation discussions, and the contingent-release problem in Appendix B.3—use the harm-only surplus $\mathcal{S}_H(h) = e^{-wh}B - L(h)$, which counts expected harm and nothing else. It is the implicit criterion of the fixed-hazard security literature and the natural reading of an expected-harm objective, and its closed forms are marginally sharper than those of Proposition 4(i). This subsection records them, and the one place they mislead. Recall that $s = w/x_D^*$ is benefit decay relative to contest search intensity and $\eta = L(0)/B$ is initial expected harm relative to release benefit. Write $\mathcal{D}_H \equiv h^*[(1 + \kappa^{-1})x_D^{\text{eff}}(h^*) - w]$ and $\bar{\eta}(s) \equiv (e^{s-1}/s)^\kappa$.

Proposition B.3. *Under the harm-only accounting, for every $\kappa > 1$: (i) Schedule surplus has at most one interior local maximum, attained at the smaller root of*

$$(\kappa + 1) \log\left(1 + \frac{z}{\kappa}\right) - sz = \log \frac{\eta}{s}. \quad (45)$$

For $s < 1$ the policy space partitions into immediate release ($\eta \leq s$), staging ($s < \eta < \bar{\eta}(s)$), and withholding ($\eta > \bar{\eta}(s)$); for $s \geq 1$ staging is never optimal; ties at the boundaries are recorded in the proof; and every interior optimum that beats withholding satisfies the ceiling (16). (ii) At an interior optimum,

$$\varepsilon_{m_0} = -\varepsilon_B = \frac{1}{\mathcal{D}_H}, \quad \varepsilon_w = \frac{wh^* - 1}{\mathcal{D}_H}, \quad \frac{\partial h^*}{\partial \log m_0} \leq \frac{\kappa}{w}, \quad (46)$$

and under quadratic costs, with $\zeta \equiv 1/(1+p)$,

$$\varepsilon_\gamma = -\varepsilon_{c_D} = \frac{1 - (1 + \zeta)\varphi}{\mathcal{D}_H} - (1 - \zeta\varphi)\left(1 + \frac{wh^*}{\mathcal{D}_H}\right), \quad \varepsilon_v = -\varepsilon_{c_A} = \frac{(1 + \zeta)\varphi}{\mathcal{D}_H} - \zeta\varphi\left(1 + \frac{wh^*}{\mathcal{D}_H}\right), \quad (47)$$

with $\varepsilon_\lambda = \varepsilon_\gamma + \varepsilon_{m_0}$, and, as $\lambda \rightarrow \infty$: $z \propto \lambda^{1/3}$, $h^* \propto \lambda^{-1/6}$, and $\varepsilon_\lambda \rightarrow -\frac{1}{6}$.

Proof of Proposition B.3. Part (i). By (7) and (10), $\mathcal{S}_H(h) = e^{-wh}B - \lambda p m_0(1 + z/\kappa)^{-\kappa}$ and $\mathcal{S}'_H(h) = x_D^* \lambda p m_0(1 + z/\kappa)^{-(\kappa+1)} - wBe^{-wh}$. Dividing by wBe^{-wh} and taking logarithms, $\mathcal{S}'_H(h) > 0$ if and only if $G(z) < \log(\eta/s)$, where $G(z) \equiv (\kappa + 1)\log(1 + z/\kappa) - sz$, with $G(0) = 0$ and $G''(z) = -(\kappa + 1)/(\kappa + z)^2 < 0$. As $h \rightarrow \infty$, benefit decays exponentially and harm polynomially, so $\mathcal{S}_H(h) \rightarrow 0^-$ and the supremum is attained.

G is strictly concave with maximiser $z_G = (\kappa + 1)/s - \kappa$ when positive, so the superlevel set $\{z : G(z) > \log(\eta/s)\}$ is an interval, nonempty if and only if $\log(\eta/s) < \max G$ strictly; hence $\mathcal{S}'_H$ changes sign at most twice—positive, negative, positive—and $\mathcal{S}_H$ has at most one interior local maximum, at the left endpoint z_1 . Since $G(0) = 0$, $z_1 > 0$ requires $\eta > s$; if $\eta \leq s$ there is no interior maximum and $\mathcal{S}_H(0) = B(1 - \eta) > 0$ when $\eta < 1$, so immediate release is optimal. If $\log(\eta/s) \geq \max G$, $\mathcal{S}_H$ is nondecreasing and negative on $(0, \infty)$ and withholding is optimal.

For the partition and the ceiling: at an interior stationary point $L = we^{-wh}B/x_D^{\text{eff}}$, so $\mathcal{S}_H(h) = e^{-wh}B[1 - w/x_D^{\text{eff}}(h)]$, positive if and only if $x_D^{\text{eff}}(h) > w$, that is $z < \bar{z} \equiv \kappa(1 - s)/s$, which requires $s < 1$ and is (16) on dividing by x_D^* . The maximiser of G satisfies $z_G - \bar{z} = 1/s > 0$, so $\bar{z}$ lies strictly left of the peak; hence $z_1 < \bar{z}$ if and only if $\log(\eta/s) < G(\bar{z}) = -(\kappa + 1)\log s - \kappa(1 - s)$, which is $\eta < (e^{s-1}/s)^\kappa = \bar{\eta}(s)$. The interval $(s, \bar{\eta}(s))$ is nonempty for every $s \in (0, 1)$ because $\log s/(s - 1) \geq 1 > \kappa/(1 + \kappa)$. Write $h_1 \equiv z_1/x_D^*$. If $\eta > \bar{\eta}(s)$, either z_1 fails to exist or $\mathcal{S}_H(h_1) < 0$. At $\eta = \bar{\eta}(s)$ exactly, $z_1 = \bar{z}$ and $\mathcal{S}_H(h_1) = e^{-wh_1}B[1 - w/x_D^{\text{eff}}(h_1)] = 0$, the withholding value: a tie. For $s \geq 1$, $\bar{z} \leq 0$ and the comparison is $\max\{B(1 - \eta), 0\}$, with $\mathcal{S}_H(0) = B(1 - \eta) = 0$ at $\eta = 1$: immediate release ties withholding there.

Part (ii). Differentiate $G(z) = \log(\eta/s)$ at z_1 , treating s and η as functions of a primitive q . With $G_z = (\kappa + 1)/(\kappa + z) - s > 0$ at z_1 and $\partial G/\partial s = -z$, $d \log z = [d \log(\eta/s) + wh^* d \log s]/\mathcal{D}_H$, where $\mathcal{D}_H = zG_z = h^*[(1 + \kappa^{-1})x_D^{\text{eff}} - w]$, using $x_D^*(\kappa + 1)/(\kappa + z) = (1 + \kappa^{-1})x_D^{\text{eff}}$. Since $h^* = z/x_D^*$, $\varepsilon_q = d \log z/d \log q - \varepsilon_q^x$ where $\varepsilon_q^x \equiv d \log x_D^*/d \log q$. Writing $y_q \equiv d \log L(0)/d \log q$ and noting $\eta/s = L(0)x_D^*/(wB)$,

$$\varepsilon_q = \frac{y_q + \varepsilon_q^x - \mathbf{1}\{q \in \{w, B\}\} + wh^*(\mathbf{1}\{q = w\} - \varepsilon_q^x)}{\mathcal{D}_H} - \varepsilon_q^x. \quad (48)$$

For $q \in \{m_0, B, w\}$, $\varepsilon_q^x = 0$ and $y_{m_0} = 1$, $y_B = y_w = 0$, giving (46); and $\partial h^*/\partial \log m_0 = h^*/\mathcal{D}_H$, which since staging requires $x_D^{\text{eff}} > w$ is at most κ/w . For the remaining elasticities at $\kappa = 2$: from (3), $d \log p = -\varphi d \log \Theta$; from $\ell = 2\gamma\lambda p/(1 + p)$, $d \log \ell = d \log(\gamma\lambda) + \zeta d \log p$; and $x_D^* = \ell/c_D$.

Hence

q	γ	λ	c_D	v	c_A	m_0	B	w
ε_q^x	$1 - \zeta\varphi$	$1 - \zeta\varphi$	$\zeta\varphi - 1$	$\zeta\varphi$	$-\zeta\varphi$	0	0	0
y_q	$-\varphi$	$1 - \varphi$	φ	φ	$-\varphi$	1	0	0

Substituting into (48) yields (47); the pairings follow because (48) is linear in (ε_q^x, y_q) , and $(\varepsilon_\lambda^x, y_\lambda) = (\varepsilon_\gamma^x, y_\gamma) + (\varepsilon_{m_0}^x, y_{m_0})$ gives $\varepsilon_\lambda = \varepsilon_\gamma + \varepsilon_{m_0}$.

For the harm limit: $p \sim (2\Theta)^{-1/2}$ with $\Theta \propto \lambda$ gives $p \propto \lambda^{-1/2}$, so $x_D^* \sim (2\gamma\lambda v/c_A c_D)^{1/2} \propto \lambda^{1/2}$, $s \rightarrow 0$, $L(0) \propto \lambda^{1/2}$ and $\eta/s \propto \lambda$. With $s \rightarrow 0$, (45) becomes $(\kappa + 1) \log(1 + z/\kappa) \rightarrow \log(\eta/s)$, so $z \propto \lambda^{1/(\kappa+1)}$ and $h^* \propto \lambda^{1/(\kappa+1)-1/2}$, which at $\kappa = 2$ is $\lambda^{-1/6}$; then $\varphi \rightarrow \frac{1}{2}$, $\zeta \rightarrow 1$, $\mathcal{D}_H \rightarrow \kappa + 1$ give $\varepsilon_\lambda \rightarrow \frac{1}{3} - \frac{1}{2} = -\frac{1}{6}$. Where instead v rises proportionally with λ , Θ is independent of λ , so p and φ are fixed, $x_D^* \propto \lambda$, $L(0) \propto \lambda$, $\eta/s \propto \lambda^2$, and $z \propto \lambda^{2/(\kappa+1)}$, giving $\varepsilon_\lambda \rightarrow 2/(\kappa + 1) - 1 = -\frac{1}{3}$ at $\kappa = 2$. $\square$

Two comparisons with the defence-inclusive accounting complete the remark closing Section 3.4. The staging region is wider on both sides: the floor is $\eta > s$ against the chargeable $\eta_c > s$ —a lower bar, since $\eta_c < \eta$ —and the withholding boundary is higher, because neither margin is charged the search bill. And the harm elasticity turns negative at large protection depth, where the defence-inclusive elasticity is exactly positive (Proposition 5): free search makes dangerous models an argument for short windows; priced search does not. The pairing $\varepsilon_\gamma = -\varepsilon_{c_D}$, exact here and false there, is the same accounting seen from the other side.

B.3 Learning and release contingent on findings

The Poisson prior of the main text has a consequence noted in Section 3.1: under $\alpha = 1$, found and unfound flaws are independent, so the posterior mean of the residual stock is m_0 times a survival probability *regardless of how many flaws have been found*. The belief state is deterministic, and under a fixed effort schedule a findings-based stopping time cannot improve expected value over the best deterministic date. This section shows that the absence of informational value is a knife edge and then prices information in a restricted stopping problem that holds one exogenous effort schedule fixed across every candidate release date; the full joint effort-and-stopping problem remains open.

B.3.1 The knife edge

Write q for the probability that a given flaw has been found by the evaluation date, $S = 1 - q$, Y for the number found, U for the number unfound, and $G(S) = \mathbb{E}[S^{N_0}]$ for the generating function of the stock.

Proposition B.4. *Under $\alpha = 1$, $\mathbb{E}[U \mid Y = y] = S G^{(y+1)}(S)/G^{(y)}(S)$; if this is independent of y on an open interval of S , then N_0 is Poisson, and conversely under a Poisson stock it equals $m_0 S$ for every y . For general α , with n^α read as zero at $n = 0$, the informational content of a first discovery is*

$\text{Cov}(N_0, N_0^\alpha) / \mathbb{E}[N_0^\alpha] - 1$, which at $\alpha = 1$ equals $\text{Var}(N_0) / \mathbb{E}[N_0] - 1$ and which for a Poisson stock is strictly negative at every $\alpha < 1$.

Proof. Under $\alpha = 1$ each flaw is found independently with probability $q = 1 - S$, so with $p_n = \Pr(N_0 = n)$, $\Pr(U = u \mid Y = y) \propto p_{y+u} \frac{(y+u)!}{u!} S^u$; since $G^{(y)}(S) = \sum_u p_{y+u} \frac{(y+u)!}{u!} S^u$ and $\frac{d}{dS} G^{(y)} = G^{(y+1)}$, the conditional mean is $SG^{(y+1)} / G^{(y)}$. If the expressions at $y = 0$ and $y = 1$ agree on an open interval, then $G'' / G' = G' / G$, which implies $(\log G)'' = 0$; using $G(1) = 1$ and $G'(1) = m_0$ gives $G(S) = e^{m_0(S-1)}$ by analyticity. The converse is independent thinning. For the first-discovery expression, to first order $\Pr(Y_t = 1 \mid N_0 = n)$ is proportional to $n^\alpha t$; the common proportionality constant cancels, giving $\mathbb{E}[U_t \mid Y_t = 1] \rightarrow \mathbb{E}[N_0^{1+\alpha}] / \mathbb{E}[N_0^\alpha] - 1$, and subtracting $\mathbb{E}[N_0]$ gives the covariance expression. For the Poisson sign at $\alpha < 1$, the log-moment difference $\log \mathbb{E}[N_0^{1+\alpha}] - \log \mathbb{E}[N_0^\alpha]$ is strictly increasing in α by the likelihood-ratio ordering of the tilted measures and equals $\log(m_0 + 1)$ at $\alpha = 1$ for Poisson(m_0); hence the corresponding ratio is strictly below $m_0 + 1$ for every $\alpha < 1$, which gives the stated negative sign. $\square$

Proposition B.4 says the uninformativeness of discoveries is a property of the stock and the search technology *jointly*, and that it fails in opposite directions on either side of the benchmark. The two effects of a discovery set out in Section 3.1 offset exactly only under a Poisson stock at $\alpha = 1$: overdispersion makes a productive evaluation bad news on net, while a capacity-constrained technology ($\alpha < 1$) makes it good news. Of the two departures, overdispersion is the cheap one: it leaves $\alpha = 1$, and with it every closed form of the main text, untouched, and it is empirically plausible, since vulnerability counts can vary across systems more than a Poisson benchmark allows.

B.3.2 An overdispersed stock

Let the latent Poisson mean $\tilde{m}$ have a Gamma distribution, so that $N_0 \mid \tilde{m} \sim \text{Poisson}(\tilde{m})$, and choose its parameters so that the resulting stock has mean m_0 and *dispersion index* $\delta_N \equiv \text{Var}(N_0) / \mathbb{E}[N_0] \geq 1$; the benchmark is $\delta_N \rightarrow 1$. Write x_t for the per-flaw search intensity and $\hat{m}_t$ for the posterior mean of the residual stock. Let $\mathcal{F}_t$ denote the observed finding history through time t .

Proposition B.5. (i) *The posterior on the latent mean is Gamma with shape $m_0 / (\delta_N - 1) + Y_t$ and rate $1 / (\delta_N - 1) + q_t$, so*

$$\hat{m}_t = \frac{(m_0 + (\delta_N - 1)Y_t)S_t}{1 + (\delta_N - 1)q_t}, \quad \frac{\partial \hat{m}_t}{\partial Y_t} = \frac{(\delta_N - 1)S_t}{1 + (\delta_N - 1)q_t},$$

and $\mathbb{E}[d\hat{m}_t \mid \mathcal{F}_t] = -x_t \hat{m}_t dt$. If the search-intensity path is deterministic, $\mathbb{E}[\hat{m}_t] = m_0 S_t$; if the path is adapted to the finding history, this fails by $\text{Cov}(\tilde{m}, S_t)$, generically nonzero and of either sign. (ii) For any fixed deterministic release date, the defender's optimal control is a belief-free search intensity: the effort profile (9), the survival law (10), and the effective protection rate (11) are unchanged by overdispersion. (iii) If the search-intensity path is deterministic, $\text{Var}(\hat{m}_t) = m_0 q_t S_t^2 (\delta_N - 1)^2 / (1 + (\delta_N - 1)q_t)$: zero at

$q_t \in \{0, 1\}$ and at $\delta_N = 1$, and single-peaked in q_t , with the peak at $q_t = \frac{1}{3}$ as $\delta_N \rightarrow 1$, moving earlier as δ_N rises.

Proof. (i) Conditional on the latent mean $\tilde{m}$, discoveries form an inhomogeneous Poisson process with cumulative intensity $\tilde{m}q_t$ and the residual stock is $\text{Poisson}(\tilde{m}S_t)$, so the likelihood of a history is $e^{-\tilde{m}q_t} \tilde{m}^{Y_t}$ times a factor independent of $\tilde{m}$ —also when the search intensity is adapted, since the path is then a function of the observed finding history $\mathcal{F}_t$. Conjugacy gives the Gamma posterior, and $\hat{m}_t = \mathbb{E}[\tilde{m} \mid \mathcal{F}_t] S_t$ the displayed form. The posterior mean of $\tilde{m}$ is a martingale, and $\dot{S}_t = -x_t S_t$ gives the drift. If the search-intensity path is deterministic, $\mathbb{E}[\hat{m}_t] = m_0 S_t$; if adapted, the identity fails by $\text{Cov}(\tilde{m}, S_t)$. (ii) Given $\mathcal{F}_t$, expected flow cost is $\hat{m}_t C_D(x)$ and expected flow reduction in loss is $\hat{m}_t x \psi_t$; $\hat{m}_t$ cancels, leaving the belief-free condition $C'_D(x) = \psi_t$ with terminal value ℓ . (iii) For a deterministic search-intensity path, q_t and S_t are constants: Y_t is negative binomial with mean $m_0 q_t$ and variance $m_0 q_t (1 + (\delta_N - 1)q_t)$, and since $\hat{m}_t$ is then affine in Y_t with nonrandom coefficients, the variance expression follows. Under an adapted search intensity, q_t and S_t are history-dependent random variables, and the calculation does not go through in this form; the warning in part (i) applies here too. With $b \equiv 1/(\delta_N - 1)$, $\frac{d}{dq} \log[q(1-q)^2/(b+q)]$ is strictly decreasing from $+\infty$ to $-\infty$; as $b \rightarrow \infty$ the root solves $1/q = 2/(1-q)$, that is $q = \frac{1}{3}$, and it moves left as b falls. $\square$

Part (ii) runs against the natural conjecture: if a productive evaluation signals that more flaws remain, surely the defence should search harder. It should not at a fixed deterministic deadline, and the reason is structural. Under $\alpha = 1$ the defender chooses an intensity per search channel, and the code supplies the channels; beliefs about how many channels there are scale the flow cost and the flow return in the same proportion and cancel from the first-order condition. The defender need not know how many flaws it is hunting to know how hard to hunt each one. Part (iii) is the instrument's budget constraint: the same search that generates information about the stock removes the thing the information is about, so the dispersion of the belief about the *residual* stock rises and then falls; a window long enough to measure the stock well has already removed most of it. Part (i)'s caveat about adapted search intensities is not decorative: under a rule that changes effort after discoveries, realised survival becomes history-dependent, and the unconditional identity fails. The exercise below therefore fixes one exogenous schedule for every candidate stopping date, pricing information while deliberately setting aside the joint effort-and-stopping problem.

B.3.3 Contingent release under a fixed effort schedule

A stopping problem requires one survival process shared by all candidate release dates. Fix, therefore, a deterministic continuous schedule $\bar{x} : [0, \infty) \rightarrow (0, \infty)$ and let $\bar{S}(t) = \exp\{-\int_0^t \bar{x}(s) ds\}$ and $\bar{q}(t) = 1 - \bar{S}(t)$. The schedule specifies the effort used at calendar date t if release has not yet occurred; it is not reoptimised when a stopping rule selects another date. Let $\mathcal{F}_t$ be the finding filtration generated under this schedule and $\hat{m}_t$ the posterior mean residual stock from Proposition B.5. Write $\mathcal{T}_{\mathcal{F}}$ for the class of $\mathcal{F}_t$ -stopping times taking values in $[0, \infty]$; each member is a *contingent release rule*. Under the harm-only accounting, with the payoff at $\tau = \infty$ normalised

to zero, its value is $\bar{\Phi}(\tau) = \mathbb{E}[(e^{-w\tau}B - \lambda p \hat{m}_\tau) \mathbf{1}\{\tau < \infty\}]$. The corresponding fixed-date surplus is $\bar{\mathcal{S}}_H(t) = e^{-wt}B - \lambda p m_0 \bar{\mathcal{S}}(t)$ for $t < \infty$, and define $\bar{\mathcal{S}}_H(\infty) = 0$. For comparison with a main-text deadline h , one convenient global extension of its optimal pre-release profile is

$$\bar{x}_h(t) = \begin{cases} \left[\frac{1}{x_D^*} + \frac{h-t}{\kappa} \right]^{-1}, & 0 \leq t \leq h, \\ x_D^*, & t > h. \end{cases} \quad (49)$$

Only the agreement on $[0, h]$, not this particular continuation, is needed below.

Proposition B.6. *Fix the schedule $\bar{x}$ above. (i) If $\delta_N = 1$, then*

$$\sup_{\tau \in \mathcal{T}_F} \bar{\Phi}(\tau) = \max \left\{ 0, \sup_{t < \infty} \bar{\mathcal{S}}_H(t) \right\} = \sup_{t \in [0, \infty]} \bar{\mathcal{S}}_H(t).$$

(ii) *If $\delta_N > 1$, every deterministic $h \in (0, \infty)$ is strictly dominated by a bounded perturbation measurable at some $t_0 < h$. (iii) Suppose $\delta_N > 1$, a fixed-date benchmark $h^* \in (0, \infty)$ has $\bar{\mathcal{S}}_H(h^*) > 0$, and choose $0 < t_0 < h^* < T < \infty$. With $Z = \hat{m}_{t_0}$, define*

$$\mathcal{J}_{t_0, T}(m) = \max \left\{ 0, \max_{t \in [t_0, T]} \left[e^{-wt}B - \lambda p m \frac{\bar{\mathcal{S}}(t)}{\bar{\mathcal{S}}(t_0)} \right] \right\}. \quad (50)$$

Then

$$\sup_{\tau} \bar{\Phi}(\tau) \geq \mathbb{E}[\mathcal{J}_{t_0, T}(Z)] > \mathcal{J}_{t_0, T}(\mathbb{E}Z) \geq \bar{\mathcal{S}}_H(h^*). \quad (51)$$

If $\bar{x} = \bar{x}_{h^*}$ from (49), the last term is exactly the main-text harm-only surplus at h^* .

Proof. Under the fixed deterministic schedule, future thinning gives $\mathbb{E}[U_t \mid \mathcal{F}_{t_0}] = \hat{m}_{t_0} \bar{\mathcal{S}}(t) / \bar{\mathcal{S}}(t_0)$ for every $t \geq t_0$. If $\delta_N = 1$, $\hat{m}_t = m_0 \bar{\mathcal{S}}(t)$ is deterministic. Hence the realised payoff from any stopping time is bounded pointwise by $\sup_{t \in [0, \infty]} \bar{\mathcal{S}}_H(t)$, including the zero value at withholding, and a constant stopping time attains that value whenever a finite date is optimal. This proves the expected-value equivalence in part (i) without requiring the stopping time itself to be deterministic.

For part (ii), fix $h > 0$, choose $t_0 \in (0, h)$, and put $Z = \hat{m}_{t_0}$ and $r(z) = z / (1 + z)$. The random variable $g(Z) = r(Z) - \mathbb{E}[r(Z)]$ is bounded, strictly increasing in Z , and has mean zero. Choose $\varepsilon \in (0, h - t_0)$ and then $c > 0$ small enough that $\tau_c = h + c g(Z) \in [h - \varepsilon, h + \varepsilon]$ almost surely. Because $\tau_c \geq t_0$ and is $\mathcal{F}_{t_0}$ -measurable, it is a stopping time. Applying the thinning identity conditional on $\mathcal{F}_{t_0}$ and differentiating at $c = 0$ gives

$$\begin{aligned} \left. \frac{d}{dc} \bar{\Phi}(\tau_c) \right|_{c=0} &= \mathbb{E} \left[g(Z) \left(-w e^{-wh} B + \lambda p Z \bar{x}(h) \frac{\bar{\mathcal{S}}(h)}{\bar{\mathcal{S}}(t_0)} \right) \right] \\ &= \lambda p \bar{x}(h) \frac{\bar{\mathcal{S}}(h)}{\bar{\mathcal{S}}(t_0)} \text{Cov}(Z, r(Z)) > 0. \end{aligned}$$

The covariance is strictly positive because Z is nondegenerate when $\delta_N > 1$ and $t_0 > 0$: for an

independent copy Z' , $2 \text{Cov}(Z, r(Z)) = \mathbb{E}[(Z - Z')(r(Z) - r(Z'))] > 0$. Thus every finite positive deterministic date is locally dominated without ever evaluating the schedule outside a bounded interval.

For part (iii), $\mathcal{J}_{t_0, T}$ is a finite maximum of the zero function and a compact supremum of affine functions of m , hence is continuous and convex. After observing Z , choosing the earliest maximiser in (50) (and choosing ∞ when zero is optimal) is an admissible $\mathcal{F}_{t_0}$ -measurable stopping rule, which yields the first weak inequality in (51). Moreover, $\mathbb{E}Z = m_0 \bar{S}(t_0)$, so releasing at h^* is feasible and gives $\mathcal{J}_{t_0, T}(\mathbb{E}Z) \geq e^{-wh^*} B - \lambda p m_0 \bar{S}(h^*) = \bar{\mathcal{S}}_H(h^*) > 0$. On the other hand, $\mathcal{J}_{t_0, T}(m) = 0$ for every $m \geq B e^{-wt_0} \bar{S}(t_0) / [\lambda p \bar{S}(T)]$. The Gamma–Poisson posterior Z has unbounded support when $\delta_N > 1$. A convex function that is positive at $\mathbb{E}Z$ and zero on a tail cannot be affine on the convex hull of that support, so Jensen’s inequality is strict. This proves (51). $\square$

Proposition B.6 isolates the value of information under a single exogenous effort technology. The value comes from convexity: after a reading date, the best fixed-schedule response to a posterior is a maximum of affine functions, so dispersion in the posterior raises expected value. It does *not* establish that the frozen schedule remains optimal once release is contingent: in the full joint control-and-stopping problem, findings would generally change effort, survival, and continuation values as well as the release date.

Three properties temper the restricted instrument. First, near the Poisson benchmark there is little spread to use: the variance of the residual-stock belief vanishes quadratically as $\delta_N \rightarrow 1$ (Proposition B.5(iii)). Second, the result does not rank contingency against withholding; the strict comparison in (51) requires a positive finite benchmark. Third, and for policy most importantly, the posterior depends on the finding count *and* cumulative search intensity: a developer seeking early release can manufacture apparent quiet by cutting the defender’s funding or otherwise inducing less search, and finding rates alone cannot distinguish a completed evaluation from a neglected one. Even in the restricted exercise, enforcement therefore requires observing effort as well as findings; a full contingent contract would additionally have to govern how that effort adapts to the history.

B.4 Extensions to staged access and implementation

These extensions relax instantaneous remediation and effective exclusivity, then turn to allocation and identification. Each proof is kept beside the result it supports.

B.4.1 Remediation lags can reverse deadline effort

The deadline effect relies on a finding becoming protection immediately. If remediation takes time, a finding made near release may arrive too late, and defensive effort can peak before the deadline or even fall throughout the window. Assumption 1 is therefore the baseline’s strongest restriction and the one the motivating episode most directly challenges: a finding there took two weeks on average merely to patch, with deployment still to follow.

To make this reversal precise, suppose a defensive discovery at t enters a *remediation pipeline* and yields a patch after an exponentially distributed delay with rate μ . While the patch is pending, the flaw remains exploitable. A found-but-unpatched flaw is then a third kind of object—gone from the defender’s search problem but still available to the adversary—so the state after a defensive find no longer matches the state after an exploit. Winner-independence fails, which is exactly where the dynamics return. Independence across flaws and the absence of shared remediation capacity keep the problems separable. Exponential remediation makes each per-flaw problem Markov without adding patch age as a state.

Work backwards from release. A flaw pending after h is searched by the adversary alone against the patch clock. Under quadratic costs, the adversary’s optimal per-flaw search intensity and the pending-flaw exploitation probability are

$$x_P^* = \mu \left(\sqrt{1 + \frac{2v}{c_A \mu}} - 1 \right), \quad p_P = \frac{x_P^*}{x_P^* + \mu}. \quad (52)$$

This is the contest of Proposition 1 with the patch clock μ standing in the defender’s place. A flaw still unfound at release is contested as before, except that continuation values now pass through the pending state: the defender’s first-order condition becomes $C'_D(x_D) = \tilde{\ell} - \gamma \lambda p_P$ and the adversary’s becomes $C'_A(x_A) = v - \tilde{g}$, where $(\tilde{\ell}, \tilde{g})$ are the per-flaw values of the lagged contest derived below. The defender’s stake is taxed twice. A find is worth less because it still leaks with probability p_P ; a flaw left unfound faces an adversary whose prize now includes a second chance. During the window, the shadow value of a live flaw therefore satisfies a Riccati equation forced by the risk that a find made at t remains pending at release, $\gamma \lambda p_P e^{-\mu(h-t)}$. The next proposition identifies when that force overturns the deadline effect.

Proposition B.7. *Under quadratic costs and exclusive access on $[0, h]$, the defender’s window search intensity is $x^*(t) = \psi_P(t)/c_D$, where*

$$\dot{\psi}_P = \frac{\psi_P^2}{2c_D} - \mu \gamma \lambda p_P e^{-\mu(h-t)}$$

with terminal condition $\psi_P(h) = \tilde{\ell} - \gamma \lambda p_P$, and it docks onto the lagged contest intensity at release. The effort profile falls into the release date if and only if

$$(\tilde{\ell} - \gamma \lambda p_P)^2 < 2c_D \gamma \lambda \mu p_P, \quad (53)$$

and it rises at the opening of the window if and only if $\psi_P(0)^2 > 2c_D \gamma \lambda \mu p_P e^{-\mu h}$. Effort therefore rises throughout when (53) fails; rises and then falls, with an interior peak, when (53) and the opening condition both hold; and falls throughout when (53) holds and the window is short. As $\mu \rightarrow \infty$, $p_P \rightarrow 0$, $(\tilde{\ell}, \tilde{g}) \rightarrow (\ell, g)$, and effort converges uniformly on $[0, h]$ to Proposition 3, though not uniformly in derivatives.

Proof of Proposition B.7 and derivation of the lagged contest. Let $\mu > 0$ be the patch-landing rate. Consider first a flaw already pending after release. The adversary alone searches it, at intensity y , and

the flaw leaves the pending state at rate $y + \mu$; its problem is $\max_y \{yv - C_A(y)\} / (y + \mu)$, whose first-order condition under quadratic costs is $\frac{1}{2}c_A y^2 + c_A \mu y - v\mu = 0$ with positive root x_P^* in (52); the exploitation probability of a pending flaw is $p_P = x_P^* / (x_P^* + \mu)$ and its value to the adversary is $g_P = v - c_A x_P^*$. A flaw pending during the window is the same object with the clock running from its discovery date, and if release has not yet occurred, the adversary's window technology applies instead; under exclusive access, the pending flaw is safe until h . Define the proof-local forcing term $Q(t) \equiv \gamma \lambda p_P e^{-\mu(h-t)}$, the probability that a find at t is still pending at h times the harm it then suffers.

For the post-release contest on unfound flaws: a defensive find yields a pending flaw, worth $-\gamma \lambda p_P$ to the defender and g_P to the adversary; an exploit yields $-\gamma \lambda$ and v . Writing $(\tilde{\ell}, \tilde{g})$ for the two parties' per-flaw values, the per-resolution problems are

$$\tilde{\ell} = \min_{x_D} \frac{x_A \gamma \lambda + x_D \gamma \lambda p_P + C_D(x_D)}{x_A + x_D}, \quad \tilde{g} = \max_{x_A} \frac{x_A v + x_D g_P - C_A(x_A)}{x_A + x_D},$$

with first-order conditions $C'_D(x_D) = \tilde{\ell} - \gamma \lambda p_P$ and $C'_A(x_A) = v - \tilde{g}$; existence and uniqueness follow as in Proposition 1 since the objectives are strictly quasi-concave and the stakes $(\gamma \lambda - \gamma \lambda p_P, v - g_P)$ are positive. Note $\tilde{\ell} - \gamma \lambda p_P$ is the defender's *net* stake—a find is worth only the harm it actually averts.

On the window, let $\psi_P(t)$ be the shadow value of a live flaw net of what a find still leaks. The Bellman equation is $-\dot{\psi}_P = \min_x \{C_D(x) - x\psi_P\} + \mu Q(t)$, which under quadratic costs is $\dot{\psi}_P = \psi_P^2 / (2c_D) - \mu Q(t)$ with $\psi_P(h) = \tilde{\ell} - \gamma \lambda p_P$; the control is $x^* = \psi_P / c_D$. The sign of $\dot{\psi}_P$ at $t = h$ is the sign of $\psi_P(h)^2 / (2c_D) - \mu \gamma \lambda p_P$, so effort is falling into the release date exactly when (53) holds, and since Q is increasing in t while ψ_P^2 is continuous, the profile rises then falls at most once on the relevant range. The two shape conditions are the signs of $\dot{\psi}_P$ at the endpoints: $\dot{\psi}_P(h) = \psi_P(h)^2 / (2c_D) - \mu \gamma \lambda p_P$, negative exactly under (53), and $\dot{\psi}_P(0) = \psi_P(0)^2 / (2c_D) - \mu \gamma \lambda p_P e^{-\mu h}$, positive exactly under the opening condition. The classification is exhaustive because $\dot{\psi}_P$ crosses zero at most once: since $Q' = \mu Q$, $d\dot{\psi}_P/dt = \psi_P \dot{\psi}_P / c_D - \mu^2 Q$, which at any zero of $\dot{\psi}_P$ equals $-\mu^2 Q < 0$, so $\dot{\psi}_P$ can only cross downward and does so once at most. Effort therefore rises throughout when (53) fails, rises then falls when both endpoint conditions hold, and falls throughout when (53) holds and the opening condition fails; as $h \rightarrow 0$ the two endpoint conditions coincide, so short windows deliver the declining case whenever (53) holds. As $\mu \rightarrow \infty$: expanding the root, $x_P^* = v/c_A - v^2/(2c_A^2\mu) + O(\mu^{-2}) \rightarrow v/c_A$, a finite limit, so $p_P = x_P^* / (x_P^* + \mu) \rightarrow 0$, $\mu p_P \rightarrow v/c_A$, and $g_P = v - c_A x_P^* \rightarrow 0$. The contest conditions converge to (2), while $Q(t) \rightarrow 0$ and $\mu Q(t) \rightarrow 0$ for every fixed $t < h$; more strongly, $\int_0^h \mu Q(t) dt = \gamma \lambda p_P (1 - e^{-\mu h}) \rightarrow 0$. Together with the convergent terminal condition, the integral form of the Riccati equation and Gronwall's inequality give uniform convergence of the effort level on the fixed interval $[0, h]$. At $t = h$, however, the forcing term satisfies $\mu \gamma \lambda p_P \rightarrow \gamma \lambda v / c_A \neq 0$; it can therefore produce a vanishing $O(1/\mu)$ boundary layer, so convergence of derivatives is not uniform at release. In the opposite limit $\mu \rightarrow 0$ the root behaves as $x_P^* \sim \sqrt{2v\mu/c_A}$, so $p_P \rightarrow 1$: a pending flaw with no prospect of a patch is exploited with

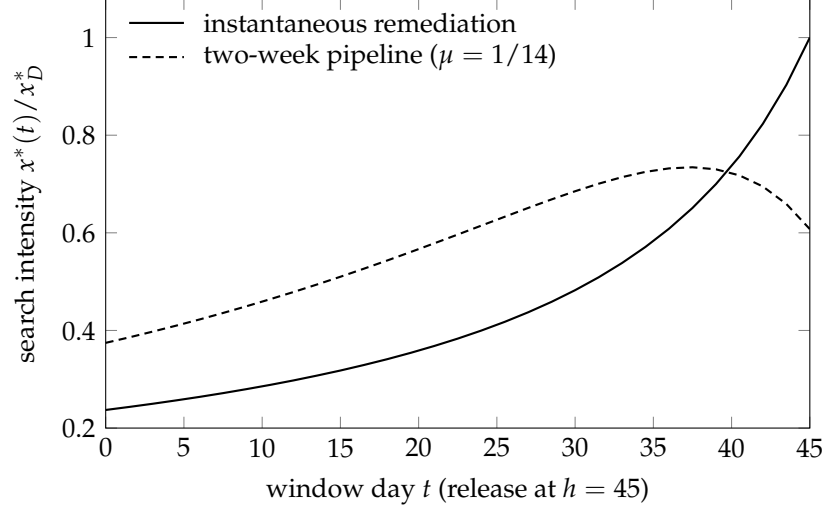


Figure 4: The effort profile with and without a remediation pipeline at the fixed release date $h = 45$ (a representative calibration: $\mu = 1/14$ per day, $\lambda = 4$, $\gamma = 0.7$, cost levels as in Figure 3). The pipeline shifts effort forward and turns it down over the final $1/\mu$: late findings can no longer be deployed by release. Intensities are normalised by the no-lag contest intensity.

certainty. □

Even at a fixed discovery intensity x , discovery followed by an independent exponential remediation time protects a flaw by date h with probability $1 - [\mu e^{-xh} - x e^{-\mu h}] / (\mu - x)$ for $x \neq \mu$: remediation changes marginal protection and its curvature rather than merely shifting the calendar. Proposition B.7 holds h fixed and signs the effort profile; it does not sign the optimal release date. When condition (53) holds, a remediation pipeline reverses the deadline effect at release and, for a sufficiently short window, can reverse it everywhere. The shadow value of a finding is the protection it delivers: an early finding is deployed with near-certainty, while one made in the final $1/\mu$ before release is likely to remain pending and is worth only $(1 - p_p)$ of a deployed patch. Effort therefore shifts forward and can fall as release approaches because the deployment channel, rather than the incentive to search, is closing. In the short-window declining case, *every* finding is at risk of arriving still pending and defensive effort is monotonically *decreasing* in calendar time—the exact reverse of the no-lag effort profile, with the same evaluation team and the same incentives—so the window does nothing that fixed-technology reasoning expects of it. If (53) fails, effort instead rises throughout. The sign of the effort trend can therefore help diagnose whether deployment or search is binding within the model, using the same logs that identify κ in Appendix B.4.4. Figure 4 draws the long-window case against the no-lag benchmark.

Separability and memorylessness do different work in the exponential-patch model: independence across flaws and the absence of shared remediation capacity preserve separability, while the exponential clock removes patch age from the state. Real queues batch, triage and congest, and congestion couples flaws through a shared server. Endogenising μ in the arrival rate, and then

choosing the date jointly with search, remains open.⁶⁵

B.4.2 Headway is valuable only against the fringe

Everything above takes exclusivity literally: before h the adversary has no discovery technology. That is rarely the institution: open-weight models, systems released in other jurisdictions, and the previous generation of frontier models remain available while the defender has the frontier model. In practice, exclusivity is a capability gap rather than a monopoly, and “exclusive access” in the main results should be read as an effective technological gap, not merely a formal restriction on access.

Let the adversary search during the window with the best openly available technology—*fringe capability*—at cost parameter $c_A^{\text{fr}} \geq c_A$, upgrading to the frontier cost c_A at release. The ratio $c_A/c_A^{\text{fr}} \in [0, 1]$ measures the fringe’s productivity relative to the frontier model and thereby indexes the *frontier–fringe capability gap*. The window is then a first contest at better odds, followed at h by the contest of Section 2 at worse ones; the defender’s shadow value solves a system coupled with the fringe adversary’s, with the contest pair (ℓ, g) as its terminal condition (as derived below).

The fringe makes the defender search harder and the developer worse off. Because a flaw missed today can now be stolen today, the shadow value of a live flaw exceeds its exclusive-window level at every pre-release date, and defensive search rises with it; a comparison argument establishes the result. That discipline does not pay. The developer’s burden is strictly higher, as follows by revealed preference when the defender faces the additional hazard.⁶⁶ The value of the window is not defensive diligence but the absence of an adversary to be diligent against. The limiting case is exact: if the fringe matches the frontier, the pre- and post-release contests coincide, expected harm equals λpm_0 for every h , and optimal headway is zero—headway has value only insofar as a frontier–fringe capability gap remains.

The open-weight implication is therefore intertemporal; it is not that open models are intrinsically more dangerous (Seger et al., 2023; Kapoor et al., 2024). Conditional on capability, open and closed releases enable the same harm. If fringe capability persists into later generations and the releasing developer does not internalise the resulting future policy costs, open release narrows the frontier–fringe capability gap and creates an intertemporal externality.

Derivation. On $[0, h)$ each live flaw is an independent problem under $\alpha = 1$. The defender’s per-flaw Bellman equation is $-\dot{\psi} = \min_x \{C_D(x) + x_A^{\text{fr}}\gamma\lambda - (x + x_A^{\text{fr}})\psi\}$, giving $C_D'(x_D) = \psi$; the fringe adversary’s is $-\dot{g}^{\text{fr}} = \max_x \{xv - C_A^{\text{fr}}(x) - (x + x_D)g^{\text{fr}}\}$, giving $(C_A^{\text{fr}})'(x_A^{\text{fr}}) = v - g^{\text{fr}}$. Under quadratic costs the system is $\dot{\psi} = \psi^2/(2c_D) - x_A^{\text{fr}}(\gamma\lambda - \psi)$ and $-\dot{g}^{\text{fr}} = x_A^{\text{fr}}v - c_A^{\text{fr}}(x_A^{\text{fr}})^2/2 - (x_A^{\text{fr}} + x_D)g^{\text{fr}}$ with controls $x_D = \psi/c_D$, $x_A^{\text{fr}} = (v - g^{\text{fr}})/c_A^{\text{fr}}$ and terminal conditions $\psi(h) = \ell$, $g^{\text{fr}}(h) = g$; setting $c_A^{\text{fr}} = \infty$ recovers Proposition 3. For the discipline claim: ψ lies strictly below $\gamma\lambda$ along any

⁶⁵The path in Figure 4 is generated directly from the displayed equations; replication code is available from the author.

⁶⁶At full compensation the burden is defence-inclusive social loss ($\Lambda = M$), so the comparison also lowers the social value of a fixed window. Under partial liability, a higher burden does not by itself rank social loss, and no monotone welfare ranking across intermediate fringe productivities is established here.

solution, so the fringe term $x_A^{\text{fr}}(\gamma\lambda - \psi)$ is positive wherever $x_A^{\text{fr}} > 0$, the fringe system's backward derivative exceeds the exclusive system's at equal states, and by the comparison theorem $\psi(t)$ —and with it $x_D(t)$ —exceeds its exclusive-window level at every $t < h$. The burden $m_0\psi(0)$ is accordingly higher. For the matched-fringe limit: at $c_A^{\text{fr}} = c_A$ the constant pair (ℓ, g) solves the system, since $\dot{\psi} = 0$ rearranges to $x_A^*(\gamma\lambda - \ell) = \ell^2/(2c_D)$, the defender's per-resolution condition, and $\dot{g}^{\text{fr}} = 0$ is the definition of g ; uniqueness of the backward solution makes both rates constant at contest levels, so the per-flaw exploit probability is p at every h and $L(h) = \lambda p m_0$ identically. $\square$

B.4.3 Who should hold a scarce window?

The strongest security team is not necessarily the best recipient of scarce early access. Holding fixed the harm sitting in code a candidate can patch, a defender too weak to search cannot use the window, while one that would win the post-release contest has little left for the window to protect. The best recipient is therefore interior in the stake–cost ratio.

To make that allocation problem precise, suppose the regulator can give the same headway h to at most K defenders, each able to patch a disjoint stock of code. For candidate i , let m_{0i} be the expected number of flaws it can patch, λ_i the harm per exploited flaw, p_i the exploited share after release, and x_{Di}^* its defensive search intensity. Giving that defender access removes expected harm

$$G_i(h) = \lambda_i p_i m_{0i} \left[1 - \left(1 + \frac{x_{Di}^* h}{\kappa} \right)^{-\kappa} \right]. \quad (54)$$

Because flaws are independent under $\alpha = 1$, these gains add across defenders. The optimal access list therefore consists of the K candidates with the largest values of $G_i(h)$. Counting defensive search costs changes the value of a grant but not the characteristics that determine this ranking.

Two of the four objects in (54) are not characteristics of the candidate at all. The exploited share p_i and the intensity x_{Di}^* are equilibrium outcomes, and by Proposition 1 both are governed by the candidate's stake–cost ratio $\Theta_i = \gamma\lambda_i c_A / (v c_{Di})$. Candidates facing the same adversary share v and c_A , so $\gamma\lambda_i / c_{Di} = \Theta_i v / c_A$ and, under quadratic costs, $p_i = p(\Theta_i)$ and $x_{Di}^* = (2v/c_A) \Theta_i p(\Theta_i) / [1 + p(\Theta_i)]$, both functions of Θ_i alone up to constants common to every candidate. Conditional on common (h, κ) , a candidate is therefore described by two candidate-specific numbers rather than four: its *patchable exposure* $\lambda_i m_{0i}$, the harm sitting in code it can fix before the contest divides any of it, and its stake–cost ratio Θ_i . The common parameters do not disappear from the ranking: because h and κ transform candidate-specific search intensity nonlinearly in (54), changing either can change the ordering.

Conditional on patchable exposure, Θ_i pulls the grant value in two directions. A defender with more at stake relative to its search cost uses the window harder because x_{Di}^* rises, but its exploited share p_i falls, leaving less harm for headway to prevent. Both effects bind at the extremes. As $\Theta_i \rightarrow \infty$, $p_i \sim (2\Theta_i)^{-1/2}$ while the bracket in (54) tends to one, so $G_i \rightarrow 0$; as $\Theta_i \rightarrow 0$, $p_i \rightarrow 1$ while $x_{Di}^* \rightarrow 0$, and again $G_i \rightarrow 0$. Since $G_i > 0$ in between, the best recipient is interior in Θ_i .

Two practical readings follow. Patchable exposure should be assigned to the party whose patch

removes the flaw rather than to the party that ultimately bears the loss, so an upstream maintainer may rank highly because one patch protects a large downstream install base. And the ordering is not a ranking on defensive capability: the instinct to hand scarce early access to the strongest security teams is right only up to the point at which their stake–cost ratio has already done the work.⁶⁷

B.4.4 Identification

This subsection asks what evaluation data could estimate κ and test the model’s account of search effort. The model supplies two independent measurements of the same parameter.

Corollary B.8. *Under exclusive access and instantaneous remediation, fix a terminal date H and write $x_H^*(t)$ and $m_H(t)$ for the corresponding optimal intensity and expected surviving stock at date t . (i) Within an evaluation with terminal date H , reciprocal search intensity is affine in time-to-release: $1/x_H^*(t) = 1/x_D^* + (H - t)/\kappa$, with slope $1/\kappa$ and intercept $1/x_D^*$. (ii) Within that evaluation, the surviving stock satisfies $m_H(t)/m_0 = [(1 + x_D^*(H - t)/\kappa)/(1 + x_D^*H/\kappa)]^\kappa$. Across otherwise comparable evaluations whose effort profiles are reoptimised for different terminal dates H , terminal survival is $(1 + x_D^*H/\kappa)^{-\kappa}$, with tail exponent κ . (iii) A model with constant effort and Gamma-distributed heterogeneity in discovery hazards reproduces this cross-deadline terminal-survival curve but predicts a flat effort path.*

Proof. Part (i) is (9). For part (ii), integrating the fixed- H effort profile from 0 to t gives $\exp\{-\int_0^t x_H^*(s) ds\}$ equal to the stated survival ratio; setting $t = H$ gives (10). Varying H across otherwise comparable evaluations, with the effort profile reoptimised for each terminal date, gives $\log(m(H)/m_0) = -\kappa \log(1 + x_D^*H/\kappa)$, so terminal survival is regularly varying at infinity with index $-\kappa$. For part (iii), if effort is inelastic at rate r and r is Gamma distributed with shape κ and scale x_D^*/κ , then $\mathbb{E}[e^{-rH}] = (1 + x_D^*H/\kappa)^{-\kappa}$, reproducing the cross-deadline terminal-survival curve; effort is constant by construction. $\square$

The first measurement comes from inputs: within an evaluation with terminal date H , regressing $1/x_H^*(t)$ on time-to-release $H - t$ gives $1/\kappa$ as the slope. The second comes from outcomes, but the empirical design must be distinguished. With cross-deadline variation, one compares terminal survival across otherwise comparable evaluations whose effort profiles are reoptimised for different terminal dates and estimates the tail exponent in (10). With data from one fixed- H evaluation, one instead fits the within-run survival expression in Corollary B.8(ii). Equation (10) is not the interim survival path of a single evaluation. Equality of the input-side and outcome-side estimates is an overidentifying restriction of the model. For the cross-deadline design, discovery counts alone are not enough, because constant effort applied to heterogeneous flaws can generate the same terminal-survival curve—the standard duration-dependence-versus-frailty problem (Lancaster, 1979; Elbers and Ridder, 1982).

⁶⁷The benefit parameters B and w help determine whether a window should exist and how long it should be; through the chosen h , they can therefore affect the access ranking indirectly. If two candidates cover overlapping code, m_{0i} should be read as the additional stock protected by admitting candidate i .

Effort data break the tie: optimising search predicts effort rising towards release, whereas the constant-effort explanation does not. The input-side data are therefore inference budgets or researcher time together with dated finding logs. The outcome-side design additionally requires a measure or structural estimate of the residual stock; its cross-deadline version requires exogenous deadline variation or evaluations that are otherwise conditionally comparable.⁶⁸

These observations identify κ , not the stock elasticity α . Estimating α requires variation in the number of remaining flaws while holding the search technology and calendar fixed; ordinary evaluation histories do not provide that variation.

B.5 Additional delegation results

The results below collect the fixed-capability elasticities, the general endogenous-capability alignment condition, the calendar bound, the participation cases, and the local welfare comparison omitted from the main statement.

Proposition B.9 (Full delegation characterisation). *Suppose development is active and all choices referred to below are interior.*

(i) *Fix capability. Conditional on building, the delegated date problem differs only by the constant $-\Psi(a)$ from the planner's problem at per-exploit harm $\gamma\lambda$, benefit βB , and full compensation. The immediate-versus-staged choice and the sharpened ceiling (40) therefore apply at those transformed primitives for every $\kappa > 1$; the build decision separately compares the transformed schedule surplus with $\Psi(a)$. As $\gamma \rightarrow \infty$, an active delegated date converges to the smaller root $h_\infty(\beta)$ of $\beta w B e^{-wh} h^\kappa = (\kappa - 1)\kappa^{\kappa-1} m_0 c_D$, provided $w h_\infty(\beta) < \kappa - 1$ and*

$$\beta B e^{-wh_\infty(\beta)} \left(1 - \frac{wh_\infty(\beta)}{\kappa - 1} \right) - \Psi(a) \geq 0. \quad (55)$$

At equality, the developer still builds at every finite rate; if the inequality is reversed strictly, or no viable root exists and $\Psi(a) > 0$, the developer exits for all sufficiently high rates.

(ii) *Continue to hold capability fixed. The delegated and mandated date conditions coincide at a common date if and only if $(1 - \beta)^{\frac{\kappa-1}{\kappa}} M = \beta(1 - \gamma)L$. At $\kappa = 2$, the aligning rate is*

$$\gamma_{\text{align}} = \frac{(1 + p)\beta}{(1 + p)\beta + (1 - \beta)(1 + z/2)}, \quad (56)$$

evaluated at the common date, where such a date exists. Hence $\gamma_{\text{align}} = 1$ at $\beta = 1$; for $\beta < 1$, $\gamma_{\text{align}} \gtrless \beta$ as $z \lesseqgtr 2p$.

(iii) *At the common fixed-capability benchmark $\beta = \gamma = 1$, the elasticity of the delegated date with*

⁶⁸Researcher hours are not themselves the model's per-flaw hazard. If measured labour is $e(t)$ and $x(t) = q(t)e(t)$, where $q(t)$ is researcher and model productivity, the reciprocal-affine restriction applies to $1/[q(t)e(t)]$, not mechanically to $1/e(t)$: hours identify κ from the input path only when $q(t)$ is known or separately estimated and stable over the window.

respect to the rate is ε_λ , while the elasticity of the mandated date is ε_γ . Under quadratic costs,

$$\frac{\varepsilon_\lambda}{\varepsilon_\gamma} = \frac{1 + \chi}{\chi} > 1. \quad (57)$$

(iv) Suppose capability is endogenous and $a^*(h, \gamma)$ is a locally unique differentiable interior response. Define $\widehat{W}(h, \gamma) = W(h, a^*(h, \gamma); \gamma)$ and $\widehat{\Pi}(h, \gamma) = \Pi(h, a^*(h, \gamma); \gamma)$. At any common regular interior mandated and delegated date, the necessary alignment condition is

$$\frac{\partial}{\partial h} \left[(1 - \beta)e^{-wh} B(a^*) - (1 - \gamma)L(h, a^*, \gamma) \right] = 0. \quad (58)$$

Equivalently,

$$(1 - \beta)we^{-wh} B(a^*) = (1 - \gamma)x_D^{\text{eff}} L + W_a a_h^*. \quad (59)$$

With a strict developer date second-order condition, these restrictions are also sufficient for local date implementation. At $\beta = 1$, they reduce to $(1 - \gamma)\widehat{L}_h = 0$.

(v) Fix $\beta < 1$ and let a finite-dimensional vector of primitives ϑ range over an open subset of $\mathbb{R}^d$. Suppose development is active, participation is slack, a^* is smooth, the mandated problem has a locally unique strict interior optimum $x^M(\vartheta) = (h^M(\vartheta), \gamma^M(\vartheta))$ with nonsingular Hessian, and the developer's date second-order condition is strict at every aligned point. Write $\rho(\vartheta) = \widehat{\Pi}_h(x^M(\vartheta); \vartheta)$. If zero is a regular value of ρ , the primitive vectors for which this selected mandated branch is exactly locally implementable form either the empty set or a codimension-one embedded submanifold, and so have empty interior and Lebesgue measure zero.

Proof. For part (i), the contest objects Θ, p, ℓ, x_D^* depend on (γ, λ) only through $\gamma\lambda$, and $M(h, a, \gamma) = m_0 \ell (1 + z/\kappa)^{-(\kappa-1)}$ is the planner's social loss at harm parameter $\gamma\lambda$ and full compensation. The developer's schedule surplus before the fixed development cost is therefore the planner's schedule surplus at harm $\gamma\lambda$ and benefit βB ; subtracting $\Psi(a)$ affects participation but not the conditional date. At the smaller limiting root, $M_\infty(h_\infty, a) = \beta B e^{-wh_\infty} w h_\infty / (\kappa - 1)$, so limiting net developer surplus is the left-hand side of (55). The conditional immediate-versus-staged comparison, ceiling, and argmax statements follow from Proposition 4 at the transformed primitives. If (55) holds with equality, $M(h, a, \gamma) < M_\infty(h, a)$ at every finite rate, so building remains strictly profitable there. With no viable root, Proposition 4(iv) gives a limiting gross supremum of zero, so $\Psi(a) > 0$ makes the limiting net maximum strictly negative. Whenever that limiting maximum is strictly negative, the compact-uniform convergence argument used for Proposition 10 implies exit at all sufficiently high rates.

For part (ii), the developer's date condition is $\beta w e^{-wh} B = \frac{\kappa-1}{\kappa} x_D^{\text{eff}} M$, while the planner's is $w e^{-wh} B = x_D^{\text{eff}} \left[\frac{\kappa-1}{\kappa} M + (1 - \gamma)L \right]$. Eliminating $w e^{-wh} B$ gives the stated condition. At $\kappa = 2$, substituting $M = \gamma L \frac{2}{1+p} (1 + z/2)$ and solving for the rate gives (56); comparison with β gives the final claim.

For part (iii), part (i) makes a change in the delegated rate identical to a change in harm severity in the full-compensation schedule problem. The delegated elasticity is consequently ε_λ , while

Proposition 5 gives ε_γ for the mandate. Substitution from (19) and the definition (31) gives (57).

For part (iv), $\Pi_a(h, a^*(h, \gamma); \gamma) = 0$, so the envelope theorem gives $\hat{\Pi}_h = \Pi_h$, and $\hat{W} - \hat{\Pi} = (1 - \beta)e^{-wh}B(a^*) - (1 - \gamma)L(h, a^*, \gamma)$. At a common regular interior mandated and delegated date, delegation gives $\hat{\Pi}_h = 0$. If developer participation is slack, mandated optimality gives $\hat{W}_h = 0$ directly. If participation binds with multiplier $\theta \geq 0$, the mandated date condition is $\hat{W}_h + \theta\hat{\Pi}_h = 0$, so the delegated condition again implies $\hat{W}_h = 0$. Thus (58) follows in either case. At the delegated date, $\Pi_h = 0$ and $L_h = -x_D^{\text{eff}}L$. Hence $W_h = -(1 - \beta)we^{-wh}B + (1 - \gamma)x_D^{\text{eff}}L$, and imposing $\hat{W}_h = W_h + W_a a_h^* = 0$ gives (59). Alternatively, the mandated date condition is $we^{-wh}B = x_D^{\text{eff}}[\frac{\kappa-1}{\kappa}M + (1 - \gamma)L] + W_a a_h^*$; substituting it into the developer's date derivative gives $\hat{\Pi}_h = x_D^{\text{eff}}[(1 - \beta)\frac{\kappa-1}{\kappa}M - \beta(1 - \gamma)L] - \beta W_a a_h^*$, which is zero exactly when $(1 - \beta)\frac{\kappa-1}{\kappa}M = \beta(1 - \gamma)L + \beta W_a a_h^*/x_D^{\text{eff}}$. This also shows why $W_a = 0$ is not an additional implementation condition: at a common (h, γ) the same response $a^*(h, \gamma)$ obtains under both regimes. At $\beta = 1$, $W_a = -(1 - \gamma)L_a$, so every form reduces to $(1 - \gamma)[-x_D^{\text{eff}}L + L_a a_h^*] = 0$. The strict developer second-order condition turns the date first-order condition into a local maximum.

For part (v), write $G(x; \vartheta) = D_x \hat{W}(x; \vartheta)$; nonsingularity of $D_x G$ and the implicit-function theorem give a locally unique differentiable map $x^M(\vartheta)$. Let $r(x; \vartheta) = \hat{\Pi}_h(x; \vartheta)$, so $\rho(\vartheta) = r(x^M(\vartheta); \vartheta)$. Under the maintained developer second-order condition, this selected mandated branch is locally implementable if and only if $\rho(\vartheta) = 0$. If zero is a regular value, the regular level-set theorem makes $\rho^{-1}(0)$ either empty or a codimension-one embedded submanifold. Such a set has empty interior and Lebesgue measure zero. The statement does not aggregate over other mandated branches; global implementation additionally requires global developer optimality, participation, and an equilibrium-selection rule. $\square$

Under quadratic costs, the composite-harm derivative in part (iv) can be written $\hat{L}_h/L = -x_D^{\text{eff}} + [\sigma_L - \frac{z}{1+z/2}\sigma_R]a_h^*/a^*$; it is negative whenever $L_a a_h^* \leq 0$. At full appropriability, local implementation below full compensation otherwise requires the adverse capability effect to offset direct protection exactly; for $\beta < 1$, the uncaptured-benefit term in (59) also enters. Geometrically, conditional on a fixed capability and on building, lower appropriability moves the developer towards longer windows and eventual exit, while a higher rate moves it along the transformed hyperbola in (15). With endogenous capability, the high-rate limiting package instead solves $\max_{h,a} \{\beta e^{-wh}B(a) - \Psi(a) - M_\infty(h, a)\}$; convergence of the joint date requires a unique active interior maximiser.

The delegated rate derivative. Let $h_D(\gamma)$ denote the delegated date and $V^D(\gamma) = \hat{W}(h_D(\gamma), \gamma)$. Then $dV^D/d\gamma = \hat{W}_\gamma + \hat{W}_h h'_D(\gamma) = W_\gamma + W_h h'_D(\gamma) + W_a(a_\gamma^* + a_h^* h'_D(\gamma))$. At an unconstrained implemented mandated optimum, $\hat{W}_h = \hat{W}_\gamma = 0$; with binding participation, the corresponding Lagrangian conditions are (36)–(37). In neither case is $W_a = 0$ a separate implementation requirement.

Proposition B.10 (Local welfare advantage of a mandate). *Let $\beta = 1$. For each $\gamma > 0$, select a global developer-optimal outcome, allowing $a_D(\gamma) = \emptyset$; when development is active, denote its package*

by $(h_D(\gamma), a_D(\gamma))$. Let $V^D(\gamma)$ be the welfare that outcome generates, with no development valued at zero. Suppose the full-compensation package is uniquely selected, active, interior, and yields strictly positive surplus. Then no rate above one improves delegated welfare over full compensation, and any attained delegated-regulator optimum lies below one.

For the local comparison, suppose in addition that, near $\gamma = 1$, $a^*(h, \gamma)$ and the mandated and delegated dates $h_M(\gamma)$ and $h_D(\gamma)$ are unique, active, interior, twice continuously differentiable, and satisfy strict second-order conditions. Define $V^M(\gamma) = \widehat{W}(h_M(\gamma), \gamma)$ locally, and let $h_0 = h_M(1) = h_D(1)$, $\mathcal{H} = \widehat{W}_{hh}(h_0, 1) < 0$, and $\widehat{L}_h^0 \equiv \partial L(h, a^*(h, 1), 1) / \partial h|_{h=h_0}$. Then $(V^D)'(1) = -L\chi < 0$ and

$$V^M(\gamma) - V^D(\gamma) = -\frac{(\widehat{L}_h^0)^2}{2\mathcal{H}}(\gamma - 1)^2 + o((\gamma - 1)^2). \quad (60)$$

Thus the local advantage is second order and strictly positive when $\widehat{L}_h^0 \neq 0$; regime or participation corners can make it first order.

Proof. Consider any $\gamma > 1$. If its globally selected developer package is active, Proposition 8 gives $V^D(\gamma) < W(h_D(\gamma), a_D(\gamma); 1) \leq \max_{h,a} W(h, a; 1) = V^D(1)$. If instead the developer exits, $V^D(\gamma) = 0 < V^D(1)$ by the strictly positive full-compensation surplus. Thus no supra-compensatory rate improves on full compensation. To evaluate the derivative at full compensation without imposing smoothness on the selected policy functions, let $P(\gamma) \equiv \max\{0, \max_{h,a} \Pi(h, a; \gamma)\}$ denote the developer's value. At $\beta = 1$, $W(h, a; \gamma) - \Pi(h, a; \gamma) = (\gamma - 1)L(h, a, \gamma)$, so in a neighbourhood of the uniquely selected active full-compensation package, $V^D(\gamma) = P(\gamma) + (\gamma - 1)L(h_D(\gamma), a_D(\gamma), \gamma)$. Danskin's theorem and uniqueness at $\gamma = 1$ give $P'(1) = \Pi_\gamma(h_D(1), a_D(1); 1) = -M_\gamma(h_D(1), a_D(1); 1)$. Consequently, $(V^D)'(1) = -M_\gamma + L = -L\chi < 0$, where the final equality uses (31). Hence full compensation is not itself optimal, and any attained delegated-regulator optimum lies below one.

For the local gap, a Taylor expansion around the mandated date at a common rate gives

$$V^M(\gamma) - V^D(\gamma) = -\frac{1}{2}\widehat{W}_{hh}(h_M(\gamma), \gamma)[h_D(\gamma) - h_M(\gamma)]^2 + o([h_D(\gamma) - h_M(\gamma)]^2).$$

At $\beta = 1$, $\widehat{W} - \widehat{\Pi} = (\gamma - 1)\widehat{L}$. Differentiating the two date conditions at $\gamma = 1$ therefore yields $h'_D(1) - h'_M(1) = \widehat{L}_h^0 / \mathcal{H}$, and substitution gives (60). $\square$

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